

The musculoskeletal system

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- The study of the structure, function , and diseases of muscles is called **myology**.
- ❖ *It consist of cells (fibers) called **myocytes***
- Muscle is a tissue whose cells have the **ability to contract, producing movements or force**.
- The forces generated move limbs, inflate the lungs, pump blood, close and open tubes, etc.
- In man, muscle tissue constitutes **40-50%** of the body mass.

The Functions of Muscles

- Movement
- Maintaining posture (Stability)
- Communication
- Control of body openings and passages
- Maintaining body temperature
(Heat production)

Architecture of muscles

is the physical arrangement of muscle fibers at the macroscopic level that determines a muscle's mechanical function.



- **Pennate muscles** are feather-like in the arrangement of their fascicles (fiber bundles): unipennate, bipennate, or multi-pennate.
- **Fusiform muscles** are spindle shaped (round, thick belly, and tapered ends).eg *biceps brachi*

- **Parallel muscles** are the fascicles w/c lie parallel to the long axis of the muscle.
- **Convergent muscles** have a broad attachment from which the fascicles converge to a single tendon.
- **Circular muscle** surrounds a body opening or orifice constricting it when contracted.

Three pennate muscles

a Extensor digitorum muscle



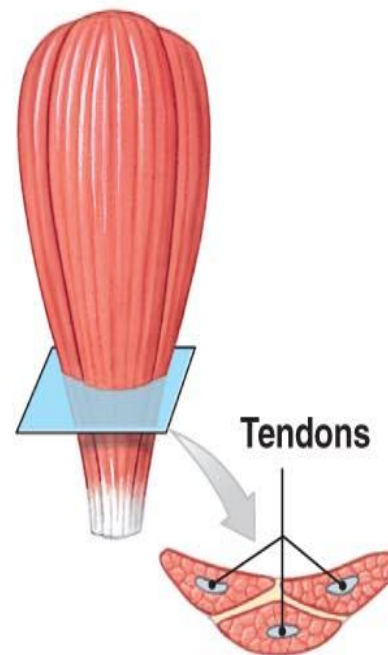
If all the muscle fibers are on the same side of the tendon, the pennate muscle is **unipennate**.

b Rectus femoris muscle

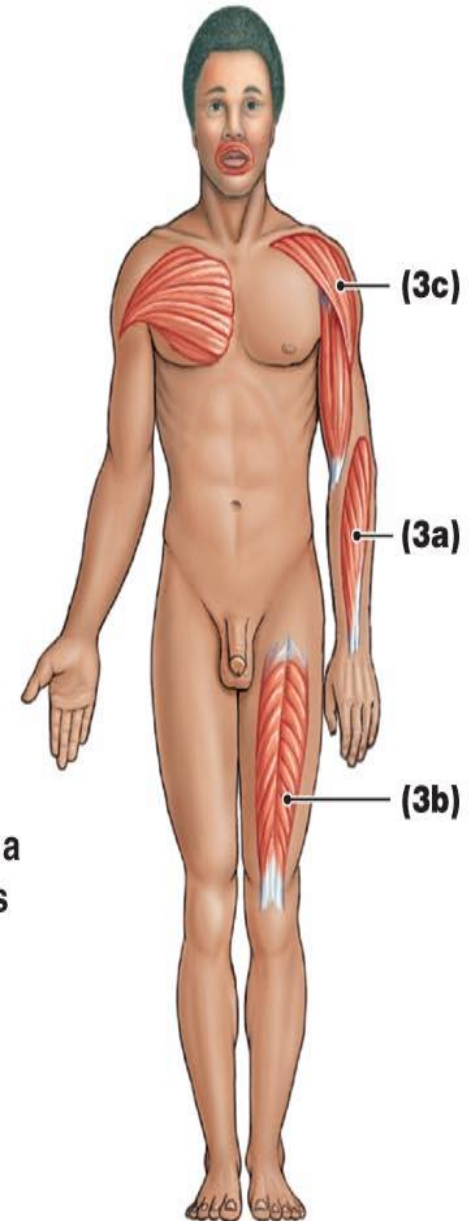


If a pennate muscle has fibers on both sides of the tendon, it is called **bipennate**.

c Deltoid muscle



If the tendon branches within a pennate muscle, the muscle is said to be **multipennate**.



Pectoralis major



(b) Convergent

Sartorius



(c) Parallel

Extensor digitorum longus



(d) Unipennate

Orbicularis oris



(a) Circular

Deltoid



(e) Multipennate

Biceps brachii

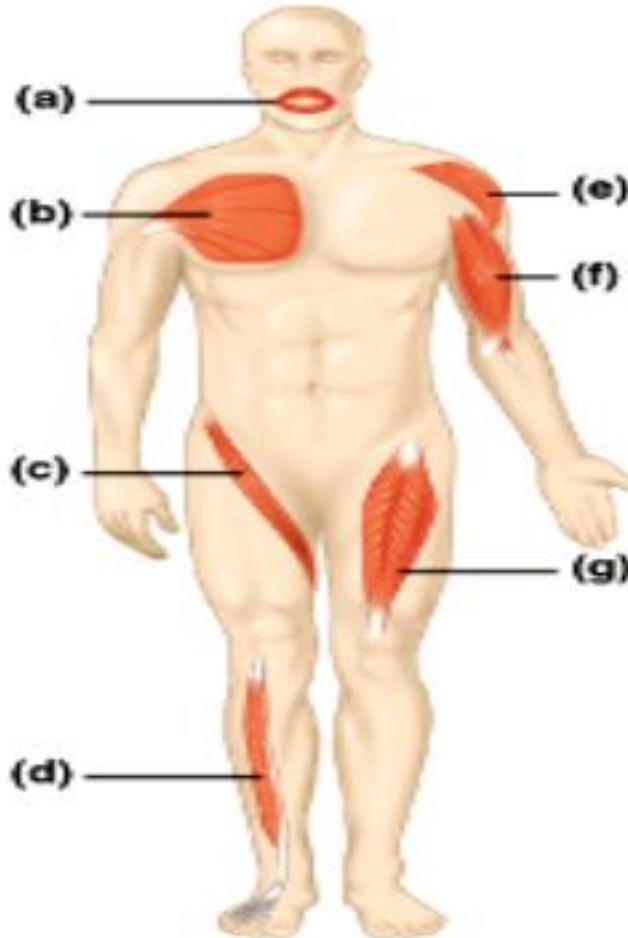


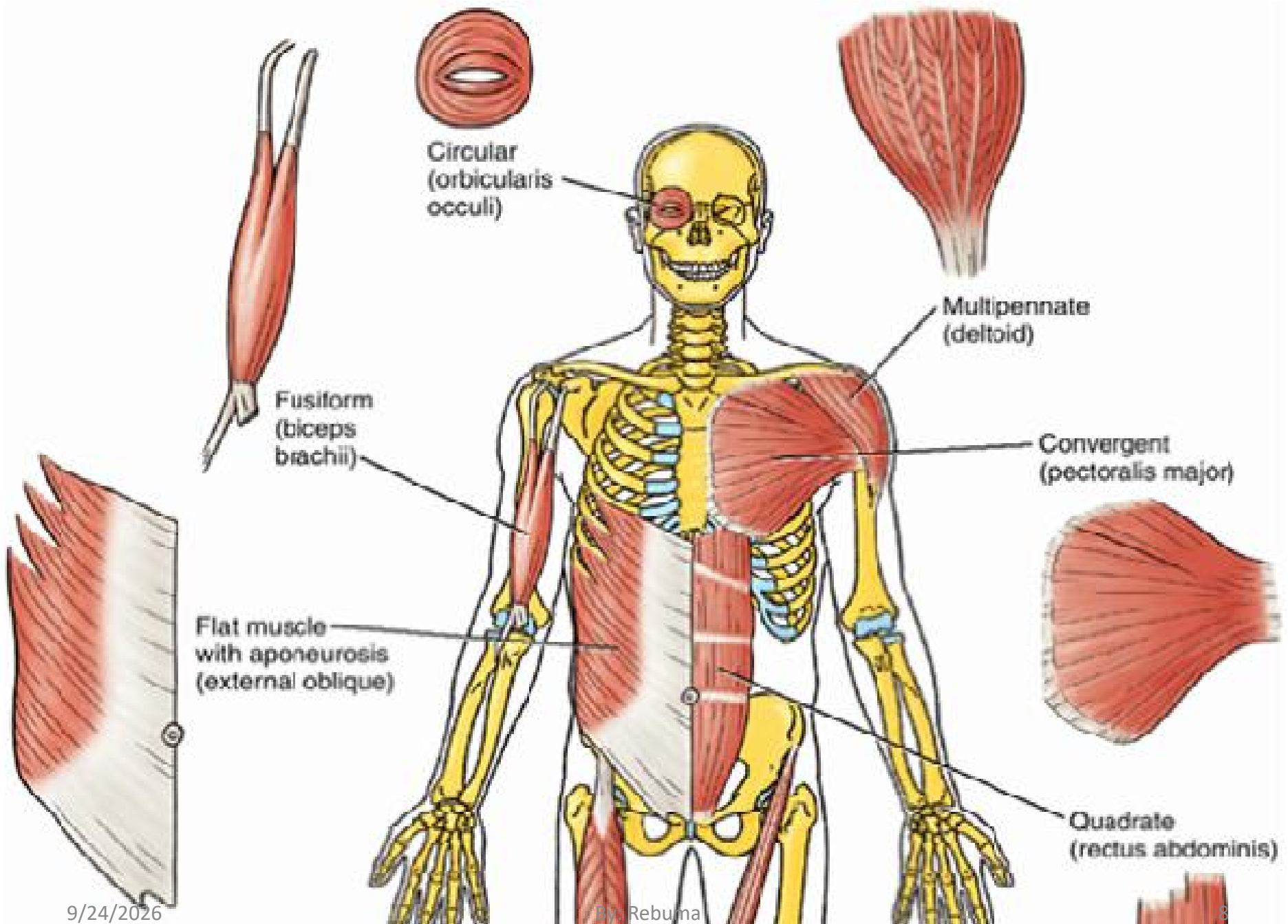
(f) Fusiform

Rectus femoris

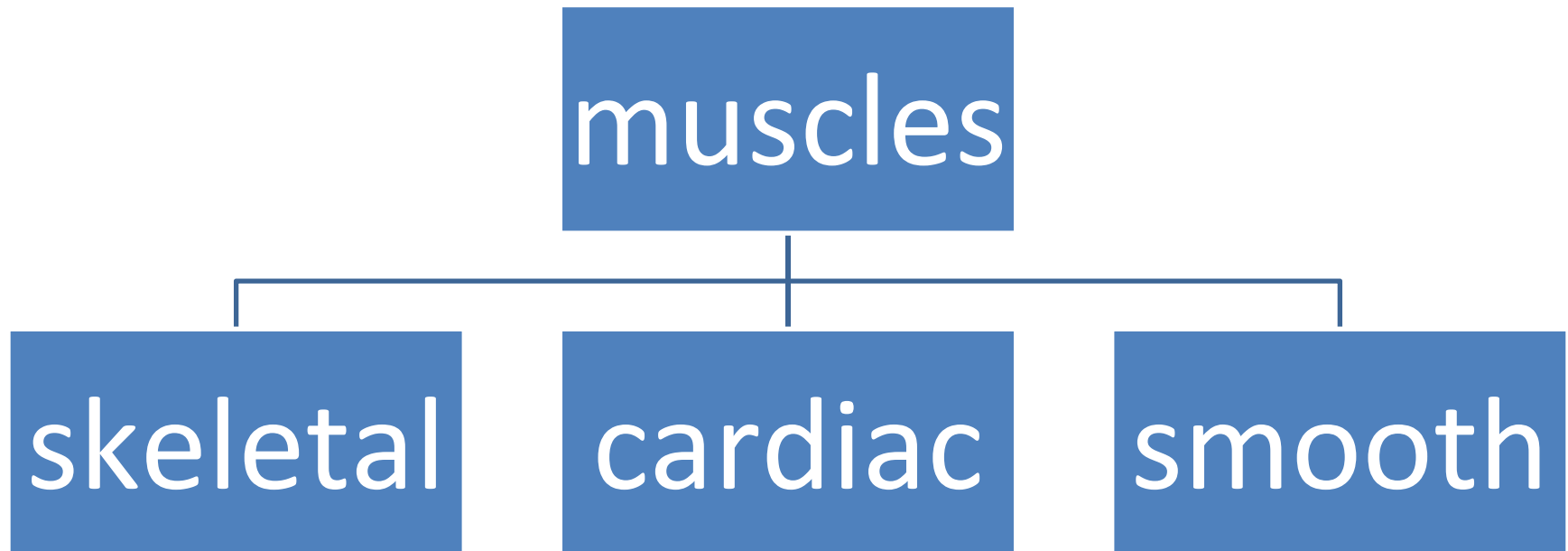


(g) Bipennate



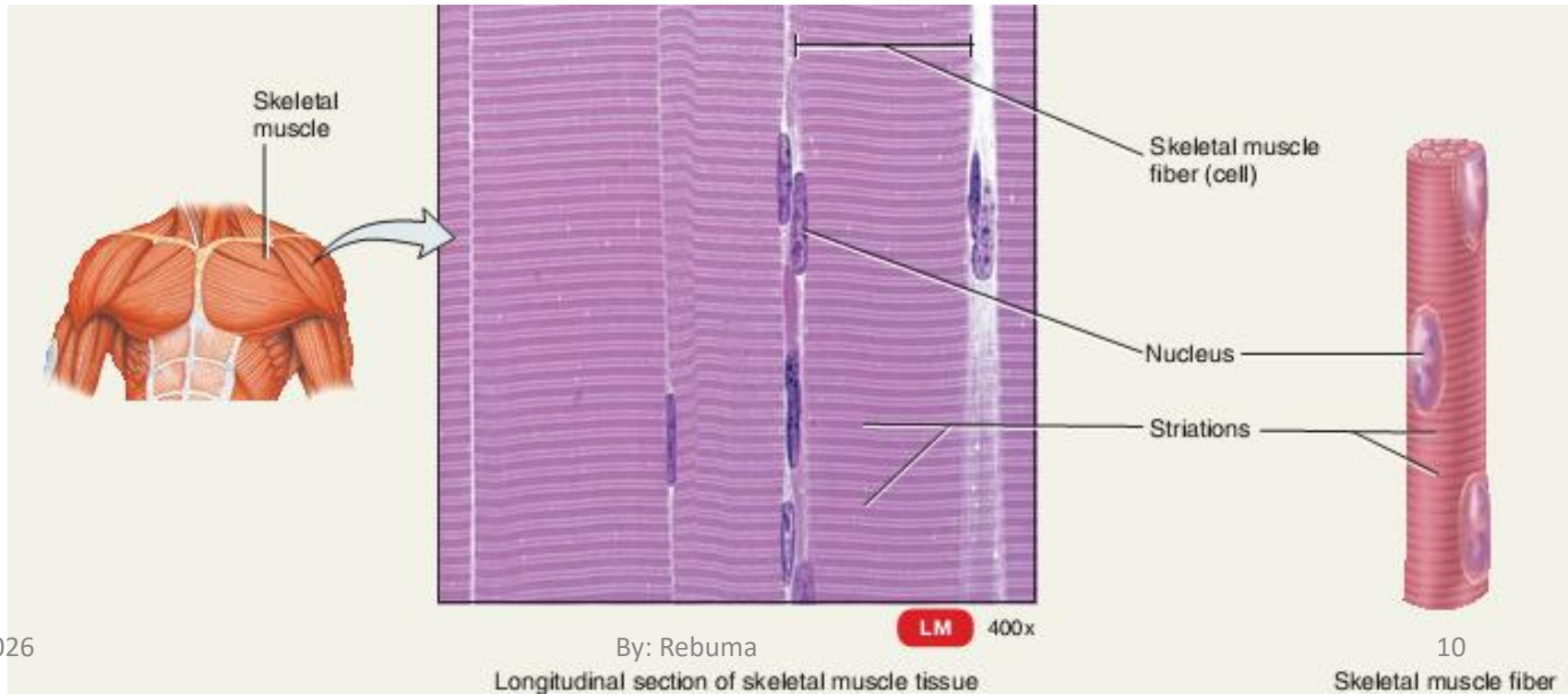


Classification of muscles based on histological structure



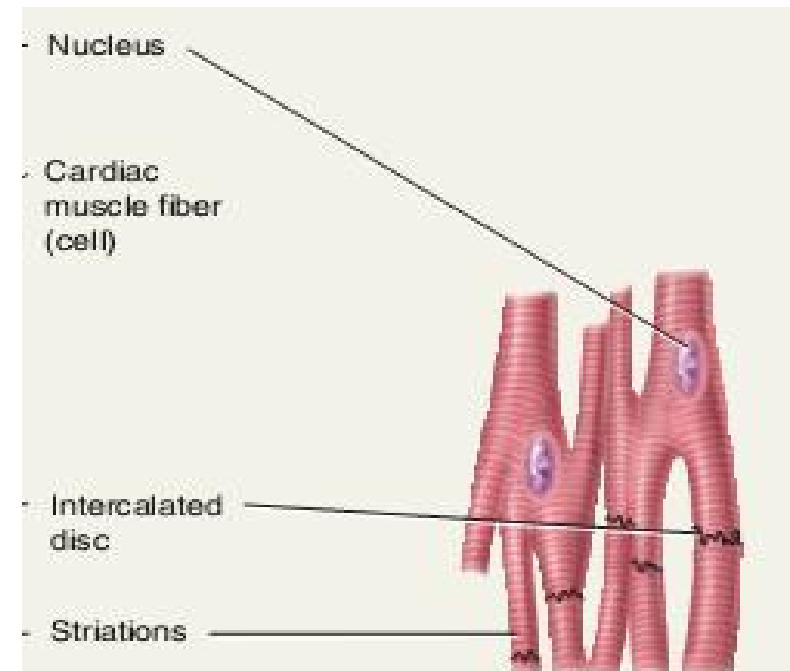
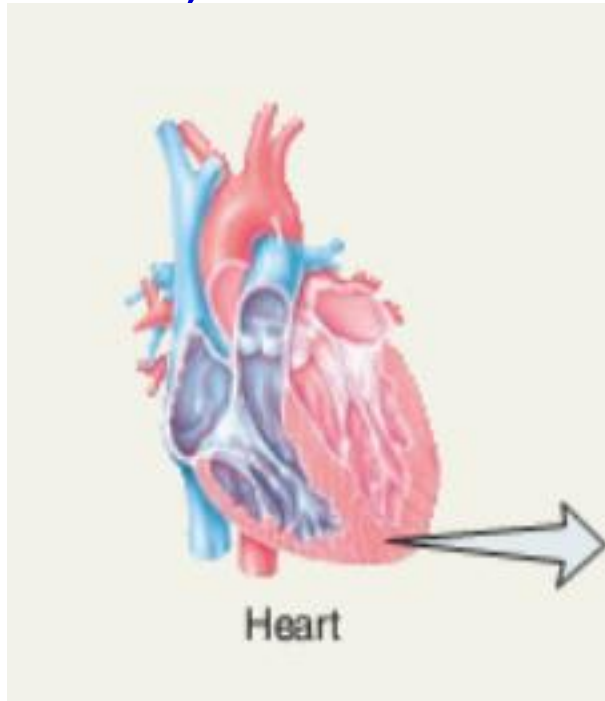
Skeletal Muscle

- *Seen attached to bones of limbs & body walls.*
- *Transverse striations –striated muscle*
- *Voluntary in nature.*
- *Long & cylindrical fibers running parallel to each other.*
- *Elongated & peripherally placed nuclei*



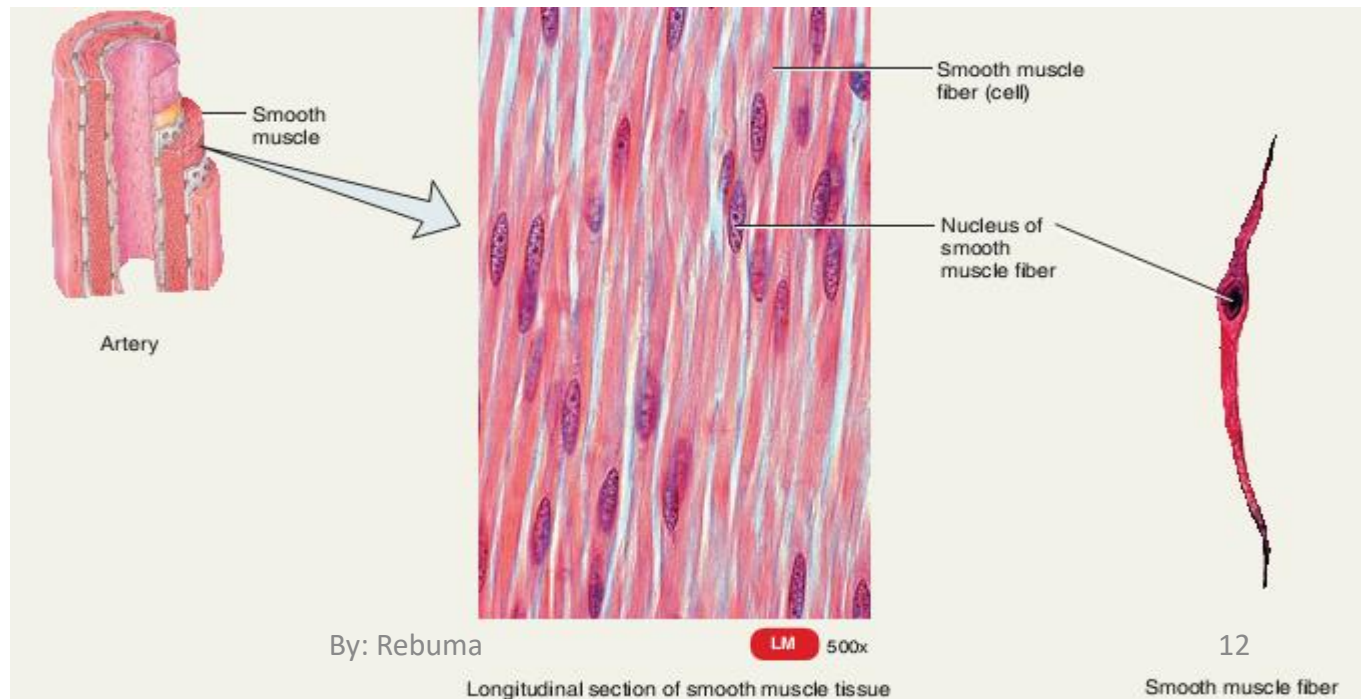
Cardiac Muscle

- *Elongated branching & anastomosing fibers*
- *Transverse striations-striated muscles*
- *Centrally located nucleus*
- *Involuntary in nature*
- *Isolated cardiac myocytes contracts spontaneously in a rhythmic manner.*



Smooth muscle

- *Seen in the walls of viscera including stomach, intestine, the urinary bladder, uterus, blood vessels.*
- *Non striated*
- *Involuntary in nature*
- *Long spindle shaped fibers.*
- *Broad central part & tapering ends*
- *Oval or elongated nucleus.*

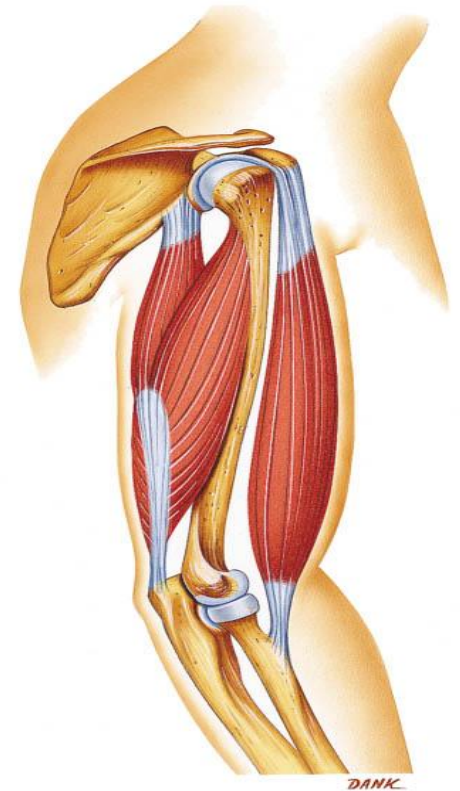


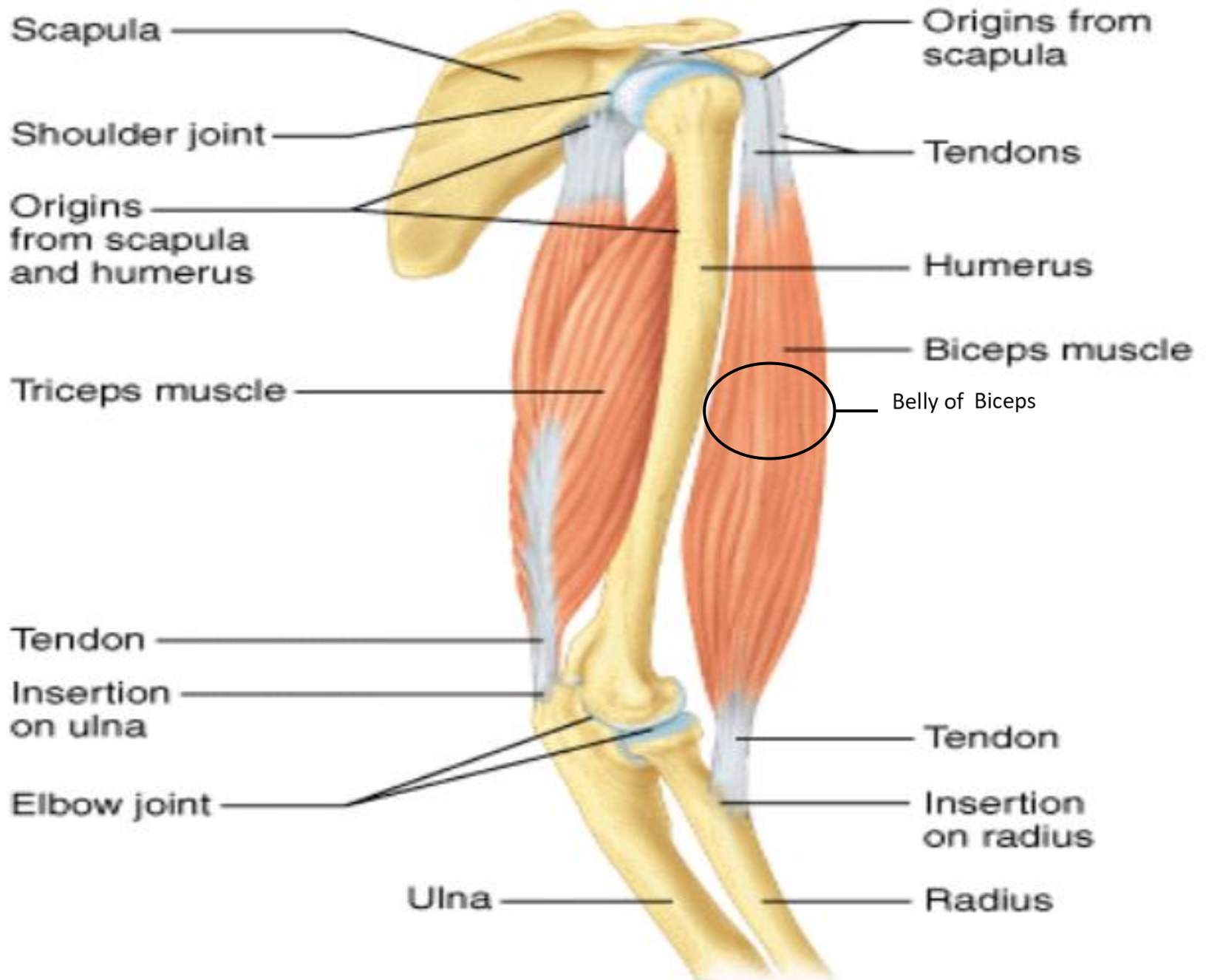
Skeletal Muscle structure

- **Bellies:** have a fleshy contractile portion composed of skeletal striated muscle
(Many muscles are narrow at the origin and insertion and have a thicker middle region)
- **Tendons:** (rounded) and aponeuroses (flat sheets) white non-contractile portion composed mainly of collagen bundles

Skeletal Muscle structure...

- **Origin:** is usually the proximal end of the muscle, which remains fixed during muscular contraction
- **Insertion:** is usually the distal end of the muscle, which is movable





Naming of muscles

- ❑ Most muscles are named on the basis of their **function** or the bones to which they are attached (abductor digiti minimi)
- ❑ On the basis of their **shape** (G. deltoid, triangular)
- ❑ On the basis of their **position** (vastuslateralis, lateral)
- ❑ On the basis of their **length** (brevis, short; longus, long),
- ❑ On the basis of their **size** (maximus, minimus)
(major, minor)

❑ on the basis of their **number of heads or bellies**

Biceps (two heads)

Triceps (three heads)

Quadriceps (four heads)

Digastric (two bellies)

❑ on the basis of **Attachment**

- Sternocleidomastoid (from *sternum and clavicle* to *mastoid process*)

- Coracobrachialis (from the *coracoid process* to the *arm*)

❖ **Muscle fiber**: the structural unit of a muscle

Classification according to the arrangement of muscle fiber

- **Unipennate muscles-**
 - The fibers are arranged in one side of the tendon
 - Eg:- Palmar interossei
- **Bipennate muscles-**
 - The fibers arranged on two sides of the tendon.
 - Eg:- Dorsal interossei
- **Multi pennate muscle -**
 - The fibers are branched & large number of fasciculi packed in a relatively small total volume.
 - Eg:- Deltoid

UNIPENNATE

Fascicles arranged on only one side of tendon.



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Example: Extensor digitorum longus muscle

BIPENNATE

Fascicles arranged on both sides of centrally positioned tendons.

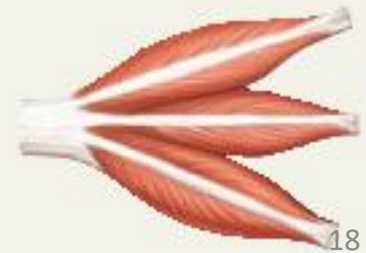


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Example: Rectus femoris muscle (see

MULTIPENNATE

Fascicles attach obliquely from many directions to several tendons.



Example: Deltoid muscle (see

Depending upon the action.

- *Prime movers [agonists]*
 - *These muscles are constantly active in initiating and maintaining a specific movement.*
 - *is the muscle acting according to the desired movement.*
- *Antagonists*
 - *These are the muscles, which oppose the movement produced by prime movers.*

- **Synergists**

- *These are the muscles, which help to prevent unwanted movements due to action of prime movers or helps action of prime mover.*

- **Fixators**

- *These are muscles, which stabilize a joint so other muscles can act on it.*
- *These are the muscles which stabilize one attachment of a muscle so that the other end may move.*

Muscle Actions during Elbow Flexion

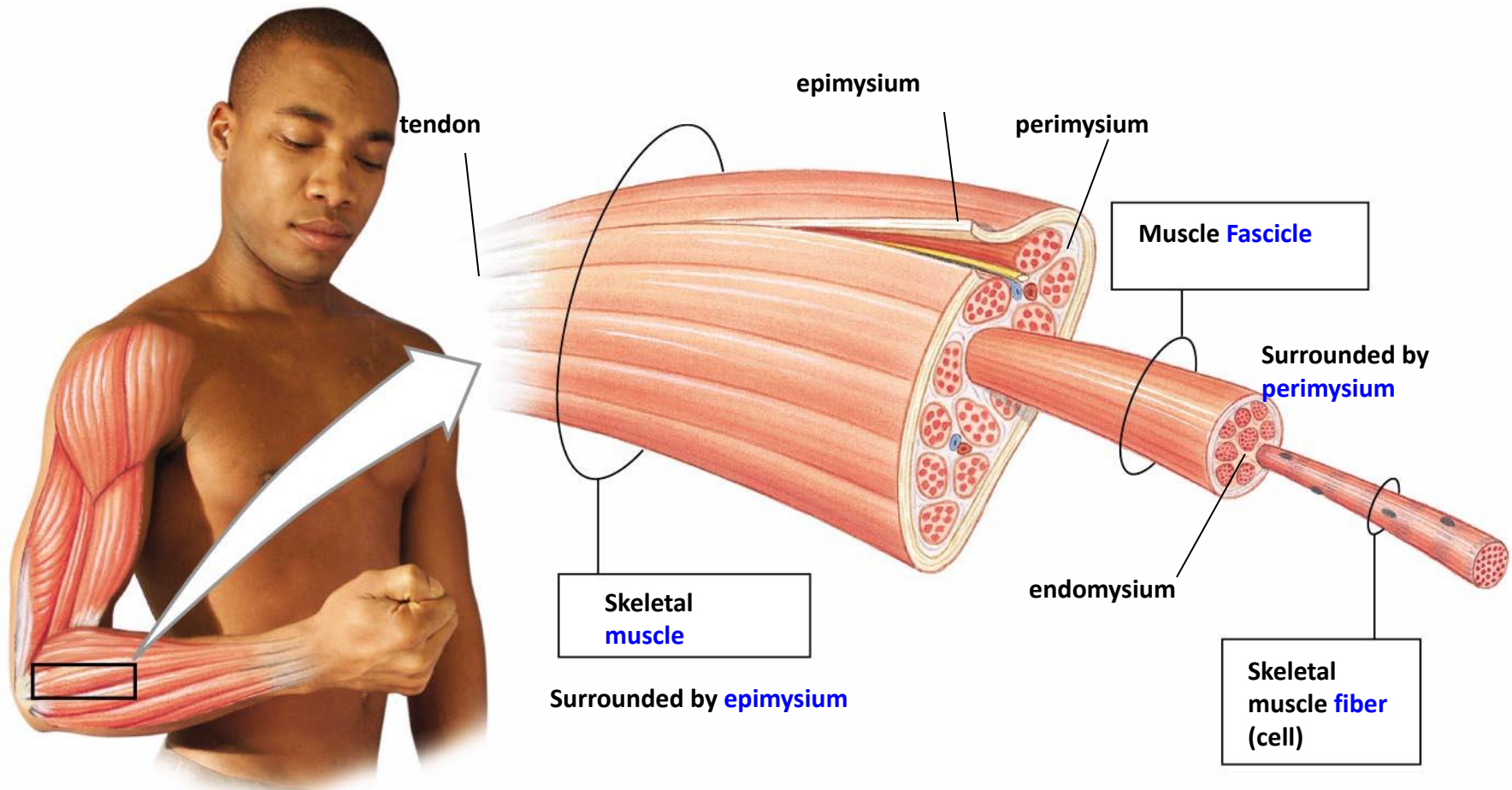


- Prime mover (agonist) = brachialis
- Synergist = biceps brachii
- Antagonist = triceps brachii
- Fixator = muscle that holds scapula firmly in place
 - rhomboideus m.

Connective tissue of skeletal muscle

- Each muscle fiber(cell) is surrounded by very delicate connective tissue called endomysium.
- The muscle fibers(cells) are arranged in bundles called fasciculi.
- Each fasciculus is bounded by another denser connective tissue layer called perimysium.
- The whole muscle is formed by many fasciculi, and enclosed in a fibrous sheath called epimysium.

Anatomy of skeletal muscles



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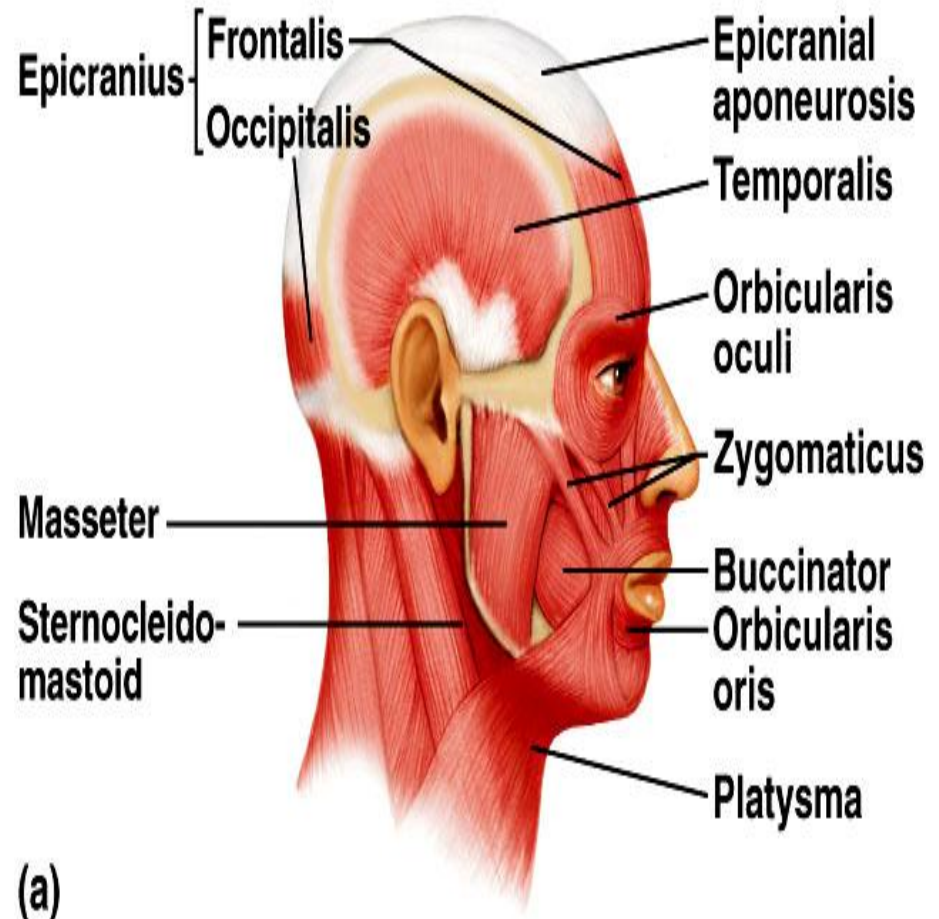
Muscle of Axial Skeletal

Muscle of Axial Skeletal

Muscles of Facial Expression

- Masseter- Chew
- Orbicularis oris- purses lips
- Risorius-sardonic-smile
- Zygomaticus major- laughing muscle

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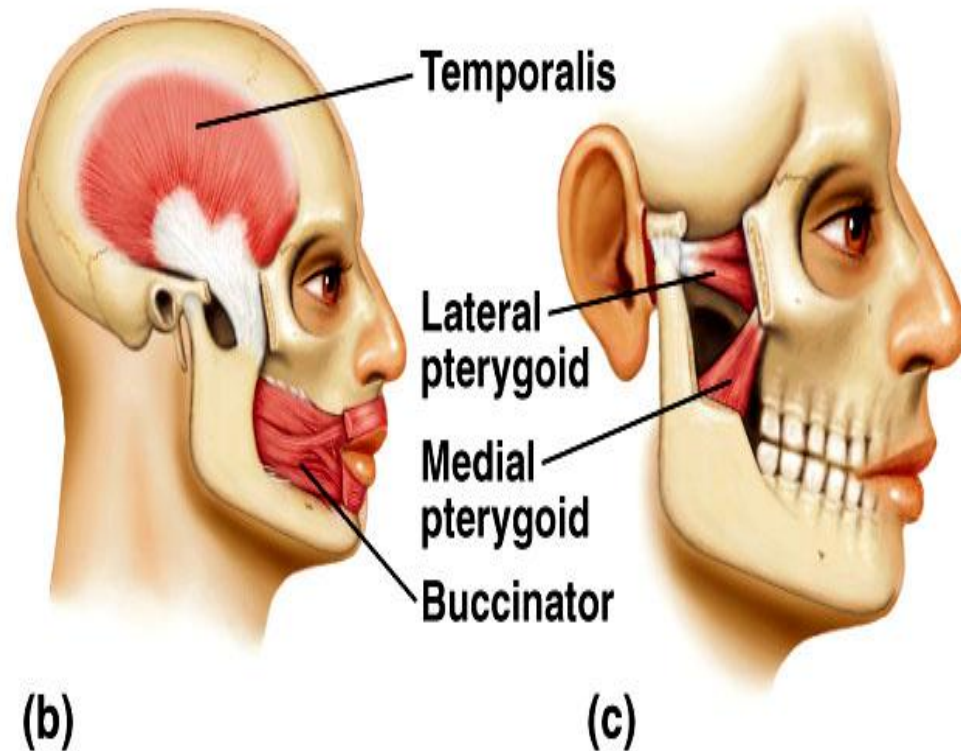


Muscle of Axial Skeletal...

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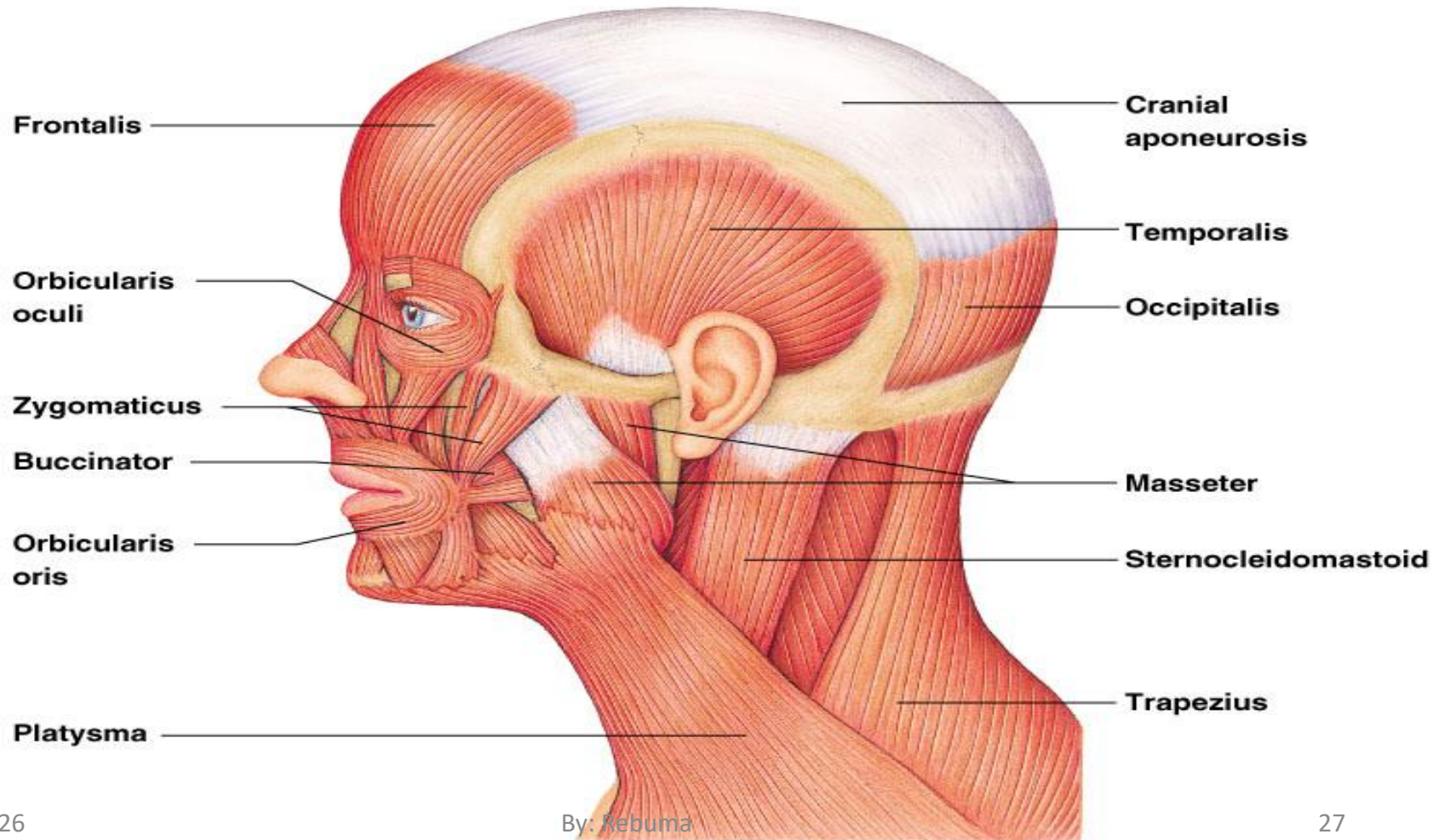
Muscles of Mastication

- Masseter
- Temporalis
- Lateral Pterygoid
- Medial Pterygoid



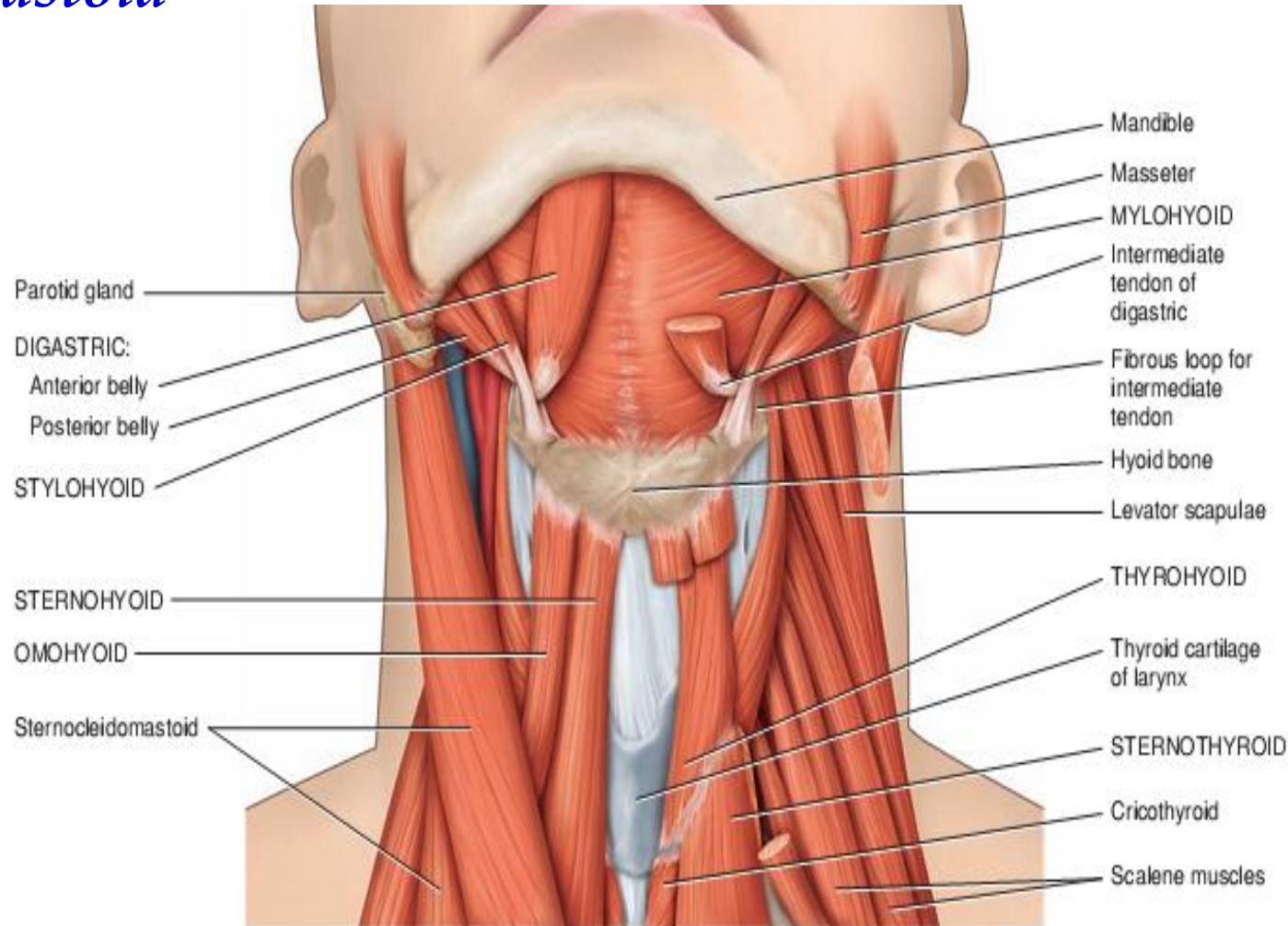
Muscle of Axial Skeletal...

Head and Neck Muscles



Muscles of the Anterior Neck

- *Sternocleidomastoid*
- *Digastric*
- *Stylohyoid*
- *Mylohyoid*
- *Geniohyoid*
- *Omohyoid*
- *Sternohyoid*
- *Thyrohyoid*



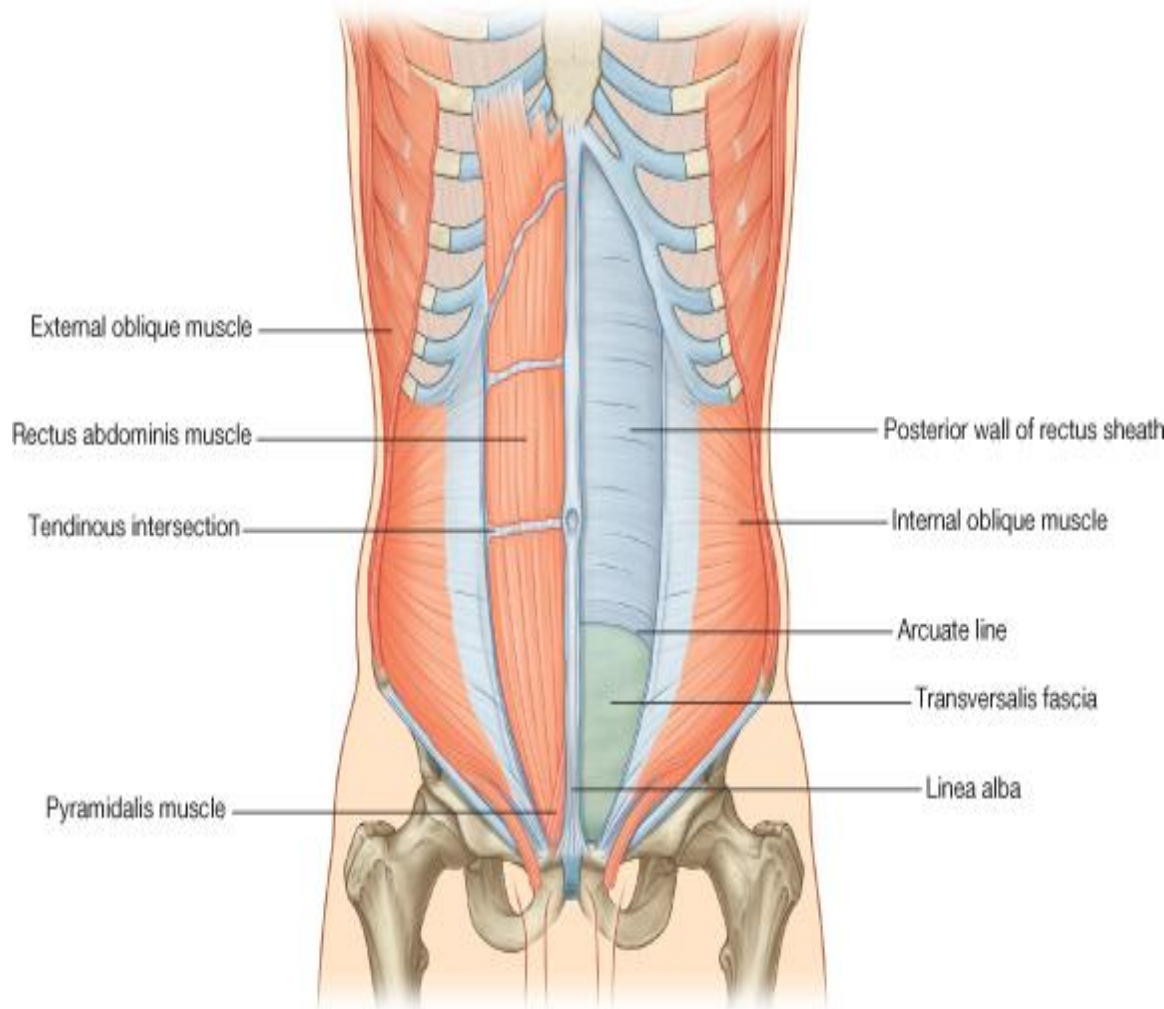
(a) Anterior superficial view

(b) Anterior deep view

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Muscles of the Abdomen

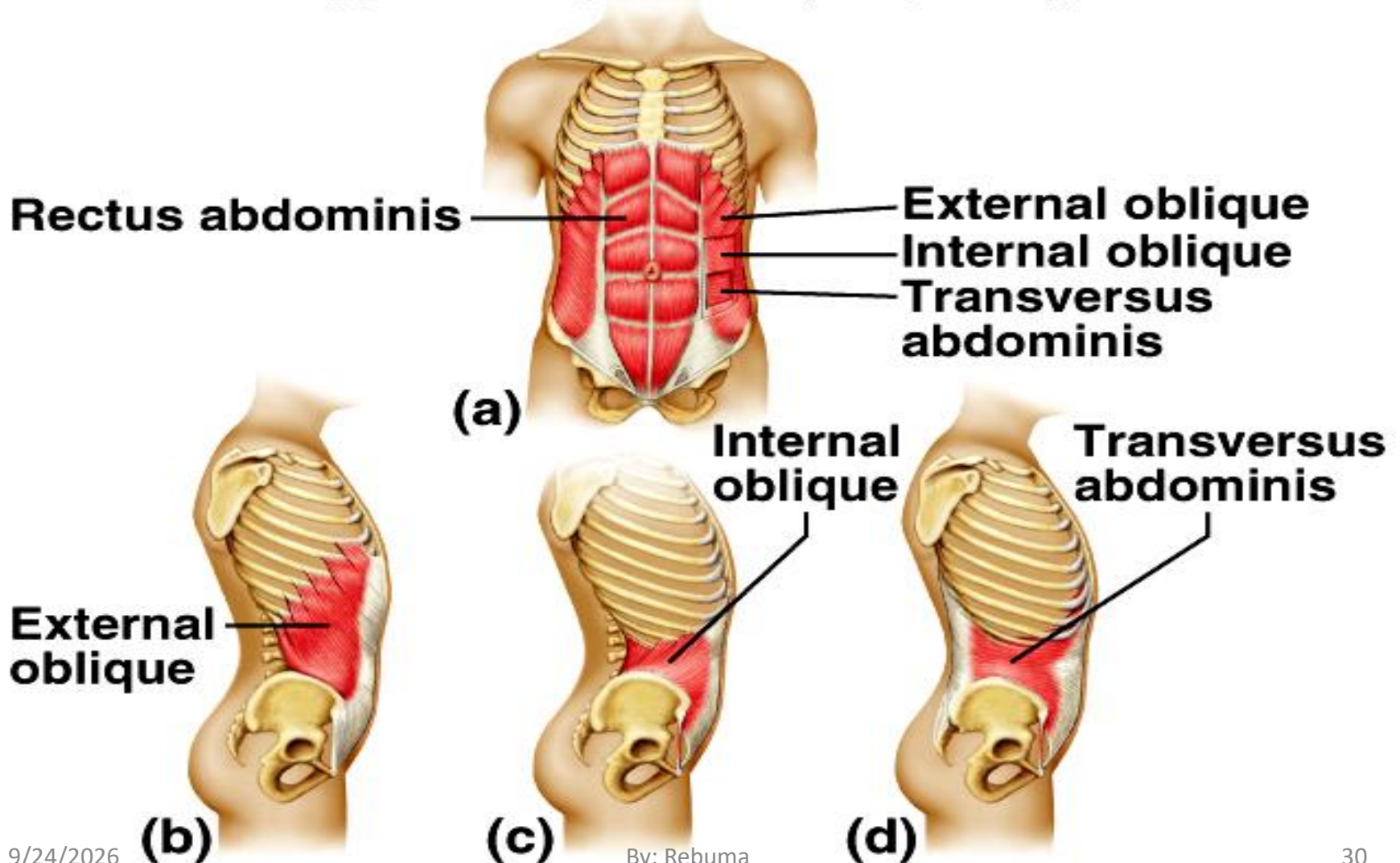
- *Rectus abdominis*
- *External oblique*
- *Internal oblique*
- *Transversus abdominis*
- *Quadratus lumborum*



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Antero-lateral Abdominal Muscles

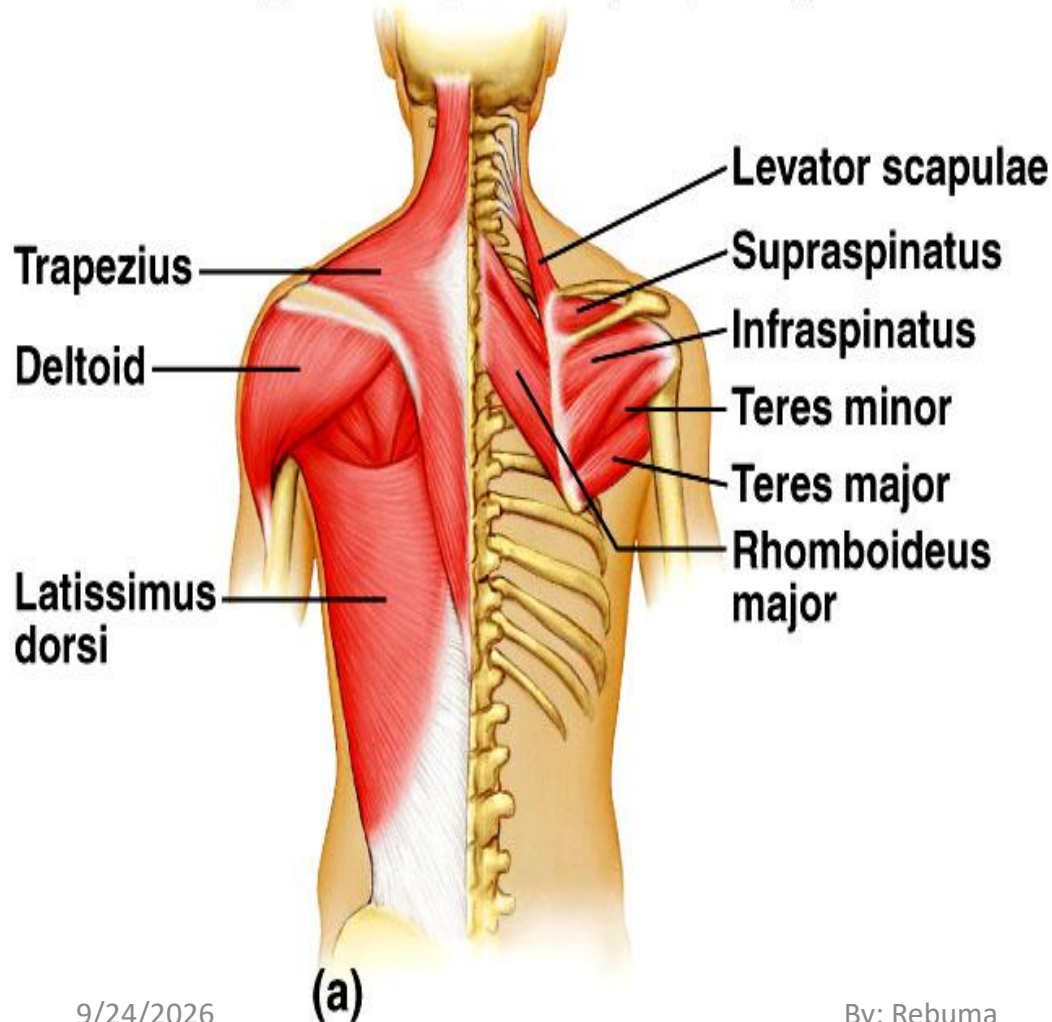
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Muscle of Appendicular Skeletal

Muscles of the Shoulder and Back

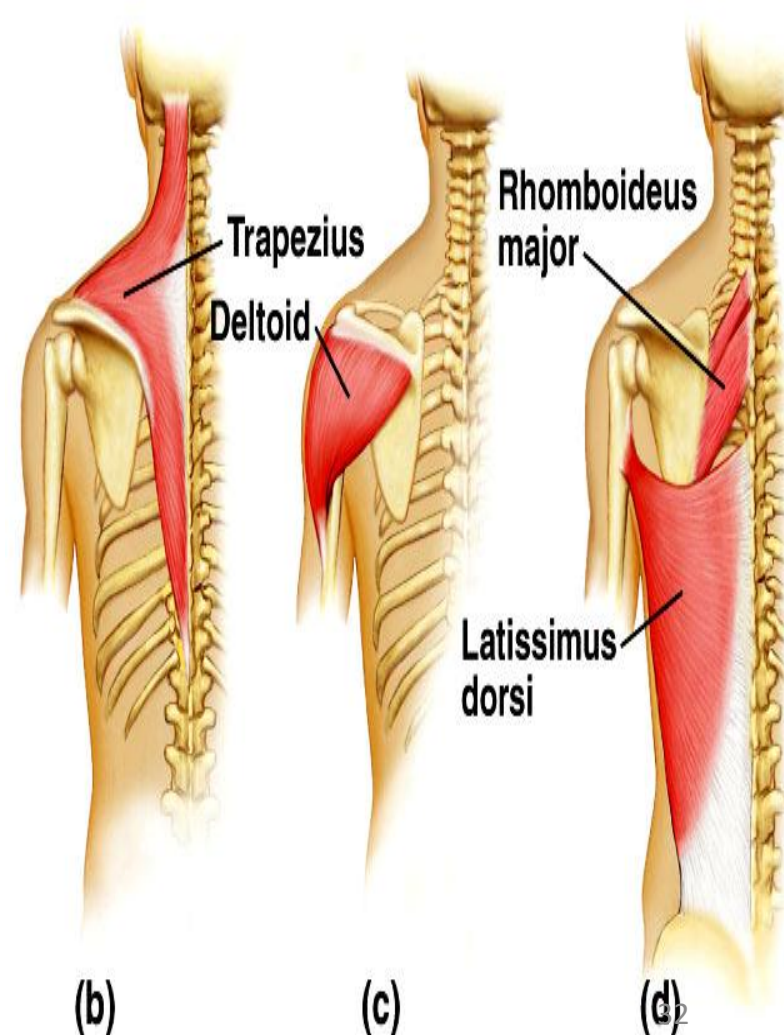
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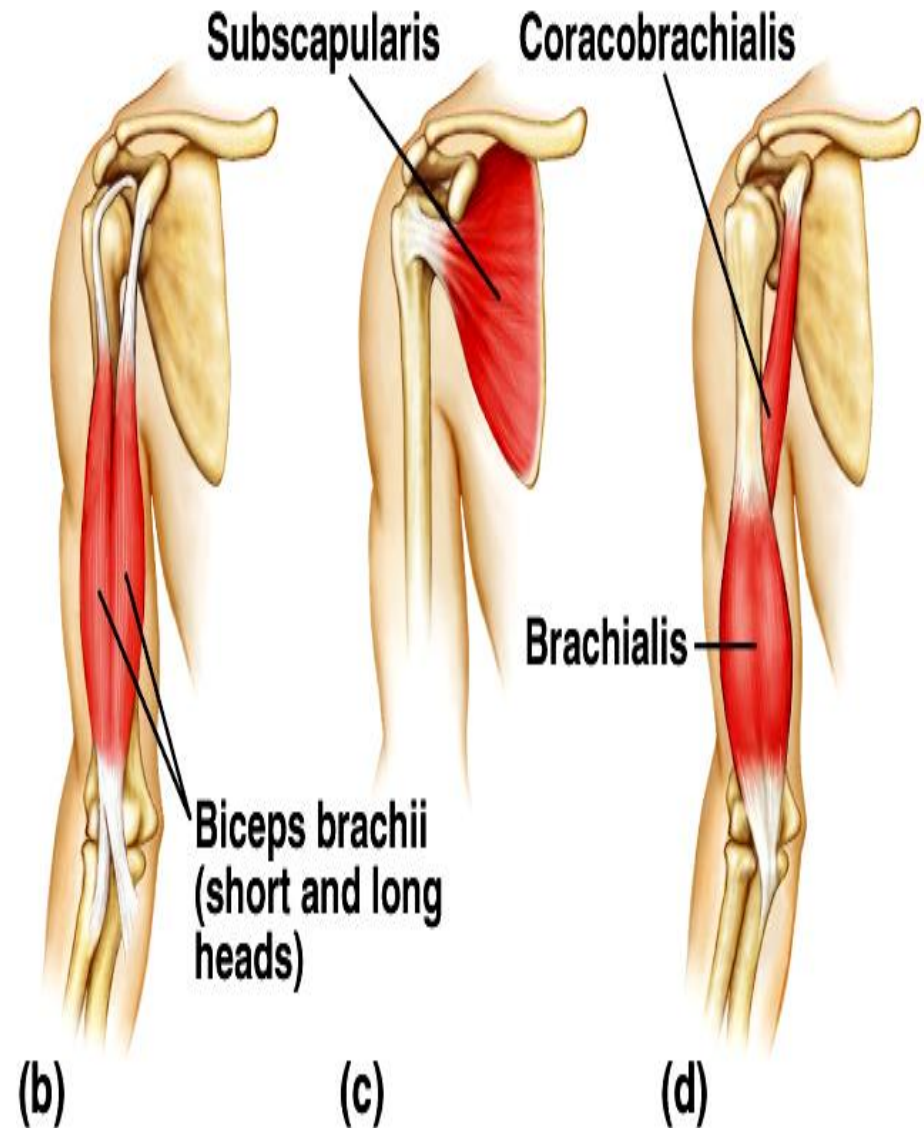
Arm muscles

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- Anterior compartment

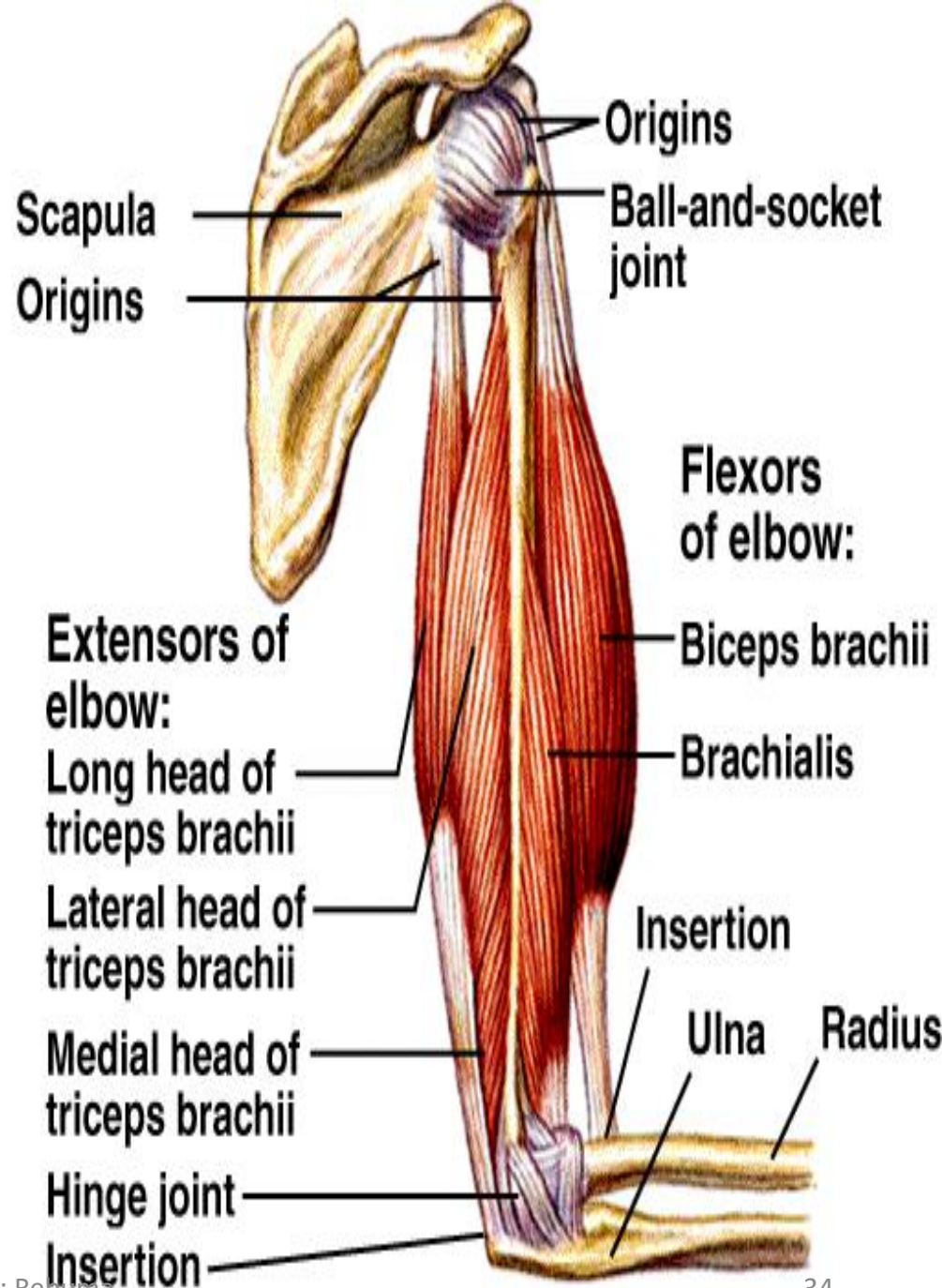
1. Biceps brachii
2. Brachialis
3. Coracobrachialis

1,2 – flexor of elbow joint
3- flexor of shoulder joint



Arm muscles

- Posterior compartment
 - Triceps brachii
 - Action: extensors of elbow



Forearm muscles

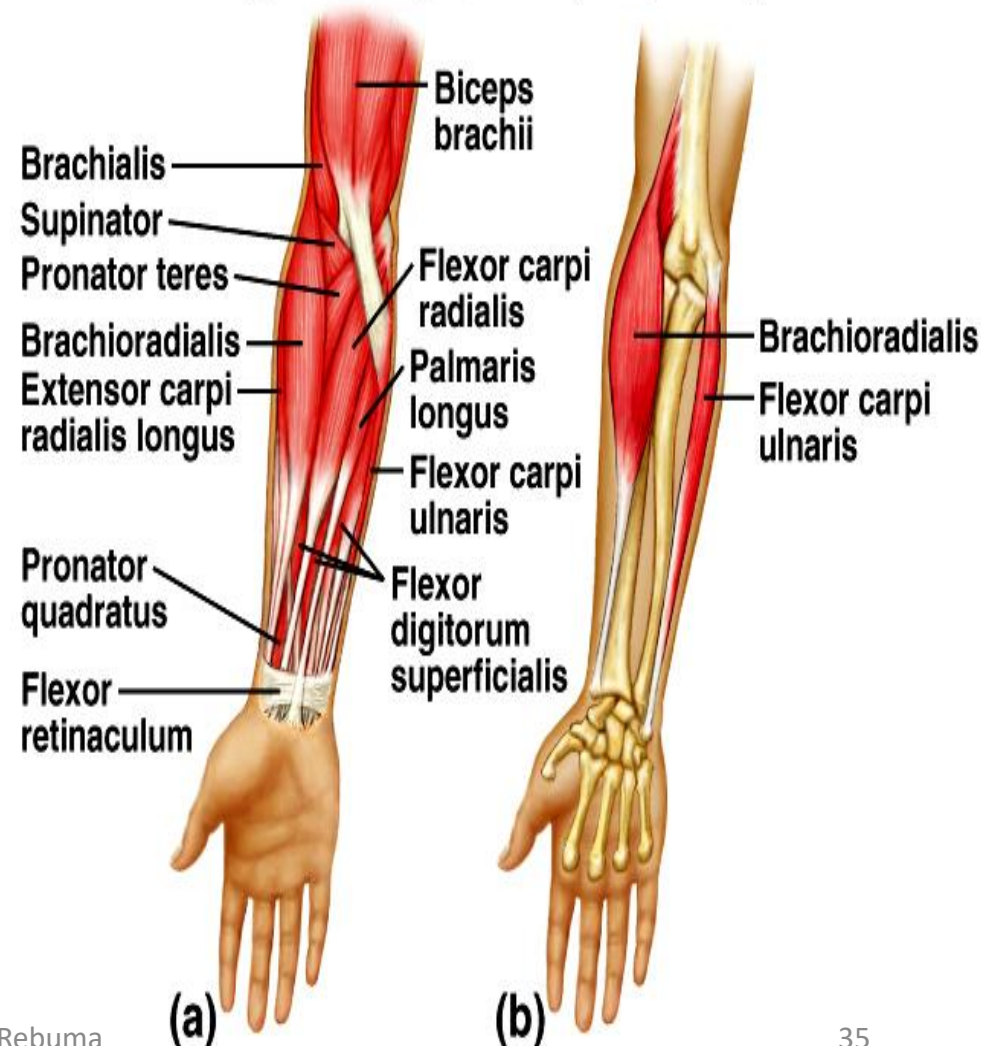
- Anterior Forearm

- Superficial groups

- pronator teres
- flexor carpi radialis
- palmaris longus
- flexor carpi ulnaris

- Predominant action of this group is:
flex wrist and digits.

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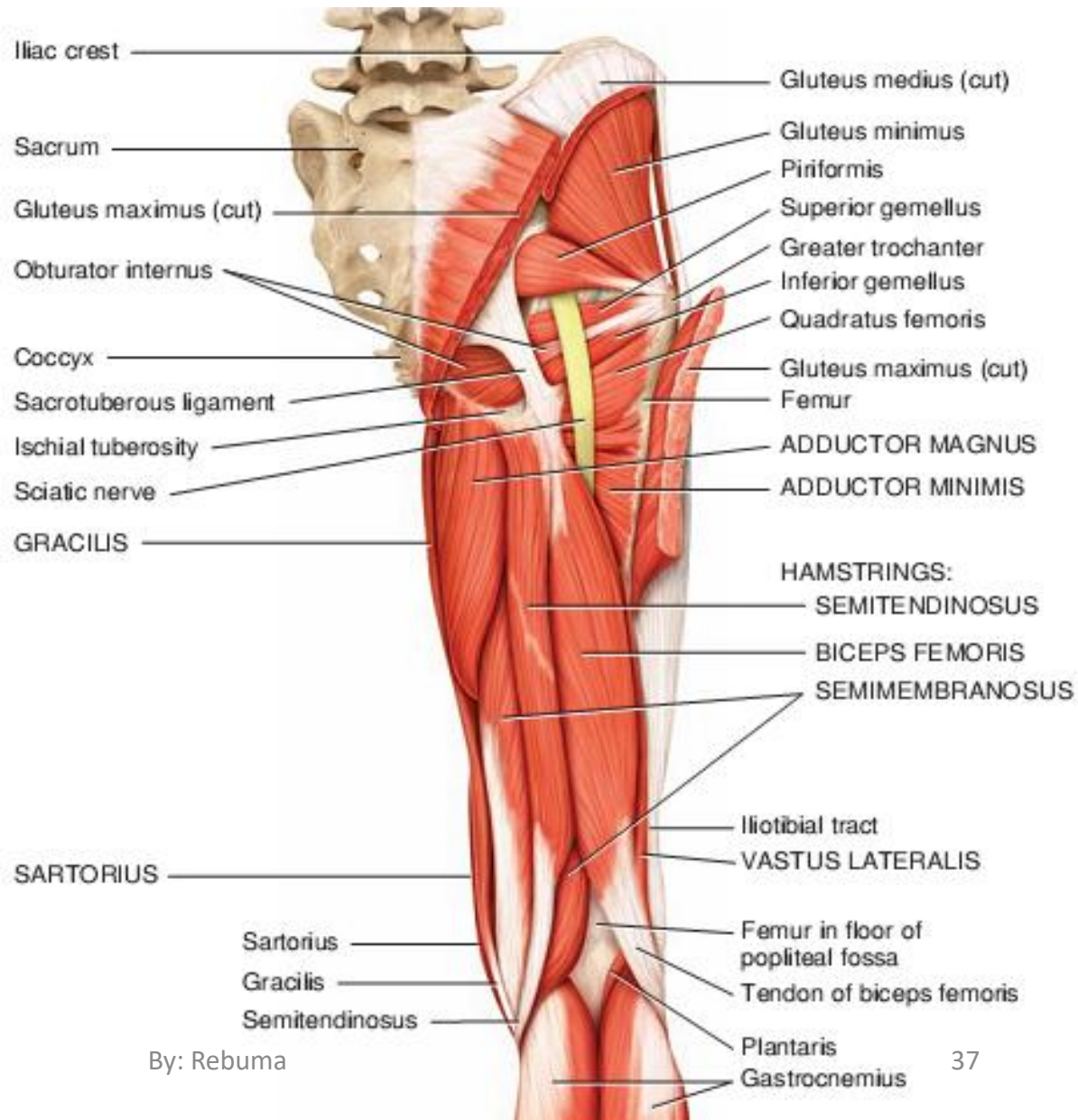
Muscles of the hand

- Thenar muscles- act on **thumb**
 - Abductor pollicis brevis
 - Flexor pollicis brevis
 - Opponens pollicis
- Hypothenar muscles- act on **little finger**
 - Abductor digiti minimi
 - Flexor digiti minimi
 - Opponens digiti minimi

Muscles of gluteal region

They are:-

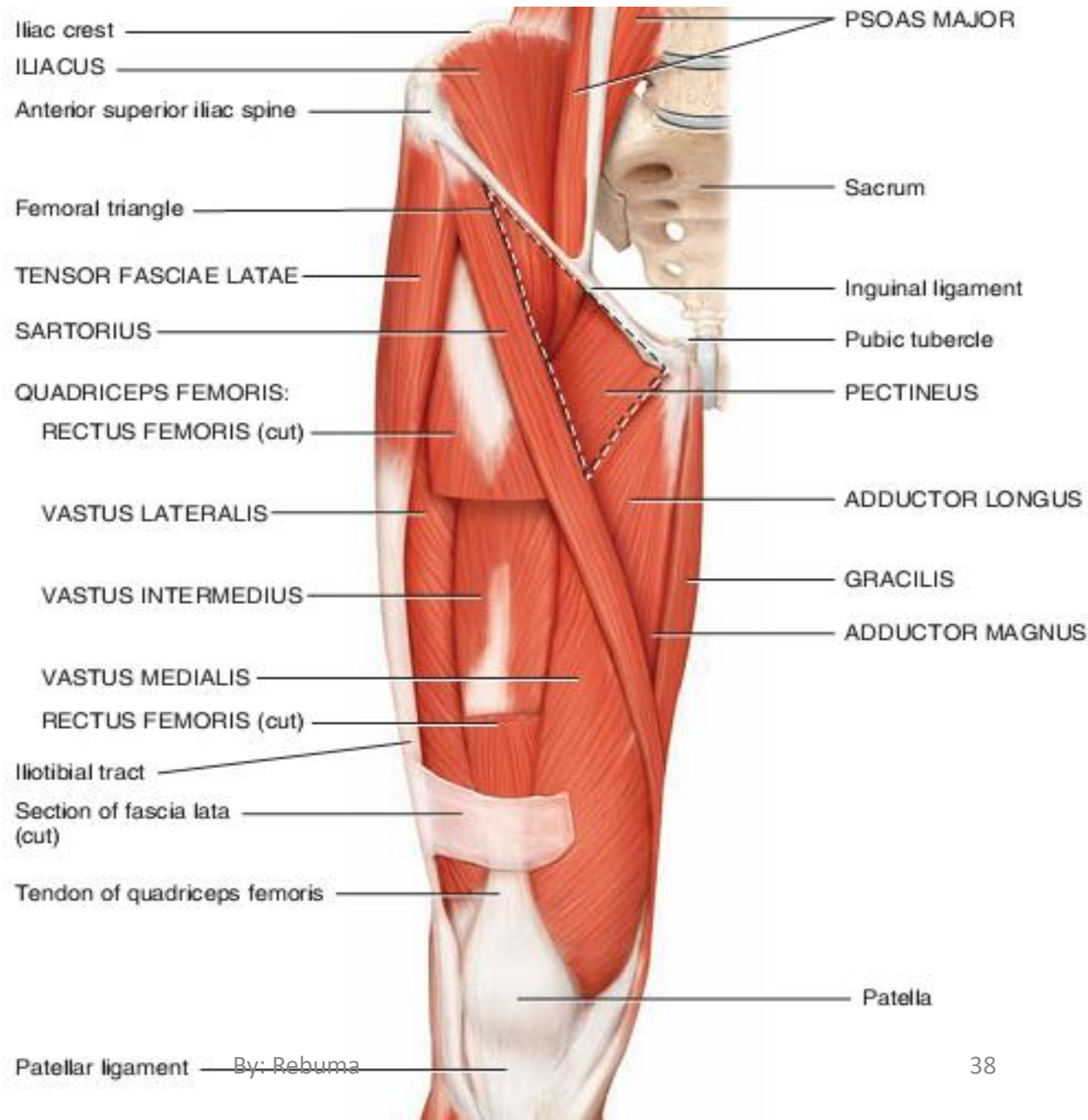
- ✓ *Gluteus maximus.*
- ✓ *Gluteus medius.*
- ✓ *Gluteus minimus.*
- ✓ *Piriformis.*
- ✓ *Superior gemellus.*
- ✓ *Inferior gemellus.*
- ✓ *Obturator internus.*
- ✓ *Quadratus femoris.*



Anterior Compartment muscle of Thigh

They are:-

- ✓ *Sartorius.*
- ✓ *Iliacus.*
- ✓ *Psoas major.*
- ✓ *Pectineus.*
- ✓ *Quadriceps Femoris.*



Medial Compartment Muscles of the Thigh

✓ *They are:-*

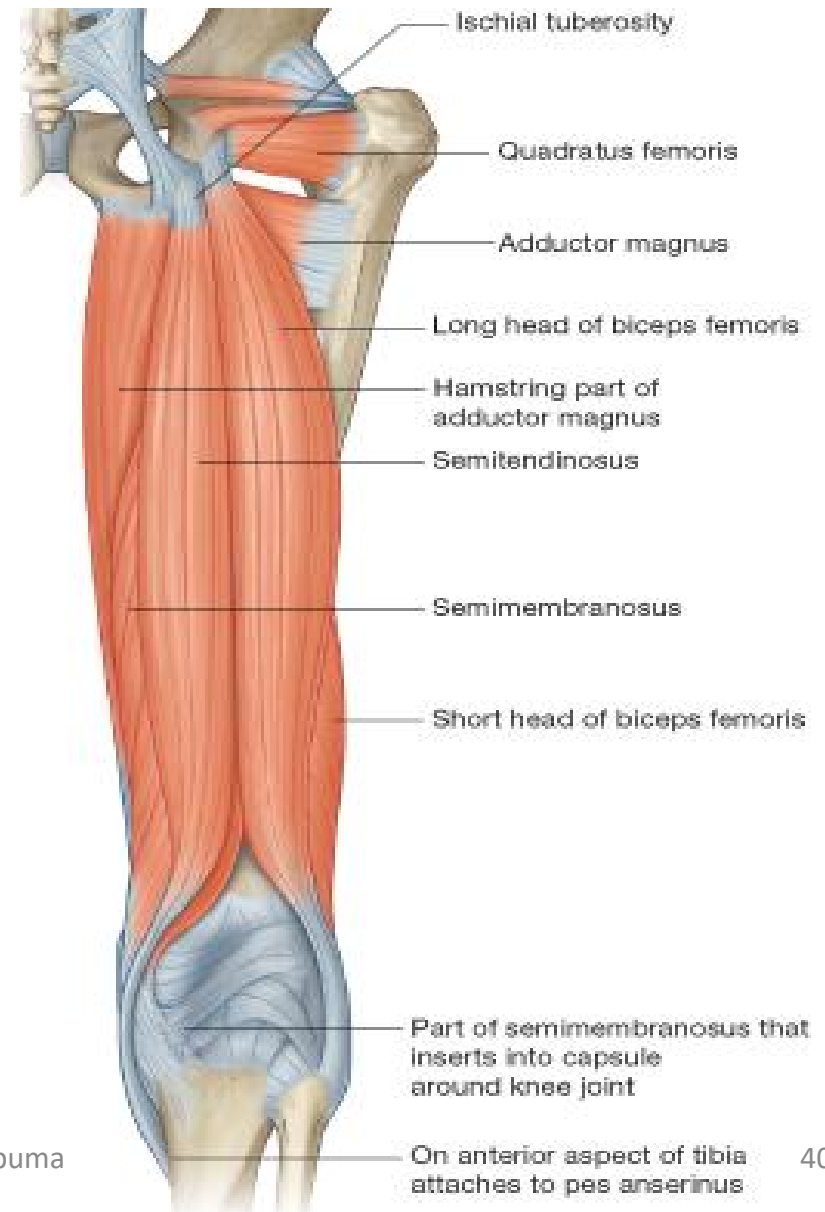
- ✓ *Gracilis.*
- ✓ *Adductor longus.*
- ✓ *Adductor brevis.*
- ✓ *Adductor magnus.*
- ✓ *Obturator externus*



Posterior Compartment Muscles of the Thigh

They are:-

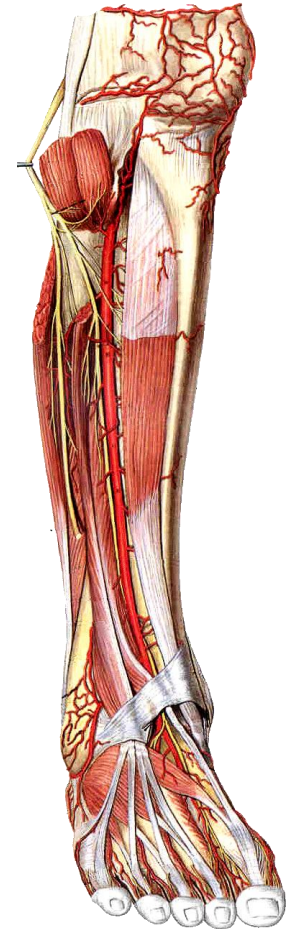
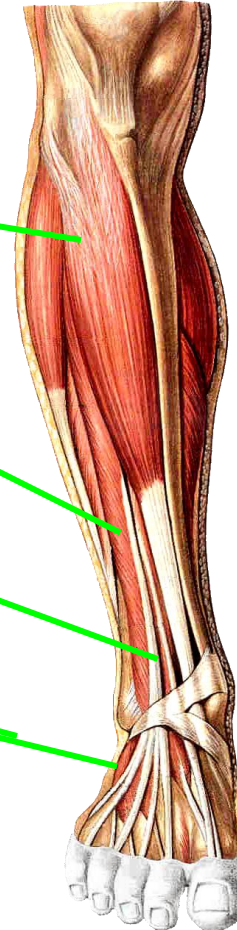
- ✓ *Biceps femoris.*
- ✓ *Semitendinosus.*
- ✓ *Semimembranosus.*



Muscles of the anterior compartment of the leg

Muscles:

- **Tibialis anterior** ;deepest & most anterior muscle.
- **Extensor digitorum longus**
-the muscle attaching lateral 4 toes
- **Extensor hallucis longus**
- the muscle attaches greater toe.
- **Peroneus tertius**: originate from distal surface fibula & enter to dorsum of 5th MC



Posterior Compartment muscles of the Leg

They are:-

✓ **Gastrocnemius.**

✓ **Soleus**

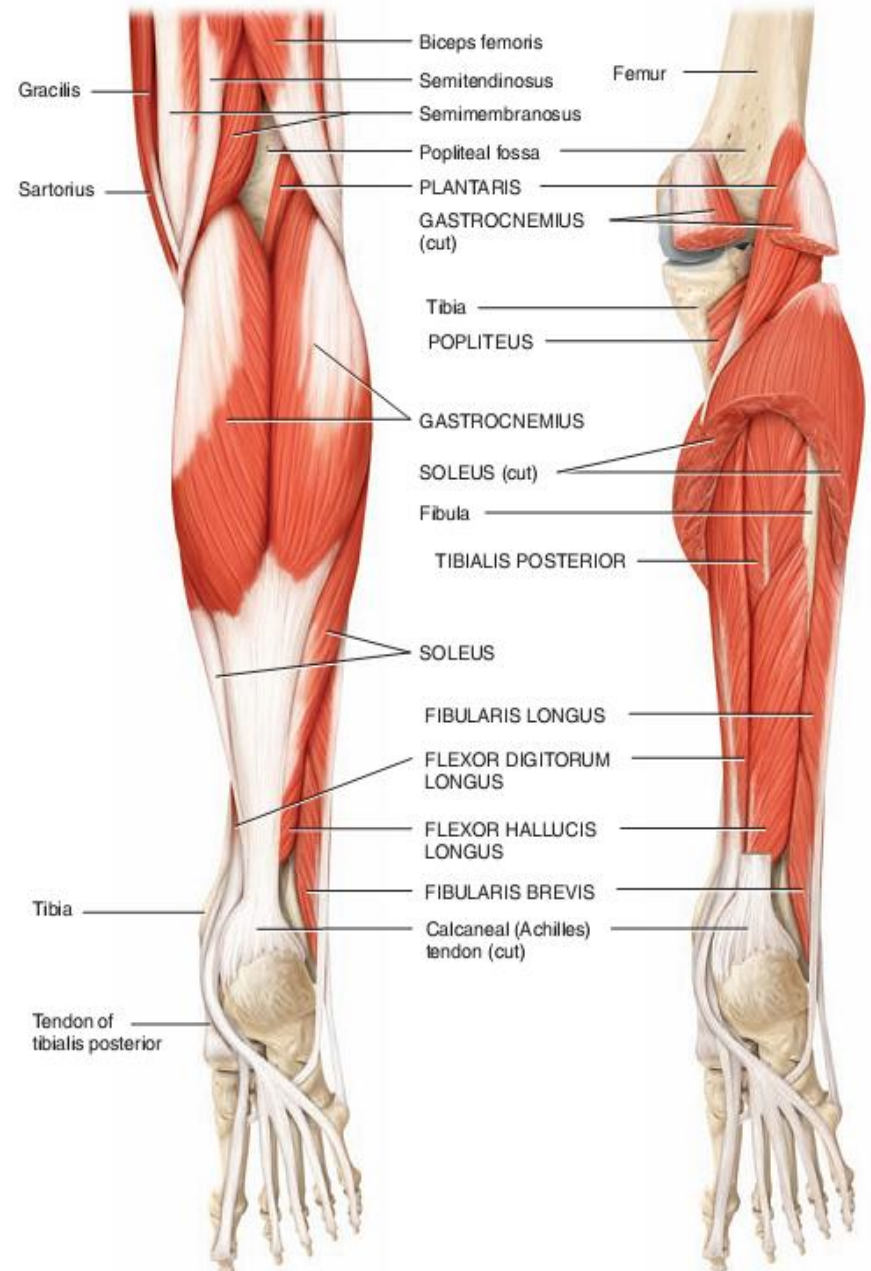
✓ **Plantaris.**

Lateral Compartment muscles of the Leg

They are:-

✓ **Peroneus longus.**

✓ **Peroneus brevis.**



10Q u so much !

Anatomy of skeletal System

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Skeleton system



Learning Objectives

- ❖ At the end of this chapter, the students should be able
- ❖ Describe organ forming skeleton system
- ❖ Discuss different bone based on its classification
- ❖ Discuss functions of bone

Outline

□ Introduction

□ Functions of bone

□ Histology of bone tissue

□ Classification of bone

- Region
- Base on shape
- Structures

□ Bone Growth and development

□ Clinical aspect

Introduction

- ❑ The entire framework of bones, cartilages, ligaments, tendons and joint constitutes the skeletal system.
- ❑ The study of the structure of bone is called osteology.
- ❑ Bone tissue makes up about 18% of the weight of the human body.

Histology of bone tissue

❖ Bone tissue

- ❑ Contains abundant matrix and few cells
- ❑ Matrix: water, collagen, crystallized mineral salts
- ❑ Cells:
 - Osteogenic cells
 - osteoblasts
 - osteocytes
 - osteoclasts

Functions bone

- ☐ Support the body
- ☐ Protect soft organs (heart, brain)
- ☐ Allow movement due to attached skeletal muscles
- ☐ Store minerals and fats
- ☐ Blood cell formation

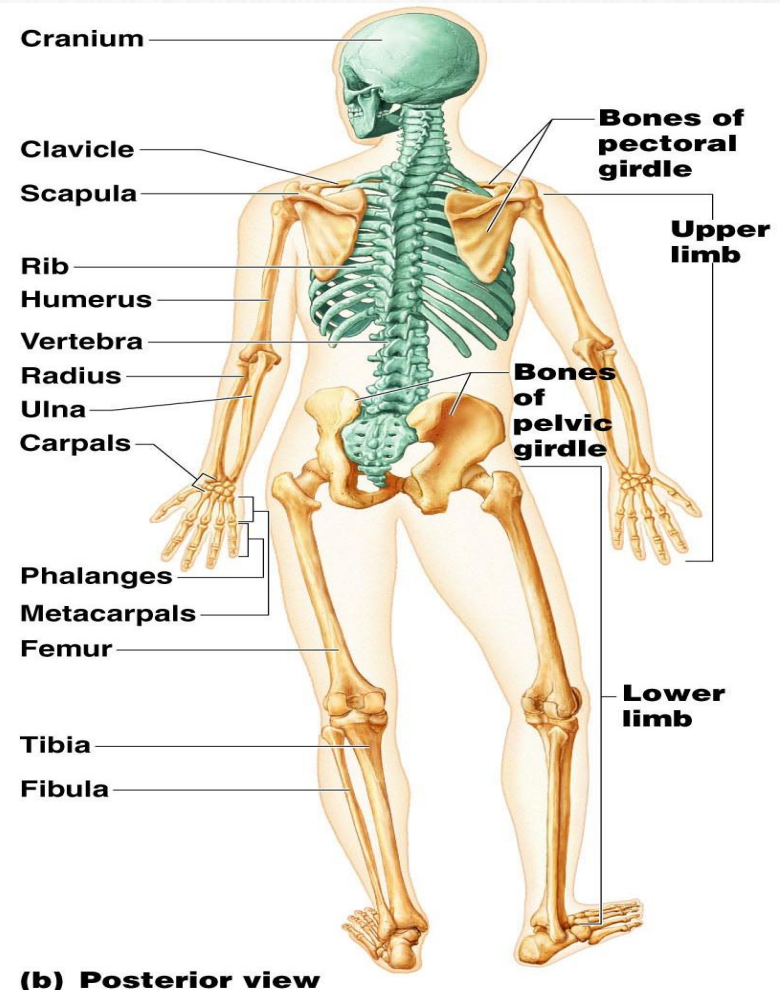
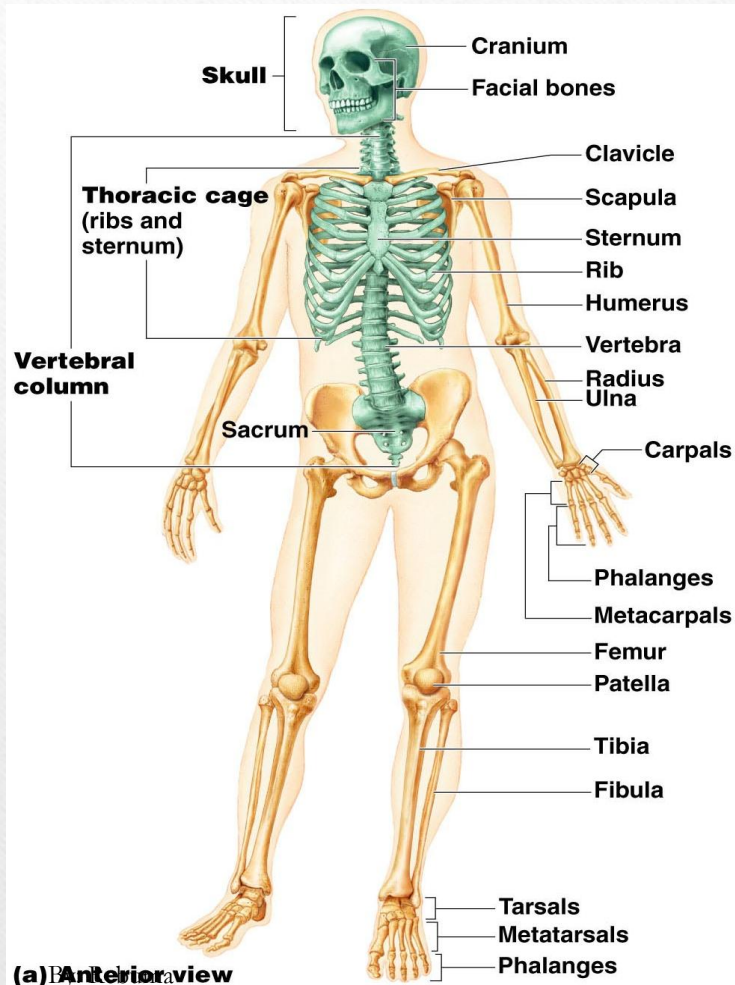
Classifications of bone: Region

Two subdivisions of the skeleton

- Axial skeleton

- Skull
- Vertebrae
- Ribs
- Ossicle
- Hyoid bone
- Appendicular skeleton
- Shoulder girdle
- Pelvic girdle
- Upper limb
- Lower Limb

Classifications of bone: Region



Classifications of bone: Region

Division of the Skeleton	Structure	Number of Bones
	Axial Skeleton	
	Skull	
	Cranium	8
	Face	14
	Hyoid	1
	Auditory ossicles	6
	Vertebral column	26
	Thorax	
	Sternum	1
	Ribs	24
		Subtotal = 80

Classifications of bone: Region

Appendicular Skeleton



Pectoral (shoulder) girdles

Clavicle	2
Scapula	2

Upper limbs

Humerus	2
Ulna	2
Radius	2
Carpals	16
Metacarpals	10
Phalanges	28

Pelvic (hip) girdle

Hip or pelvic bone	2
--------------------	---

Lower limbs

Femur	2
Patella	2
Fibula	2
Tibia	2
Tarsals	14
Metatarsals	10
Phalanges	28

Subtotal = 126

Total in an adult skeleton = 206

The Axial Skeleton

- ❑ Forms the longitudinal axis of the body
- ❑ Divided into three parts

-
- Skull
 - Vertebral column
 - Bony thorax

The Skull

- ❑ Two sets of bones
 - Cranium
 - Facial bones

Divisions of the bones of the skull

A. Cerebral / cranial bones / brain case (8 bones)

Unpaired (4)

1. occipital
2. frontal
3. sphenoid
4. ethmoid

Paired (4)

1. parietal
2. temporal

B. Facial or visceral cranium

Paired (12)

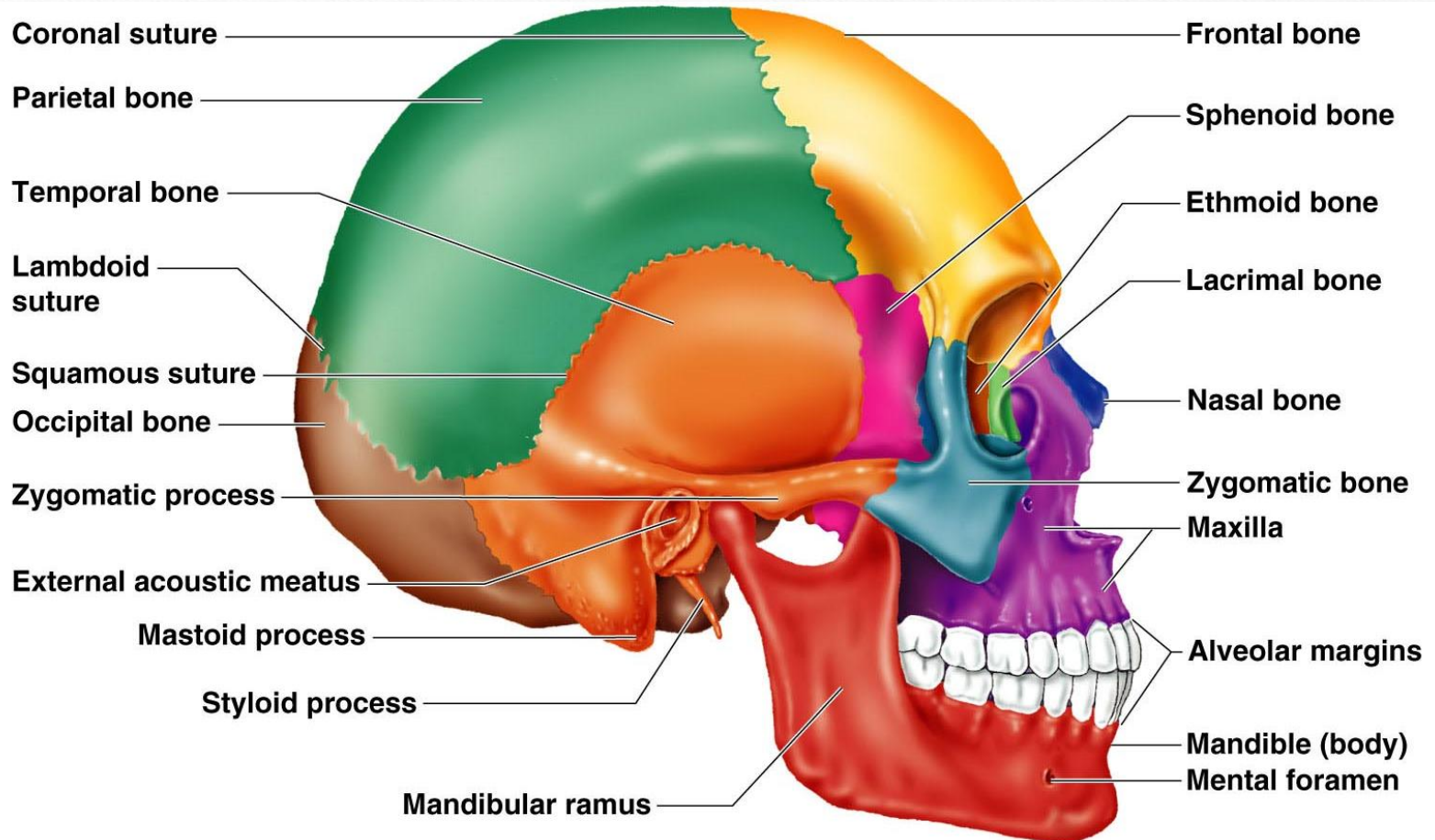
- a. Nasal
- b. Lacrimal
- c. Maxilla
- d. Zygomatic / malar / cheek bones
- e. Palatine
- f. Inferior nasal concha or turbinate

Unpaired (2)

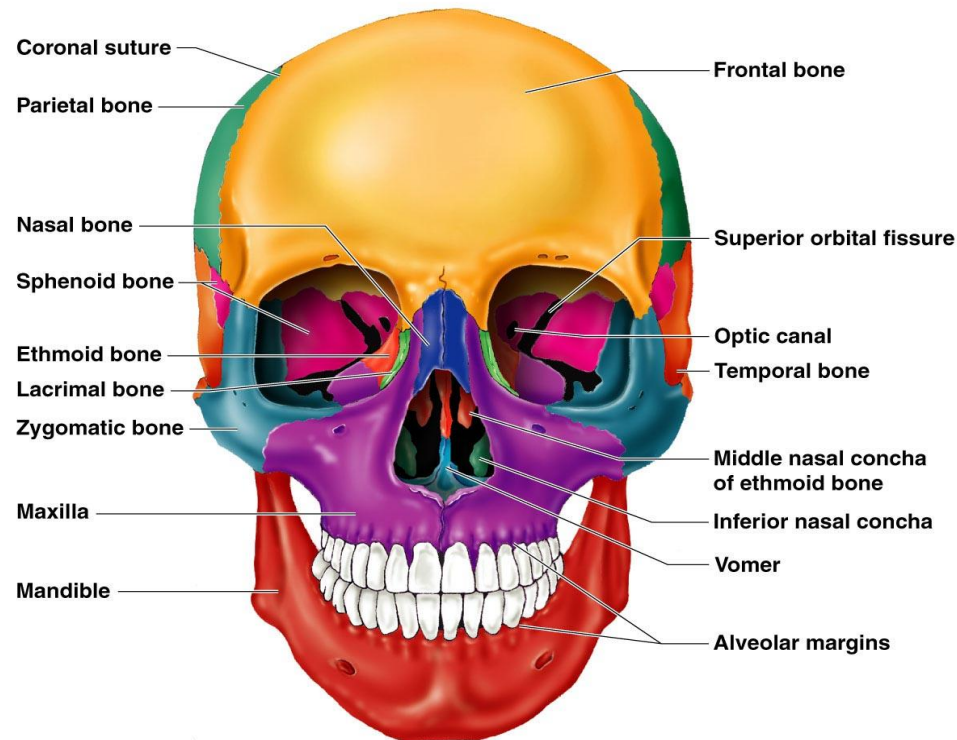
- a. Vomer

- b. Mandible

Human Skull, Lateral View

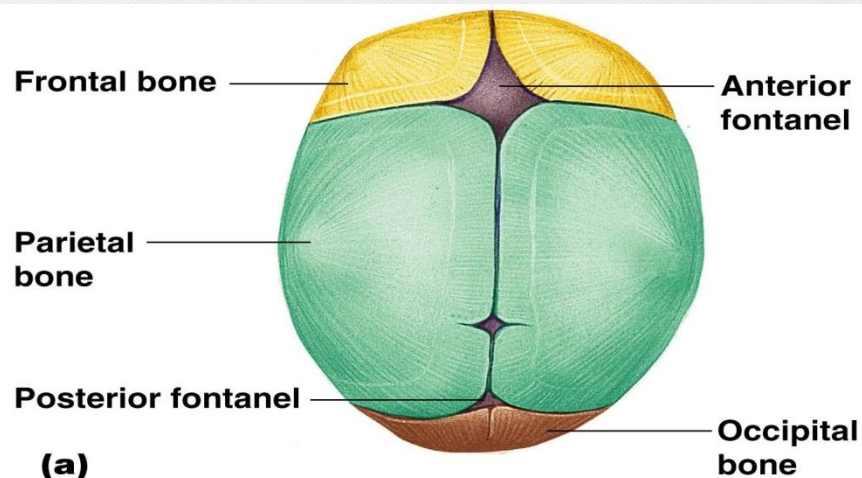


Classifications of bone: Region

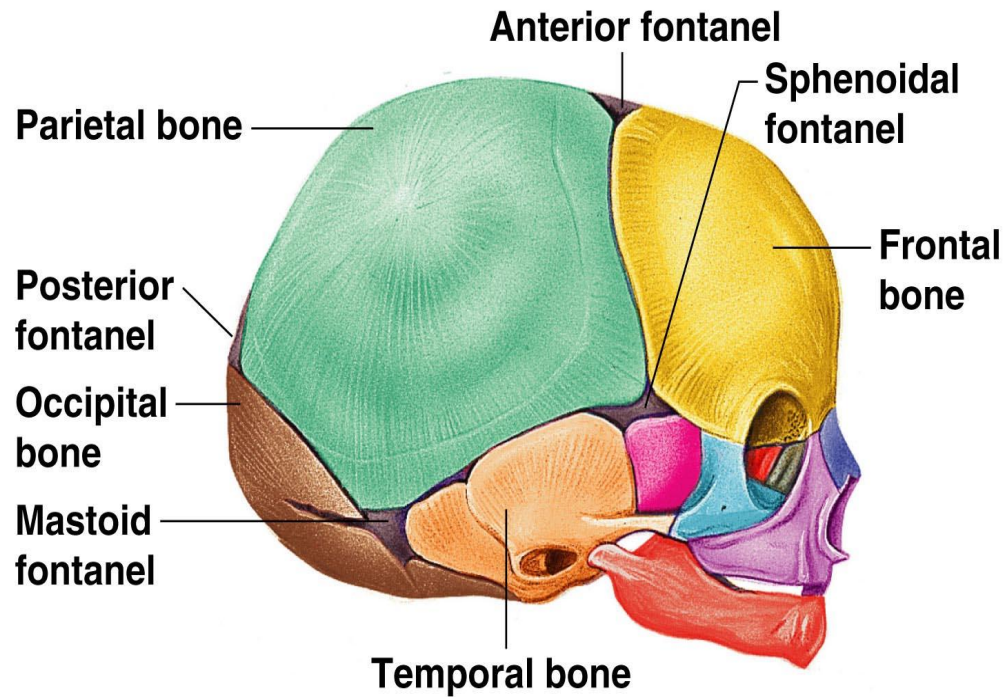


Fetal Skull

- The fetal skull is large compared to the infant's total body length
- Fontanels—fibrous membranes connecting the cranial bones
 - Allow the brain to grow
 - Convert to bone within 24 months after



Fetal Skull



(b)

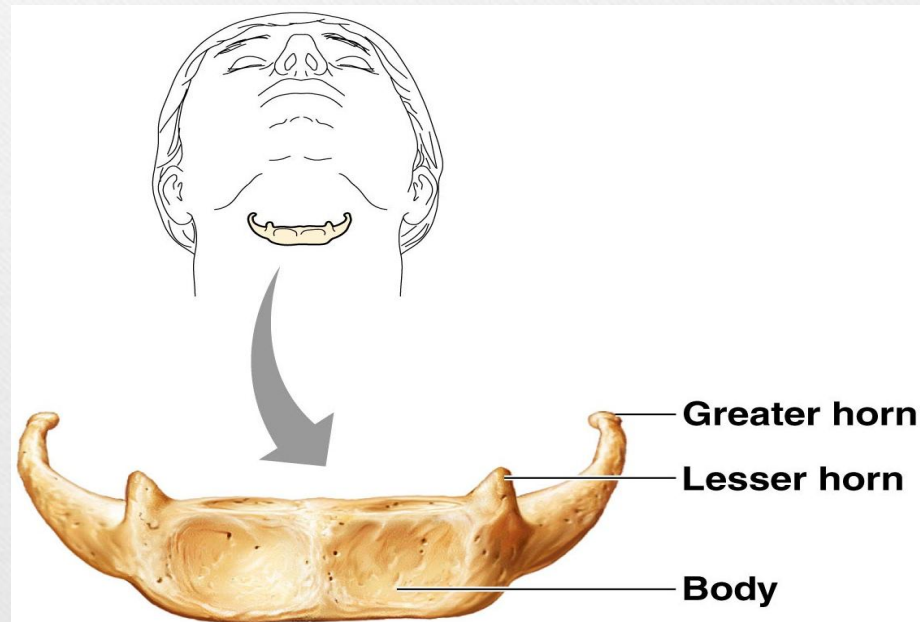
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Sutures

- ❑ are immovable joint found between skull bones. There are four main sutures in the skull.
- ❑ Coronal suture: between the frontal & the two-parietal bone.
- ❑ Sagittal suture: between the two parietal bones.
- ❑ Lambdoidal suture: between parietal & occipital bone.
- ❑ Squamosal suture: between parietal bone and temporal

Axial skeleton: Hyoid Bone

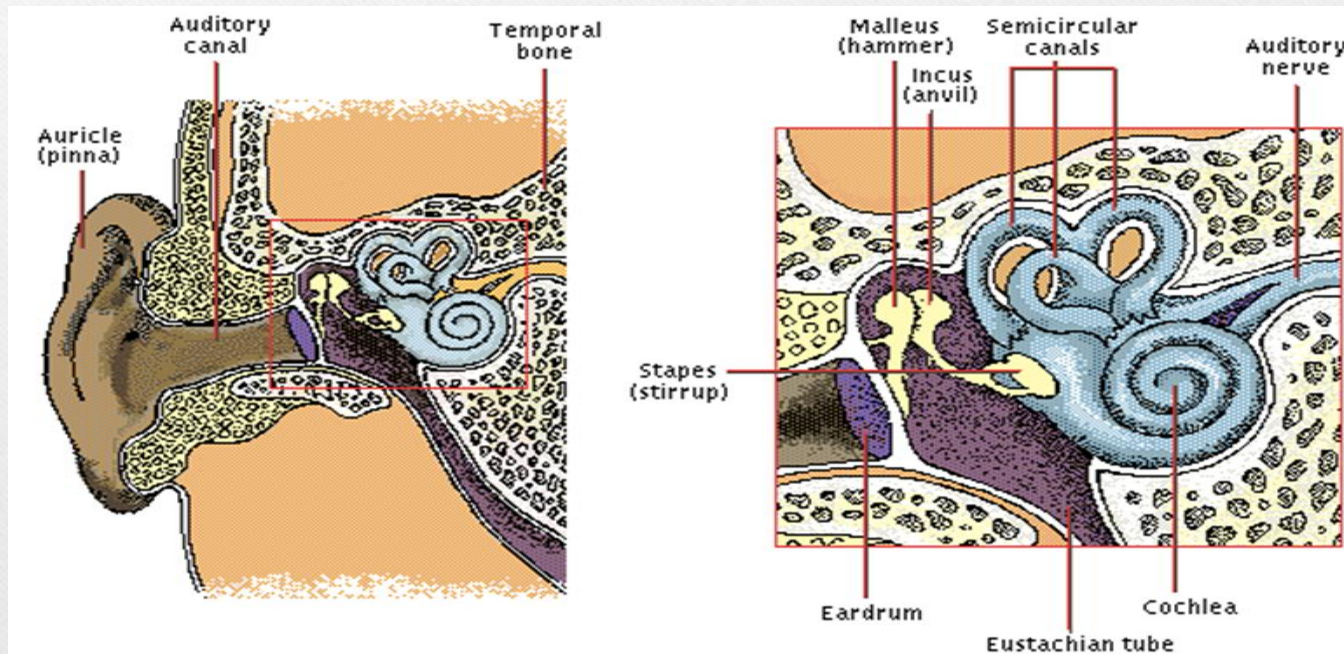
- ❑ The only bone that does not articulate with another bone
- ❑ Serves as a moveable base for the tongue
- ❑ Aids in swallowing and speech



Axial skeleton: Ossicles

□ Are small bones of the ear

- Stapes (stirrup) 2
- Incus (anvil) 2
- Malleus (hammer) 2



Axial skeleton: Vertebral column

- ☐ Curved, slightly movable
- ☐ United together by cartilage & ligaments
- ☐ 71 – 75 cm. Long
- ☐ Formed by series of bones vertebrae

Function:

- ☐ Support of the trunk
- ☐ Contains & protects the spinal cord & nerves

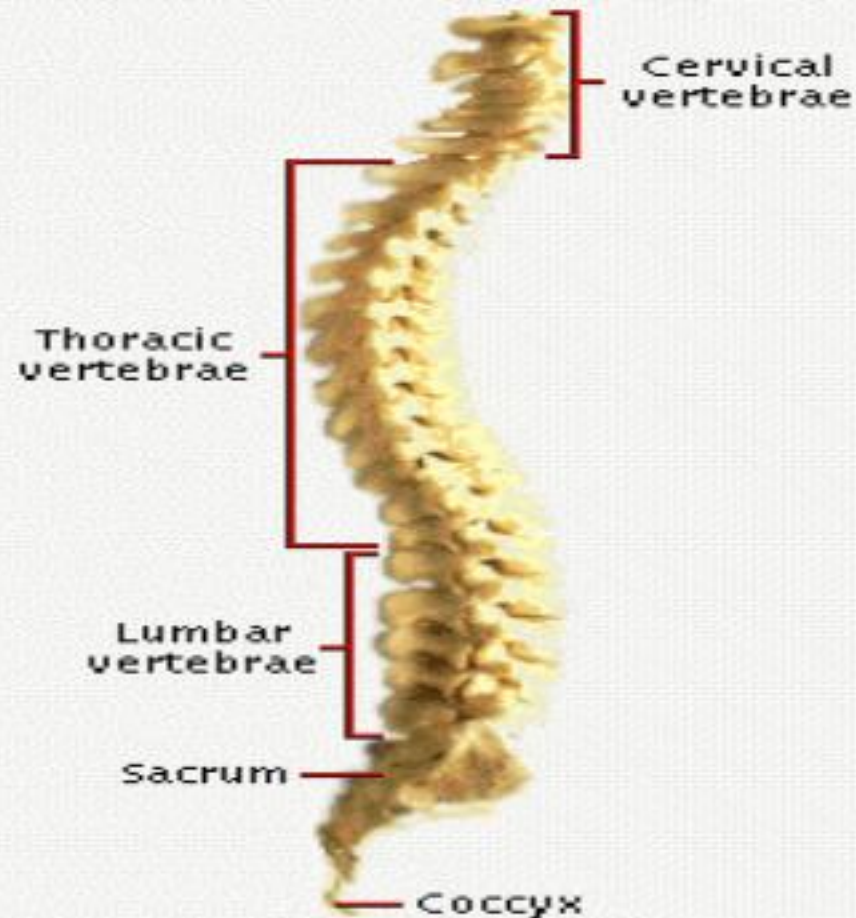
Axial skeleton: Vertebral Bone

Classification of vertebra	young adult	
cervical	7	7
thoracic	12	12
Lumbar	5	5
Sacral	5	1
coccygeal	4	1
Total	33	26

- Intervertebral discs = flattened plates of fibrocartilage that
- Are interposed between the adjacent surfaces of the
- Bodies of vertebrae

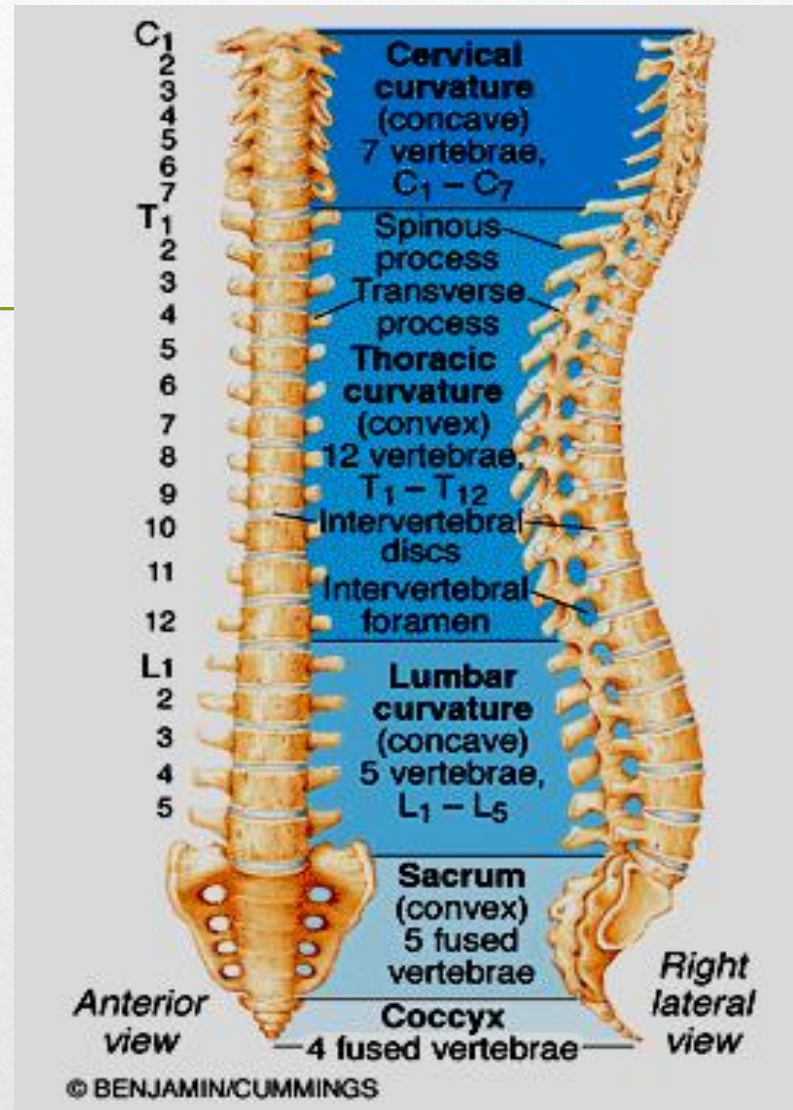
FUNCTION: 1. uniting medium between vertebrae
2. give flexibility & movement to the whole
vertebral column

Axial skeleton: Vertebral column



Axial skeleton: Vertebral column

- Cervical (C1-C7)
- Thoracic(T1-T12)
- Lumbar (L1-L5)
- Sacral(5 fused)
- Coccygeal(3-4 fused)



Axial skeleton: Sternum (breast bone)

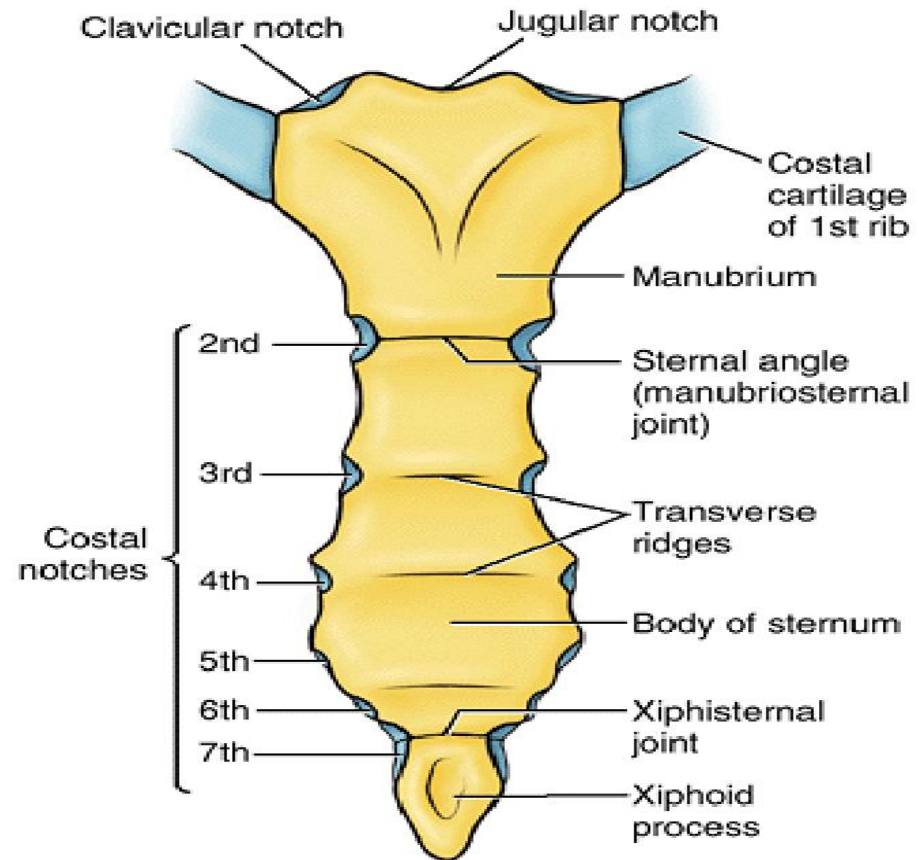
❑ Flat bone, found -- anterior thoracic wall

3- parts:

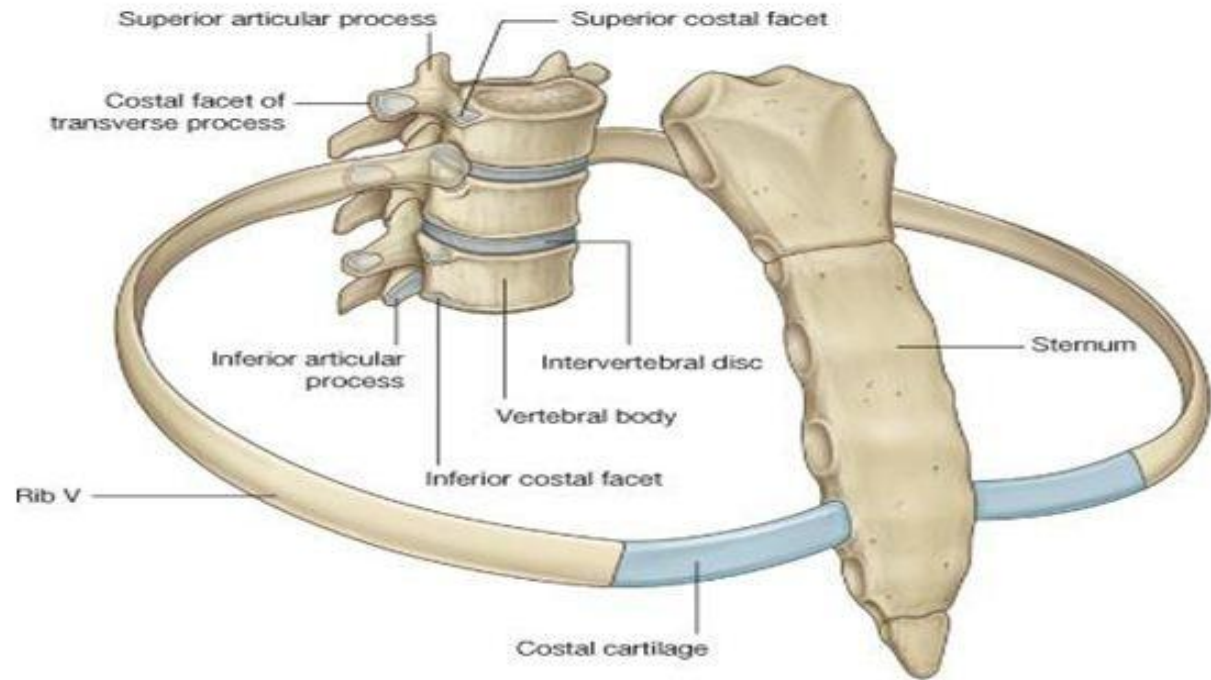
a. Manubrium

b. Body

c. Xiphoid process



Axial skeleton



Axial skeleton: Ribs

- ❑ Ribs (12 pairs)
- ❑ Narrow arched flat bones with 2 ends
 - Vertebral - Posterior; attaches with thoracic
 - Sternal - Anterior; attaches with costal cartilages

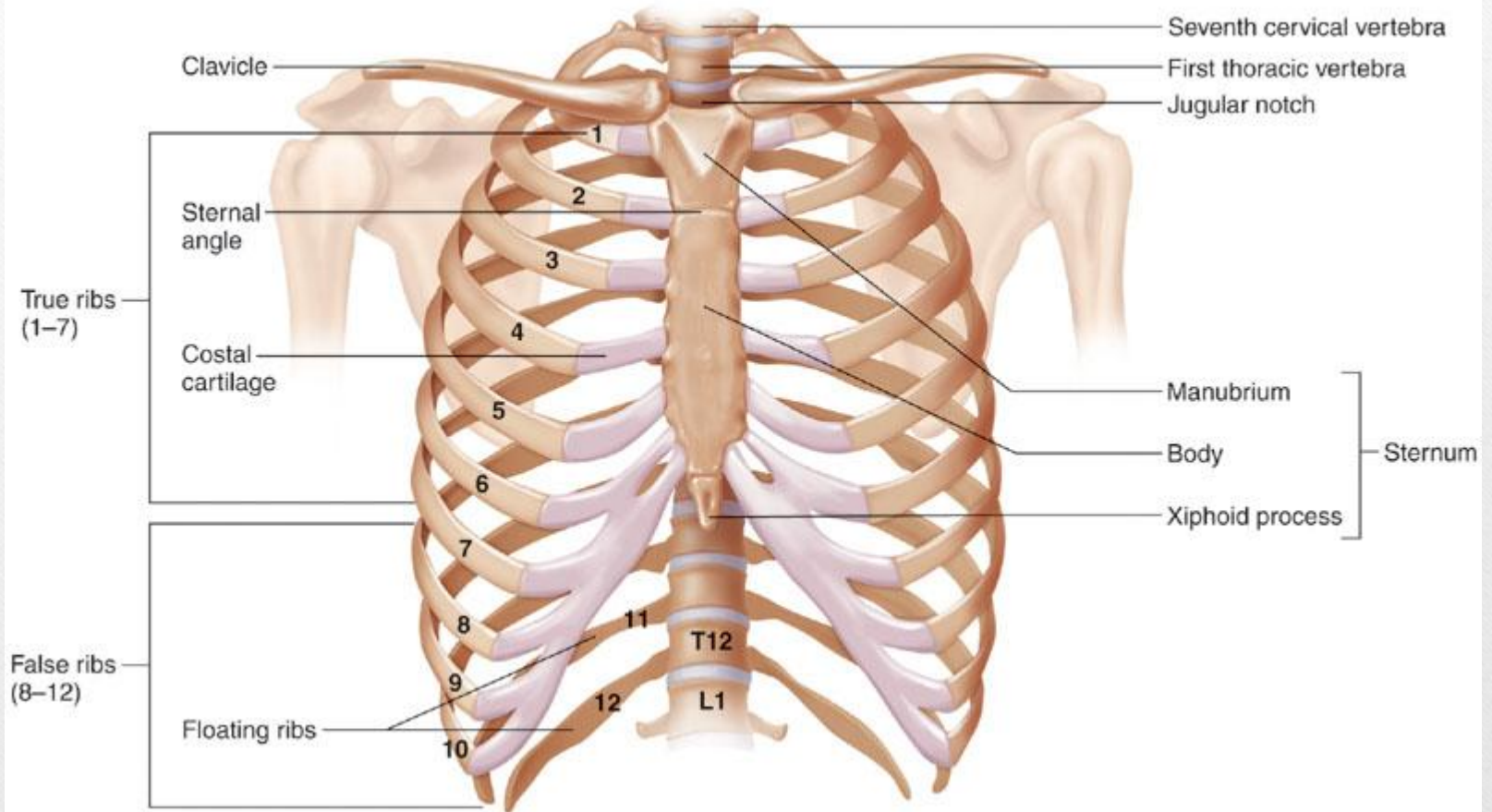
❑ Classification of ribs:

1. Sternal or true ribs (1st to 7th)
 - Ribs whose costal cartilages are directly attached to sternum
2. False ribs (8th to 12th)
 - ribs whose costal cartilages are not attached directly to the sternum but to 7th

Subdivisions: 1. False rib proper - 8th, 9th, 10th ribs
2. Floating or hanging ribs – 11th & 12th

Axial skeleton: Ribs

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Appendicular skeleton: UL

1. Clavicle (collar bone)
2. Scapula (shoulder blade) :- articulates with
— Humerus & clavicle —
3. Humerus (arm bone) :- longest & largest bone of UL articulates with scapula (above) radius & ulna (below)
4. Radius :- lateral bone of the forearm
5. Ulna :- principal bone of the forearm; longer & larger than radius

Appendicular skeleton: UL

6. Carpals (wrist bone) - 8 bones arranged into 2 rows

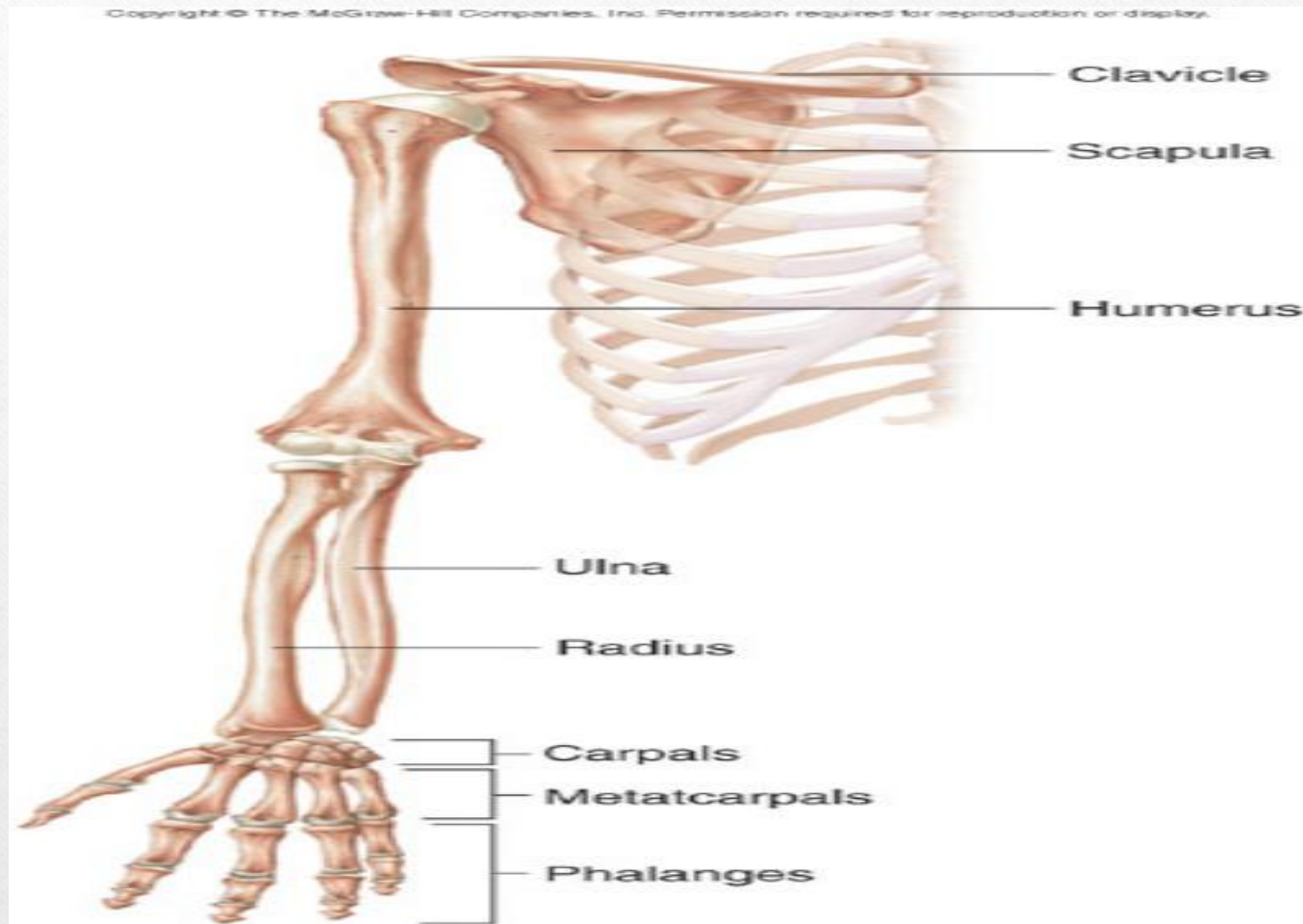
- proximal & distal rows

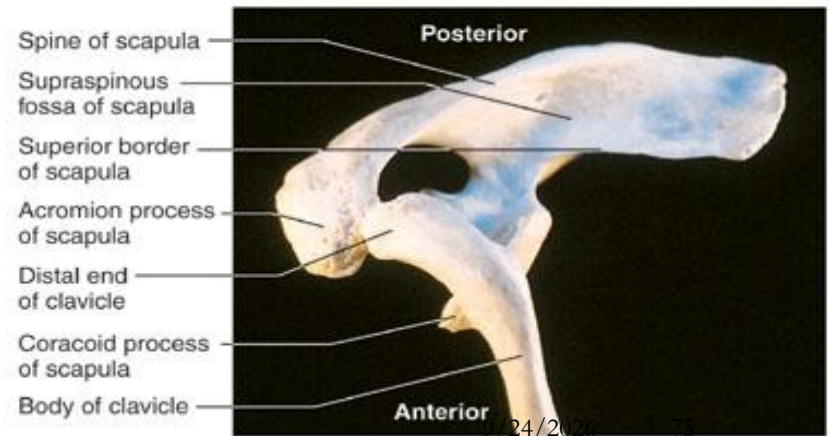
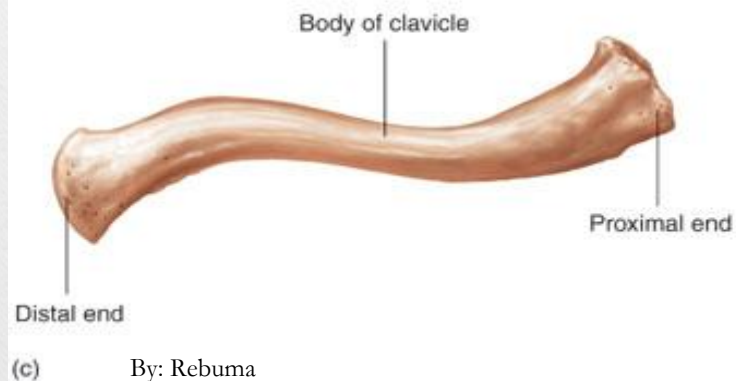
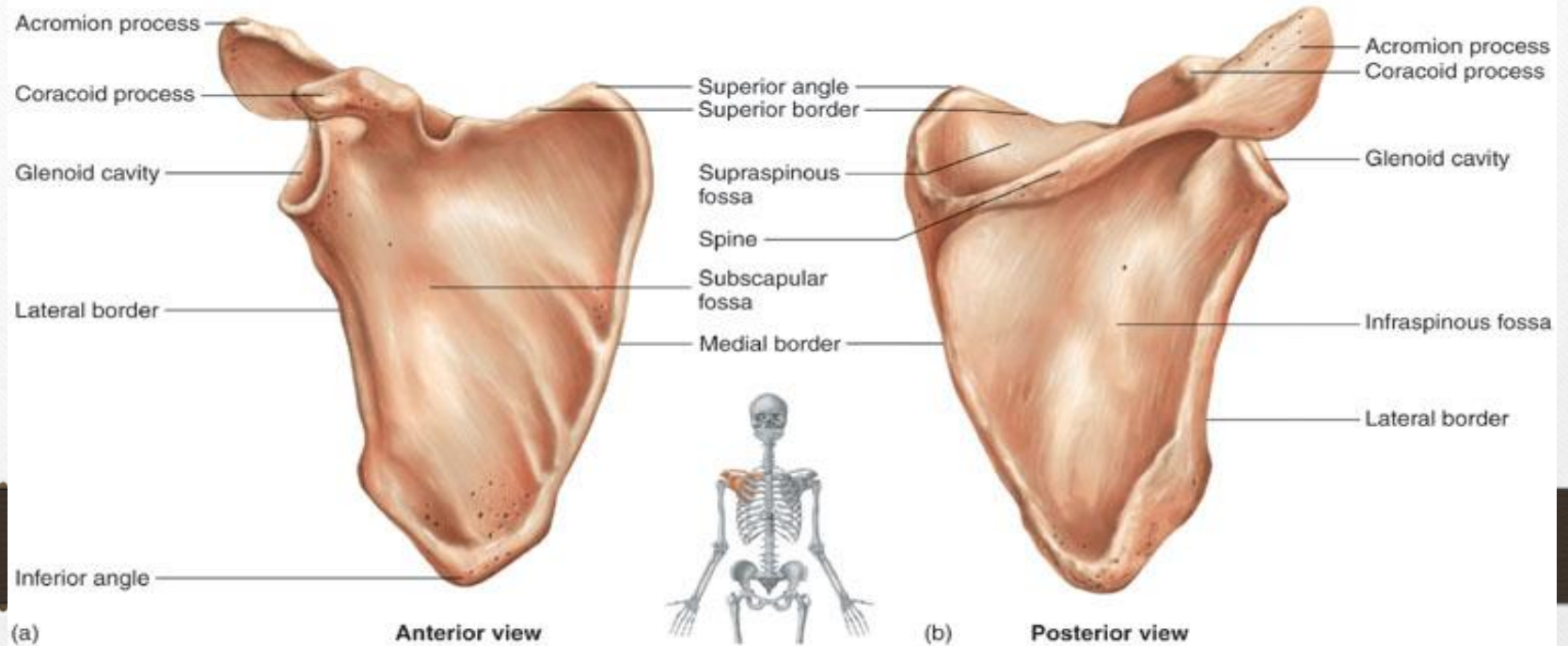
7. Metacarpals (bones of the hand) - 5 long bones placed between carpals & phalange

- Numbered from lateral to medial

8. Phalanges (bones of the fingers) - 14 long bones of the fingers - 3 bones except thumb - 2 bones

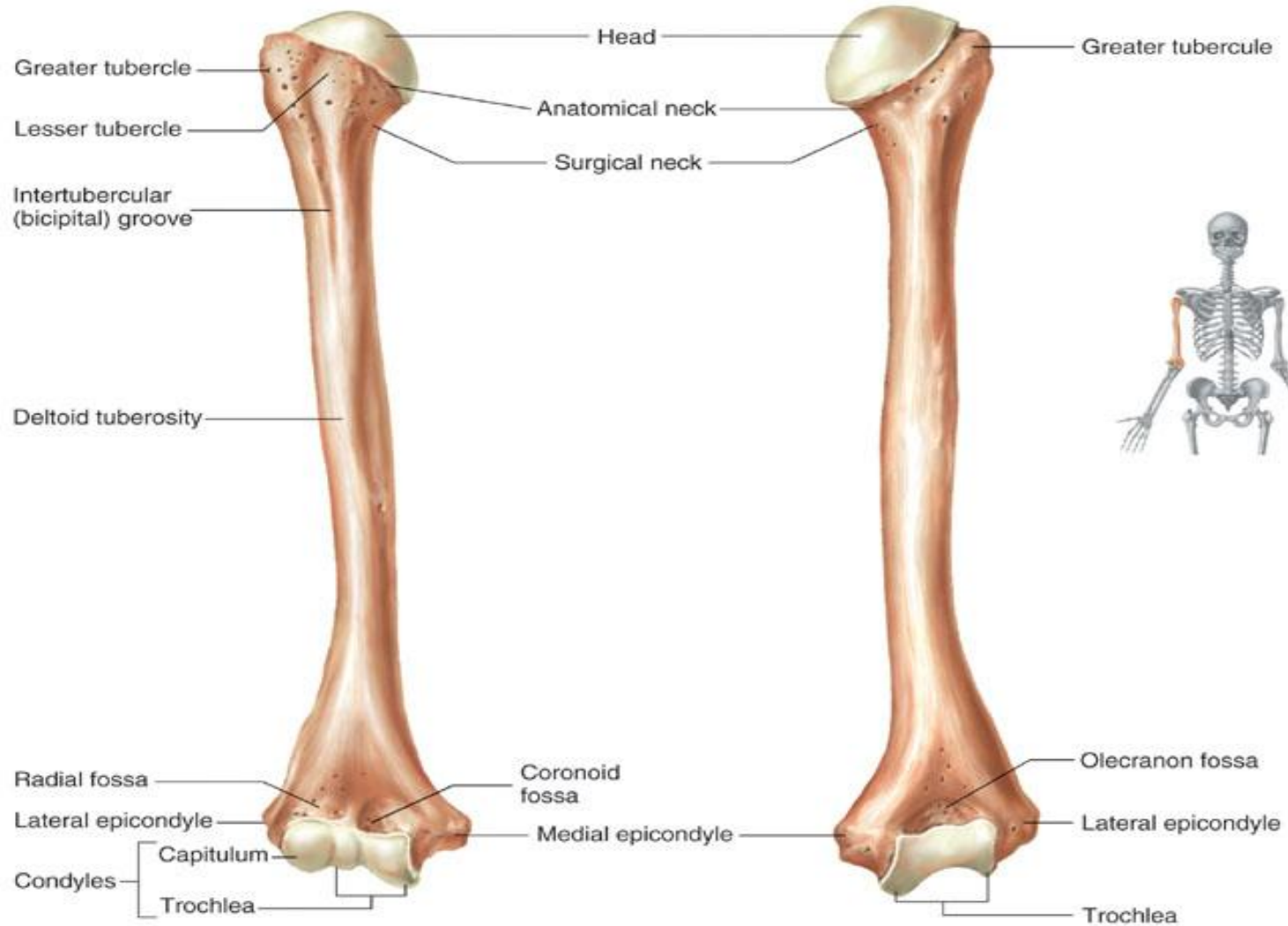
Appendicular skeleton: UL





By: Rebuma

24/20



(a)

By: Rebuma

Anterior

(b)

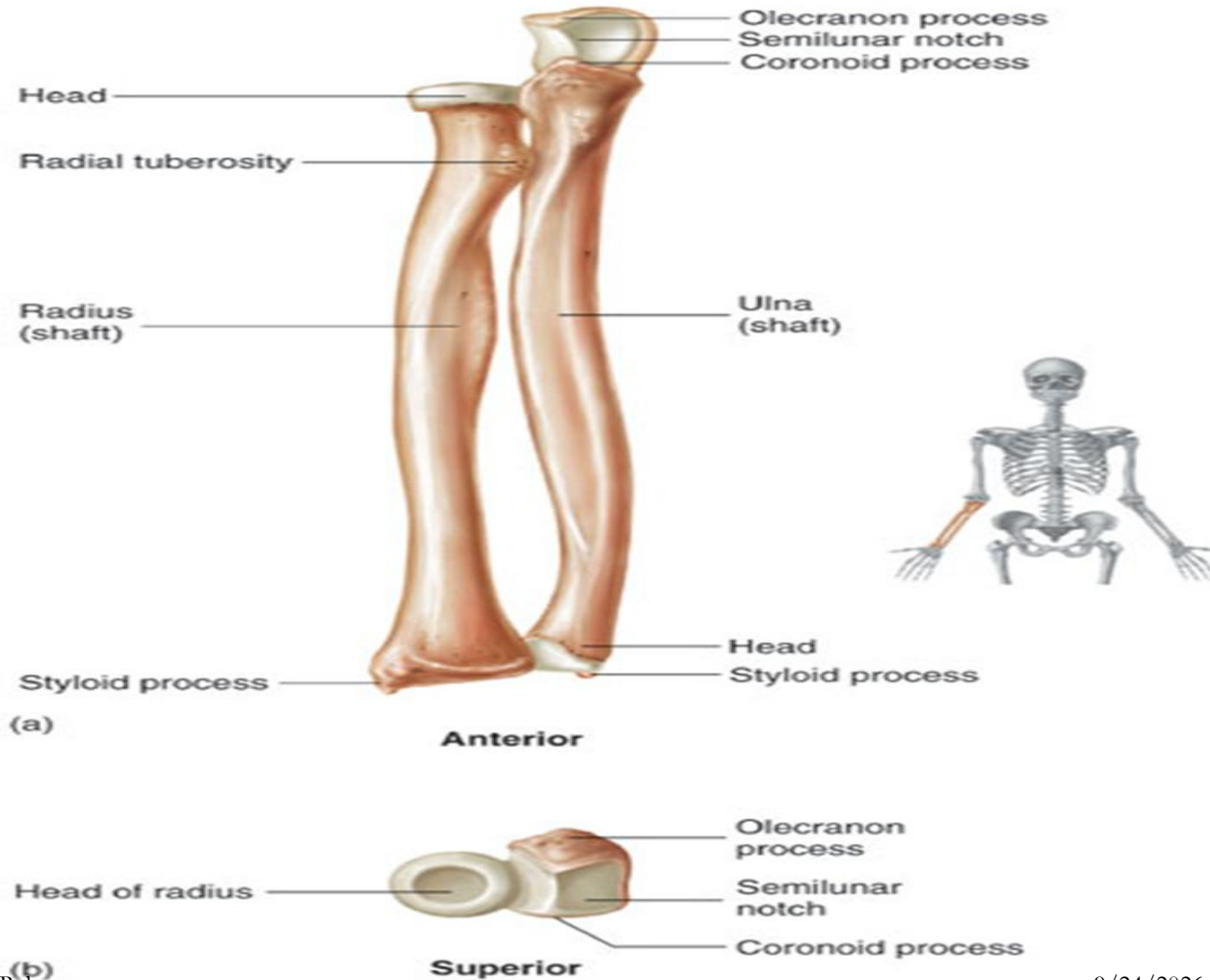
Posterior

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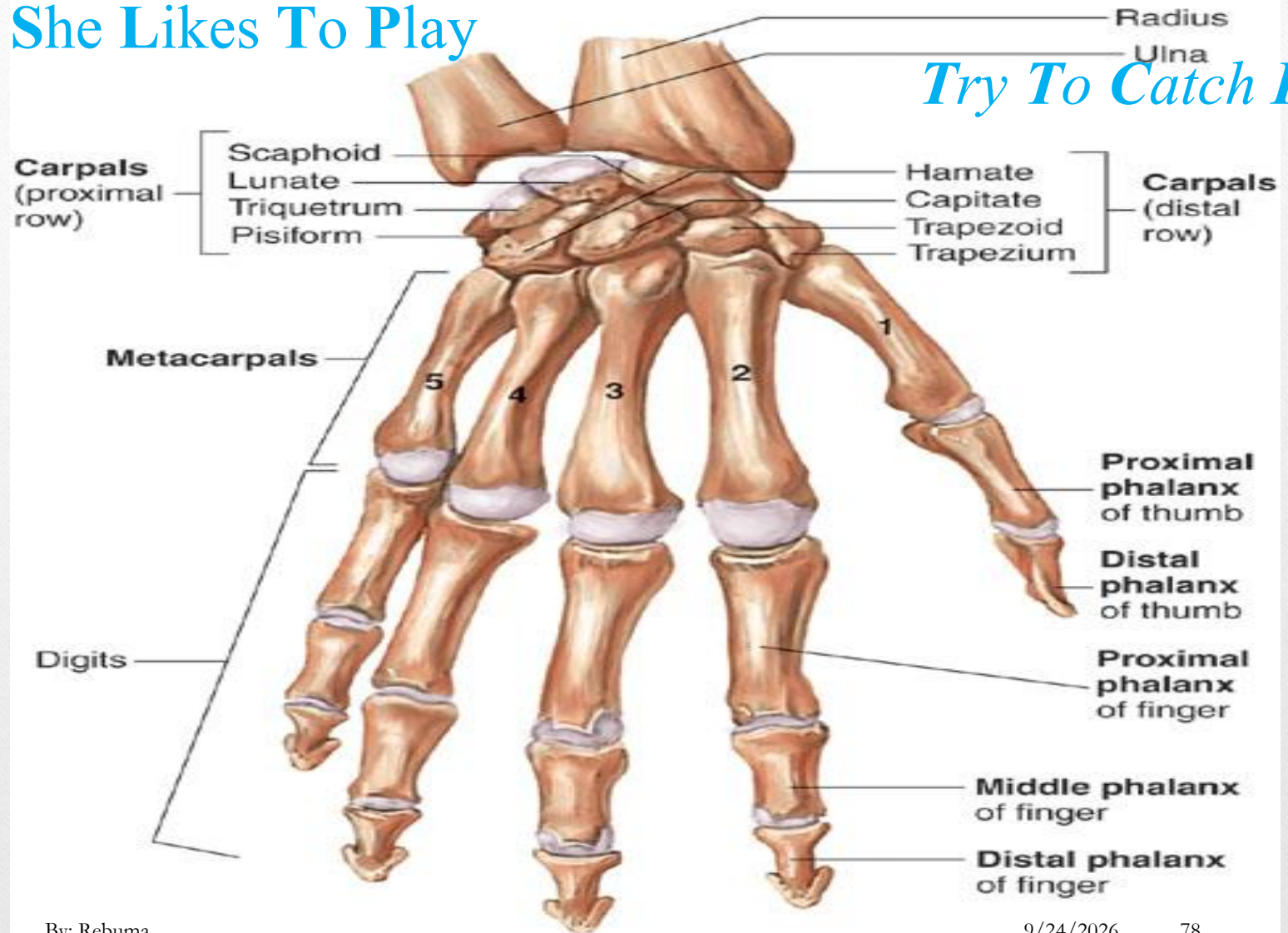
Appendicular skeleton: UL

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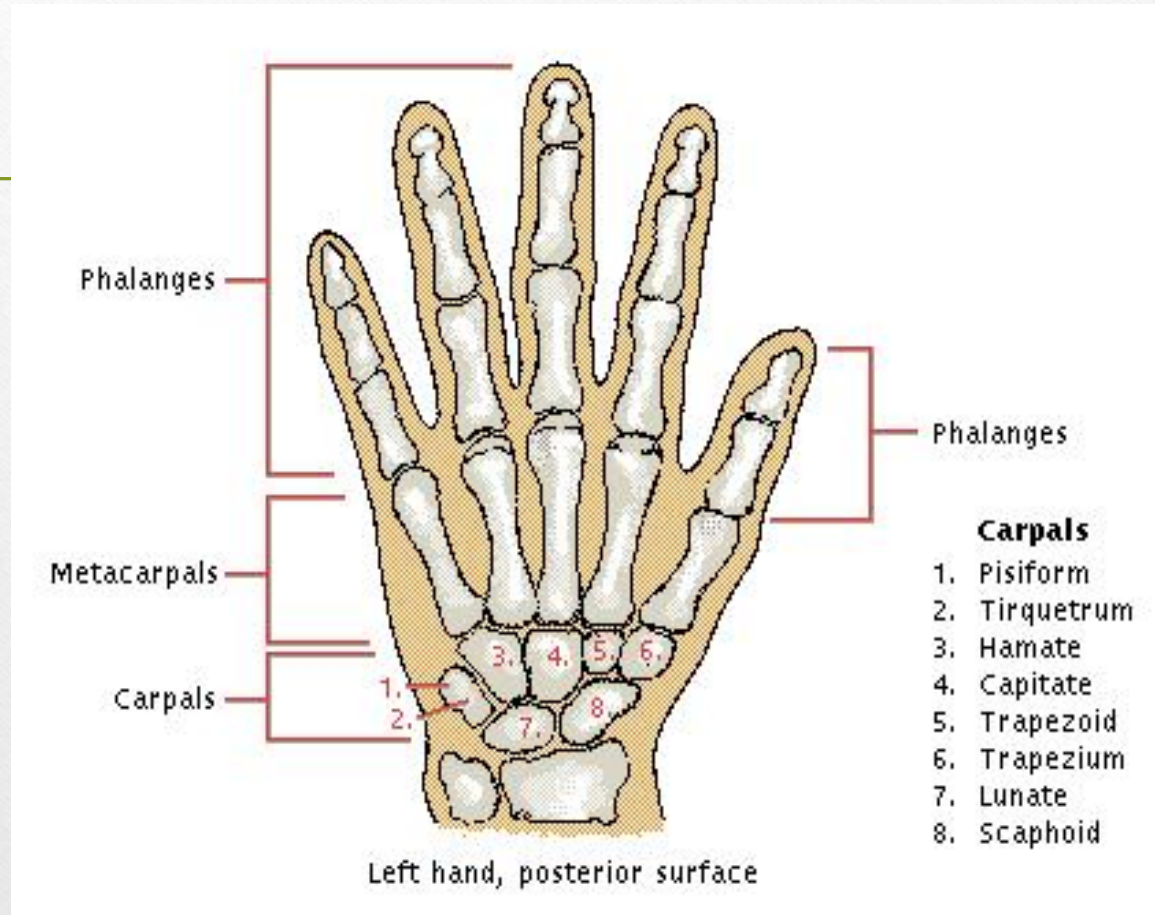


She Likes To Play

Try To Catch Her



Appendicular skeleton: UL



Appendicular skeleton: LL

1. Hip bone (innominate bone)

3 bones: 1. ilium*

2. ischium*

3. pubis*

2. Femur (thigh) Longest, strongest, largest bone in the body.

3. Tibia (shin bone) :- long bone; anterior, medial & larger of the 2 bones of the leg

4. Fibula (peroneal bone) :- Long slender bone placed parallel with the tibia but located laterally

Appendicular skeleton: LL

5. Tarsals (ankle bone): short bones;

2 rows: internal & external rows

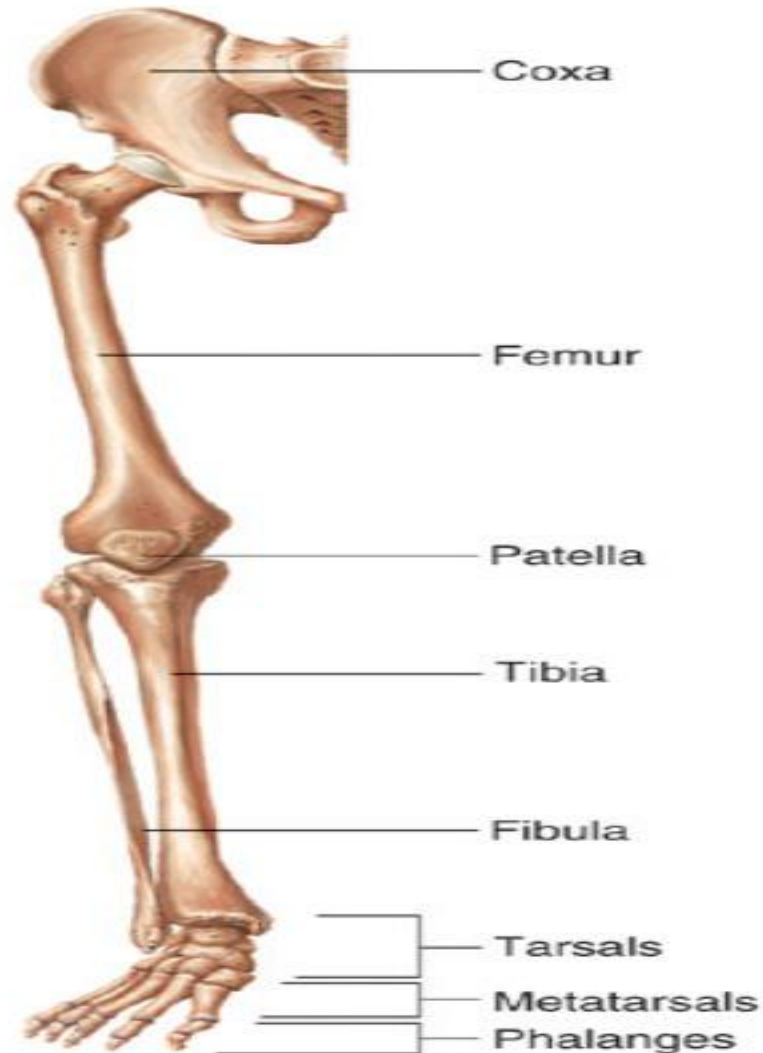
6. Metatarsals (bones of foot):- 5 long

bones numbered from medial to lateral

7. Phalanges (bones of toes) :- similar to
bones of the fingers

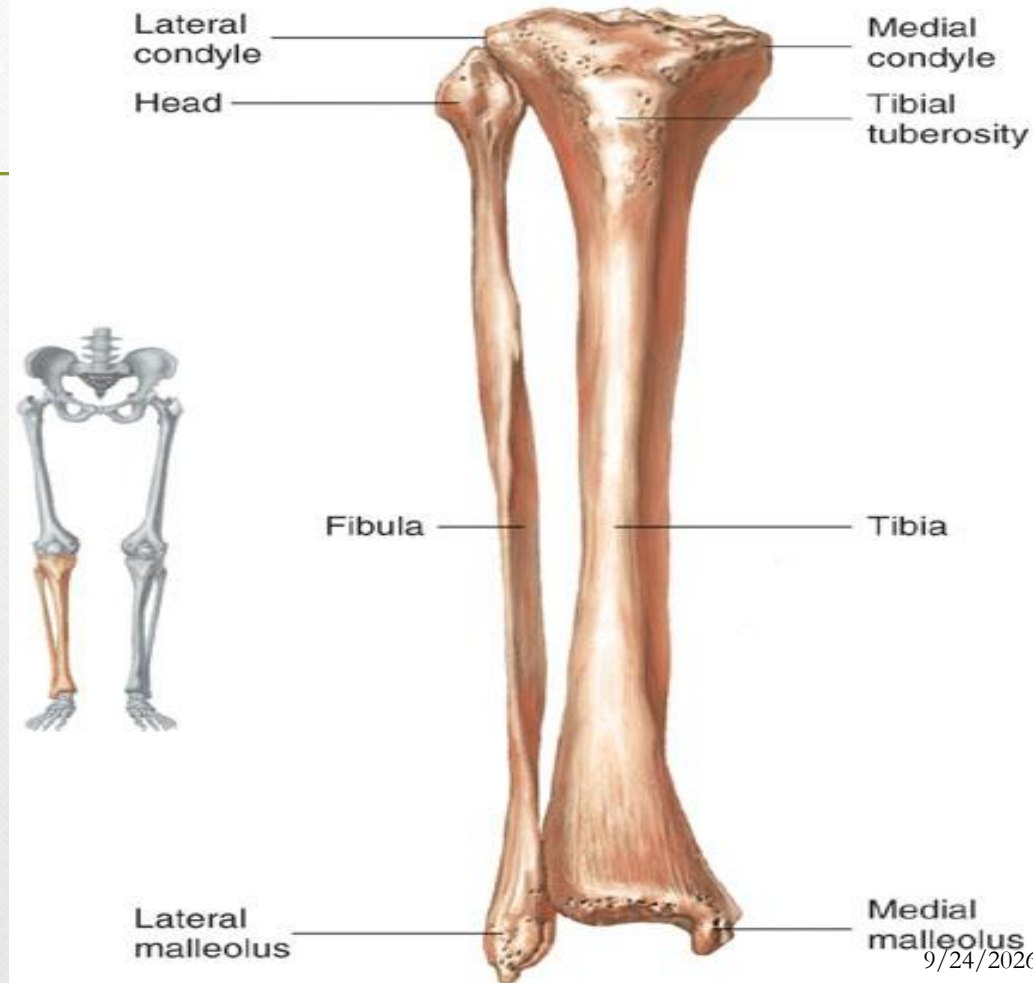
Appendicular skeleton: LL

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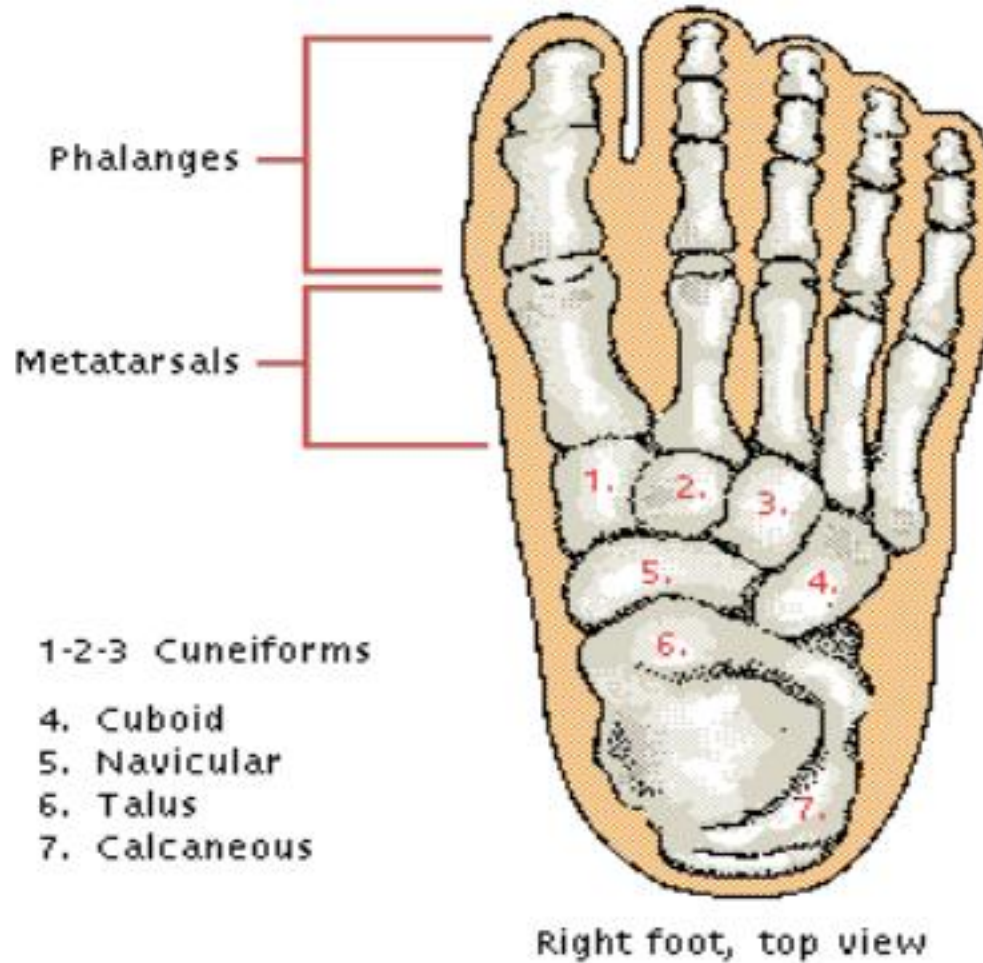


Appendicular skeleton: UL

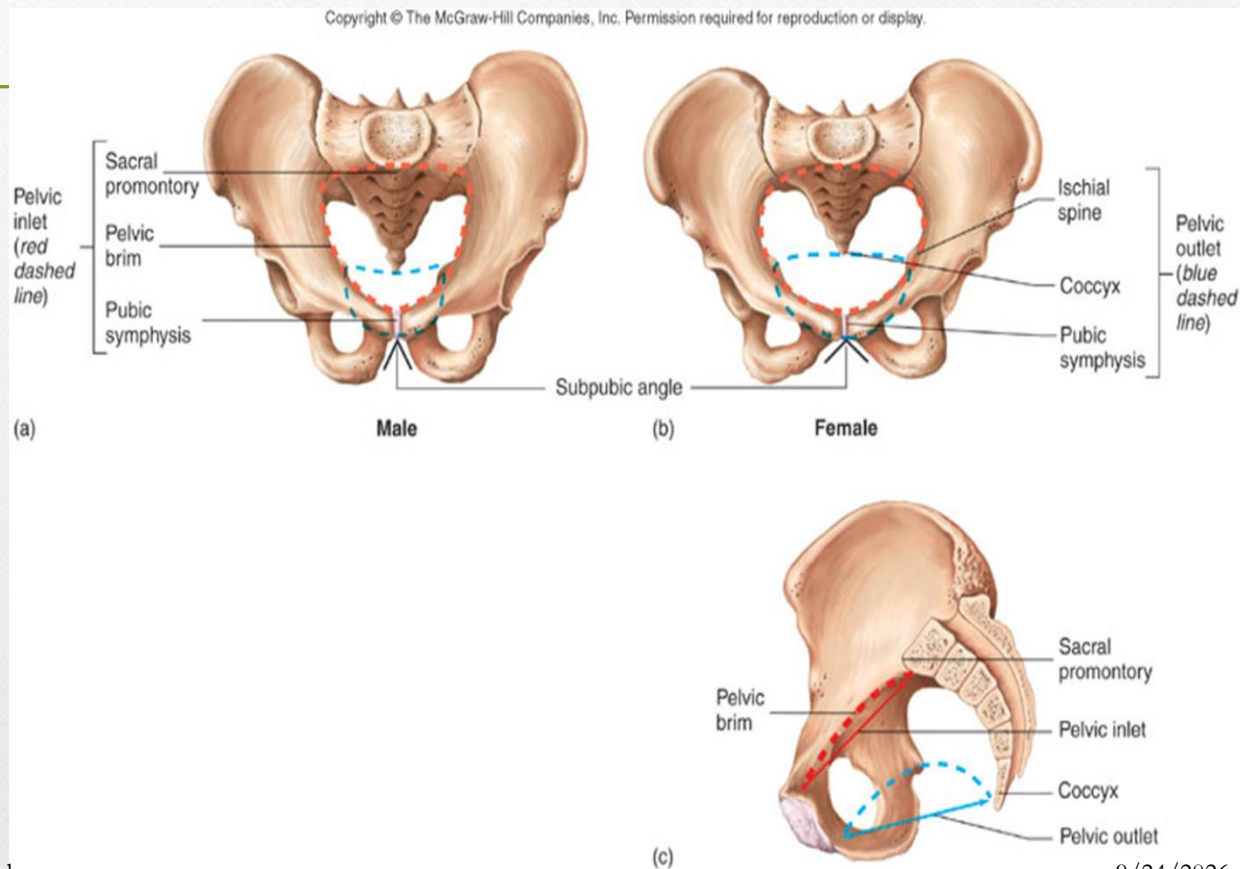
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Appendicular skeleton: UL



Appendicular Skeleton: Pelvic girdle



Bony pelvis	male	female
General structures	Thick and heavy	Thin and light
Greater pelvis (pelvis major)	Deep	Shallow
Lesser pelvis (pelvis minor)	Narrow and deep, tapering	Wide and shallow, cylindrical
Pelvic inlet (superior pelvic aperture)	Heart shaped, narrow	Oval and rounded; wide
Pelvic outlet (inferior pelvic aperture)	Comparatively small	Comparatively large
Pubic arch and sub pubic angle	Narrow ($< 70^{\circ}$)	Wide ($> 80^{\circ}$)
Obturator foramen	Round	Oval
Acetabulum	Large	Small

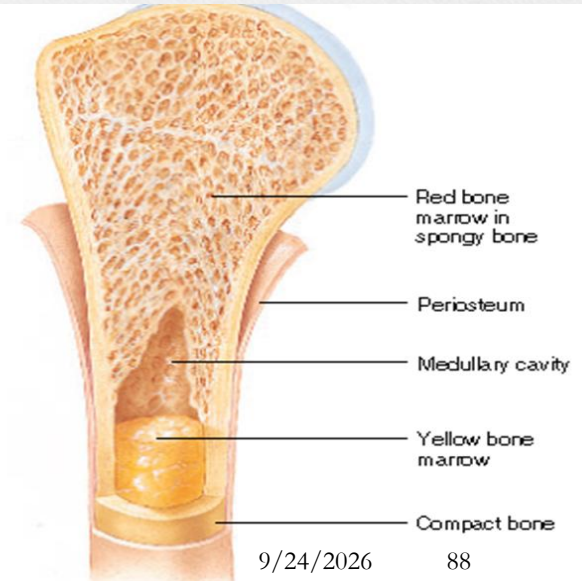
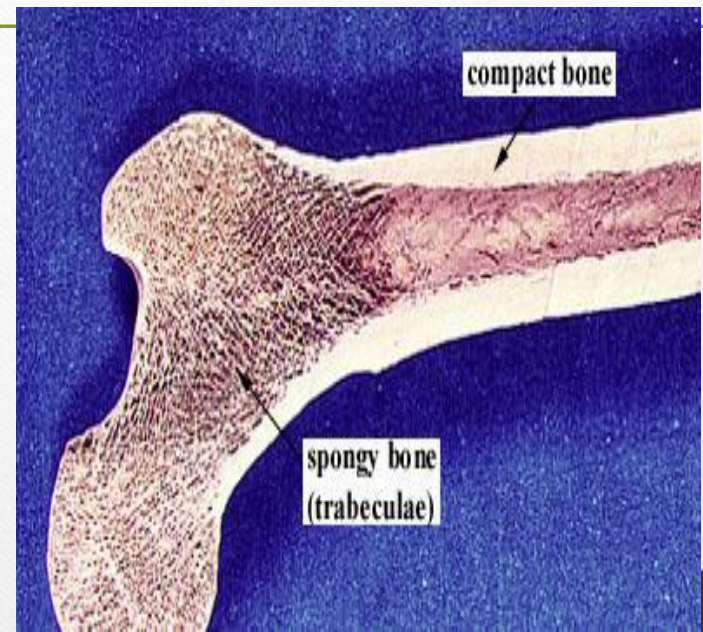
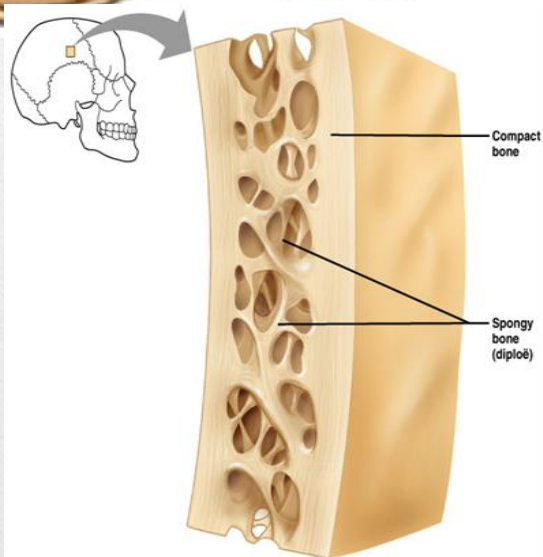
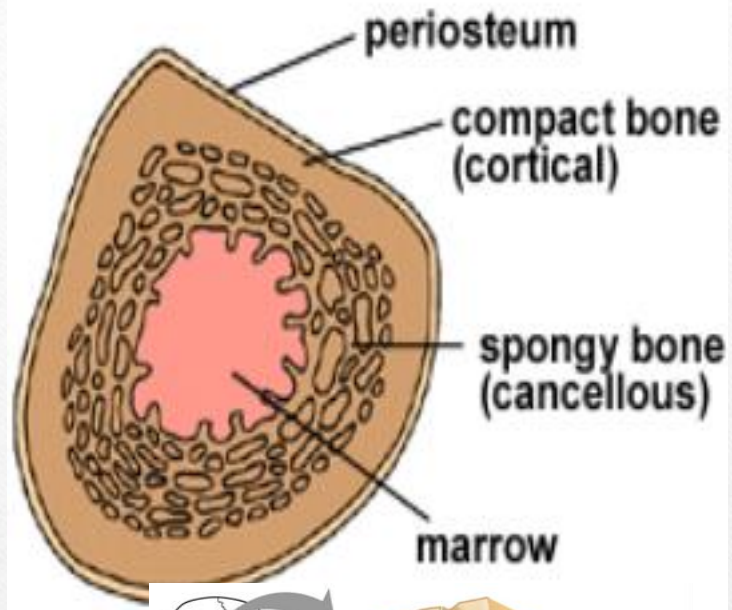
Classification of bones: Structure

□ **Compact(80%)**:- Solid mass; dense & hard
forms the outer layer of bone structure

- Functional unit- Haversian system

□ **Spongy(20%)**:- contain spaces filled with
bone marrow

- Incomplete Haversian system



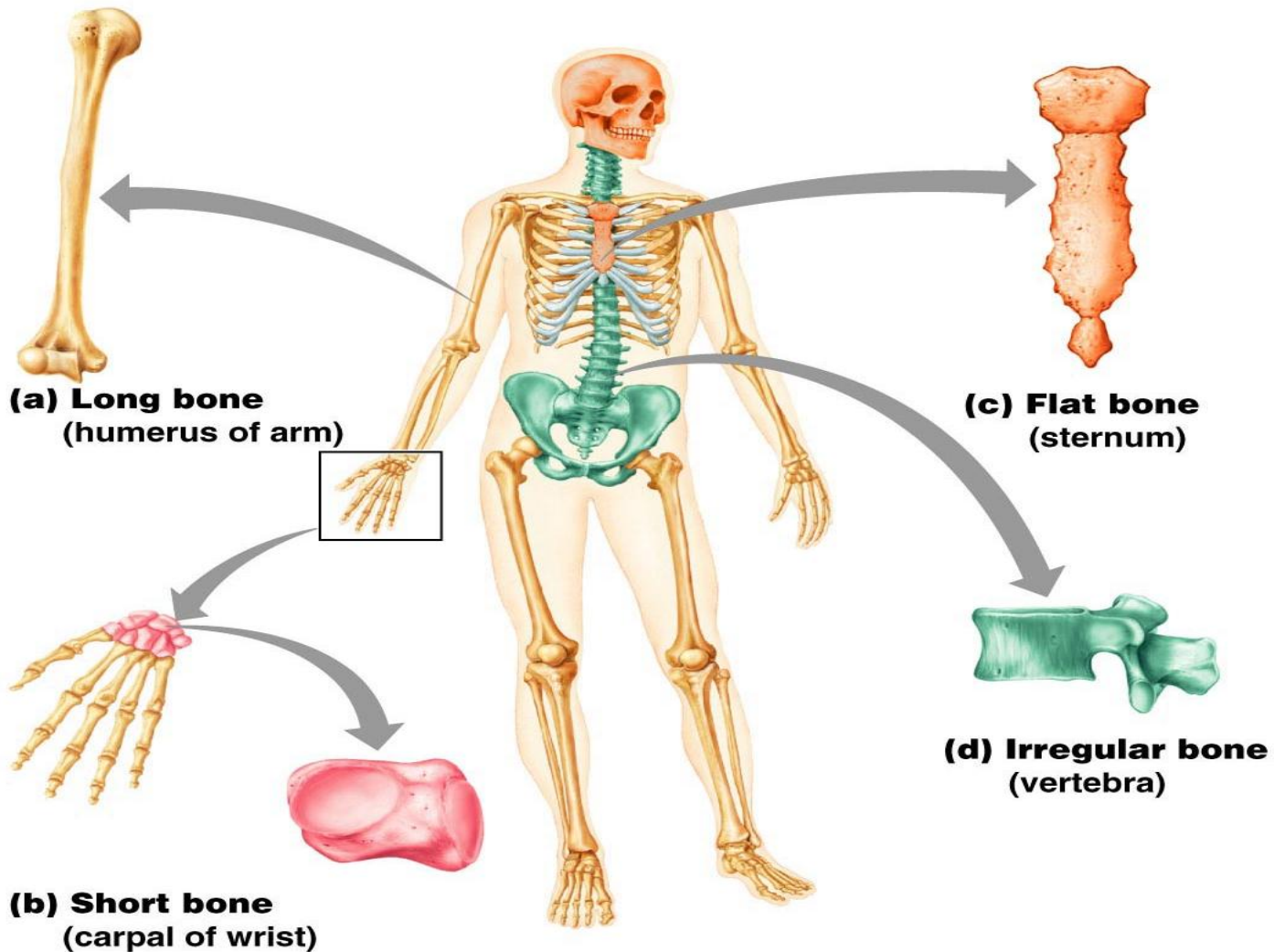
By: Rebuma

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Classification of Bones on the Basis of Shape



Formation of the Human Skeleton

- ❑ In embryos, the skeleton is primarily hyaline cartilage
- ❑ During development, much of this cartilage is replaced by bone
- ❑ Cartilage remains in isolated areas
 - Bridge of the nose
 - Parts of ribs
 - Joints

Formation of the Human Skeleton

All bones derive from embryonic connective tissue) by two processes.

1. Intramembranous ossification

- Mesenchymal model of bones formed during **embryonic period**.

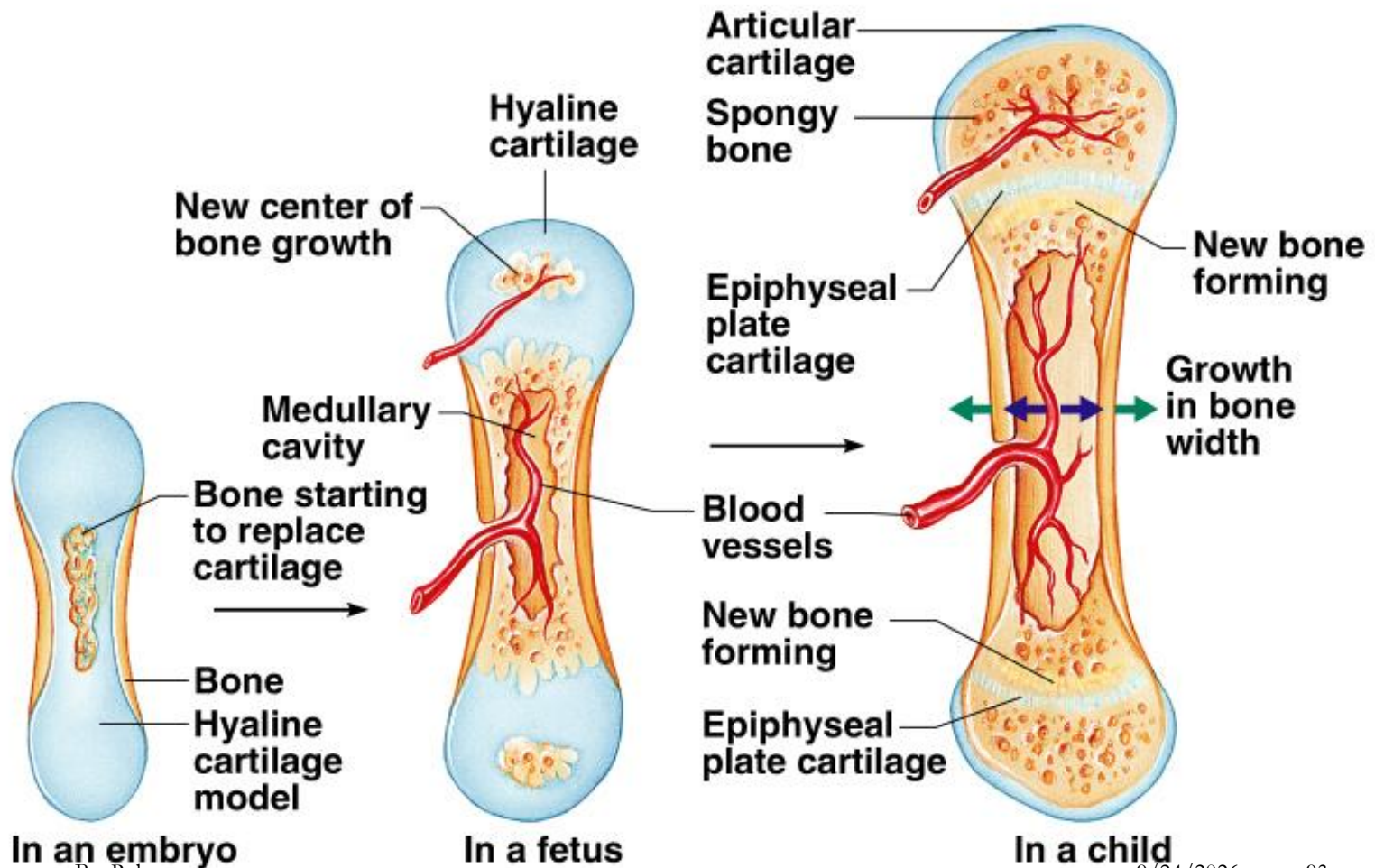
2. Endochondral ossification:-

- Cartilage models of the bones formed from mesenchyme during **fetal period**.

Bone Growth

- ❑ Epiphysial plates allow for growth of long bone during childhood.
- ❑ New cartilage is continuously formed
- ❑ Older cartilage becomes ossified
 - Cartilage is broken down
 - Bone replaces cartilage

Bone Growth



Clinical condition: Common Fractures

☐ Greenstick fracture

- The bone does not break all of the way through
 - Common in children
-

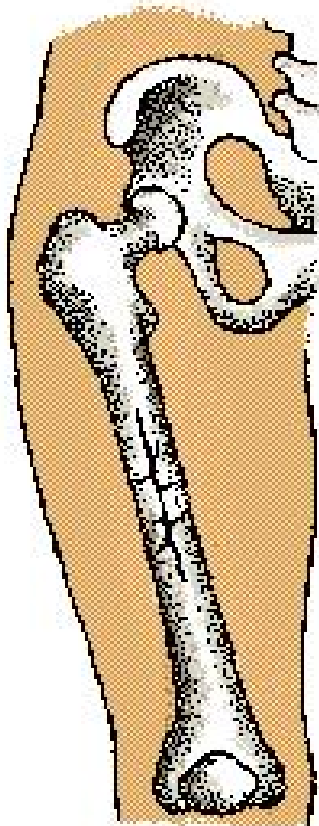
☐ Simple, or closed

- When the bone breaks but the skin does not

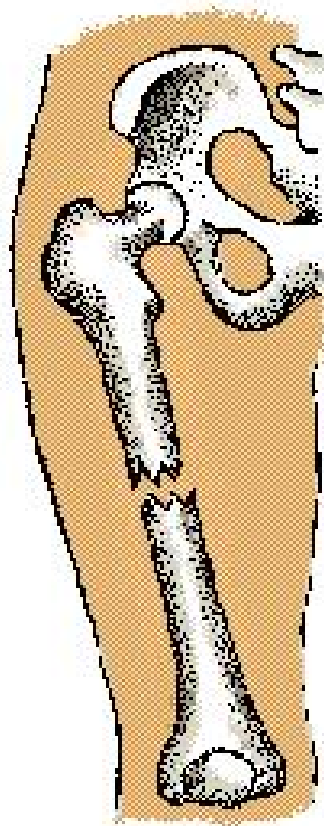
☐ compound, or open

- When the broken bone tears through the skin, introducing the dangerous possibility of infection.

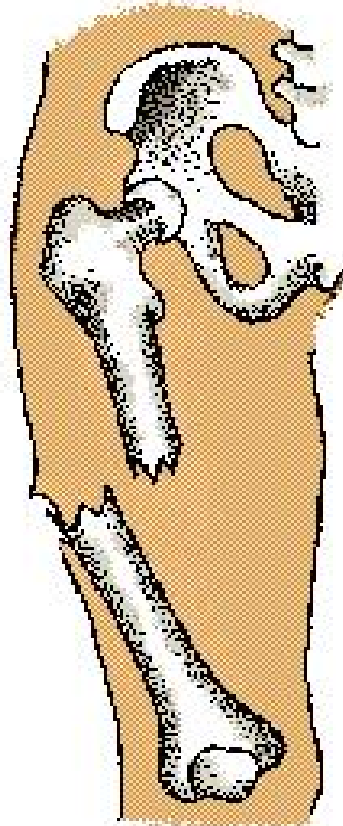
Clinical condition: Common Fractures



Greenstick



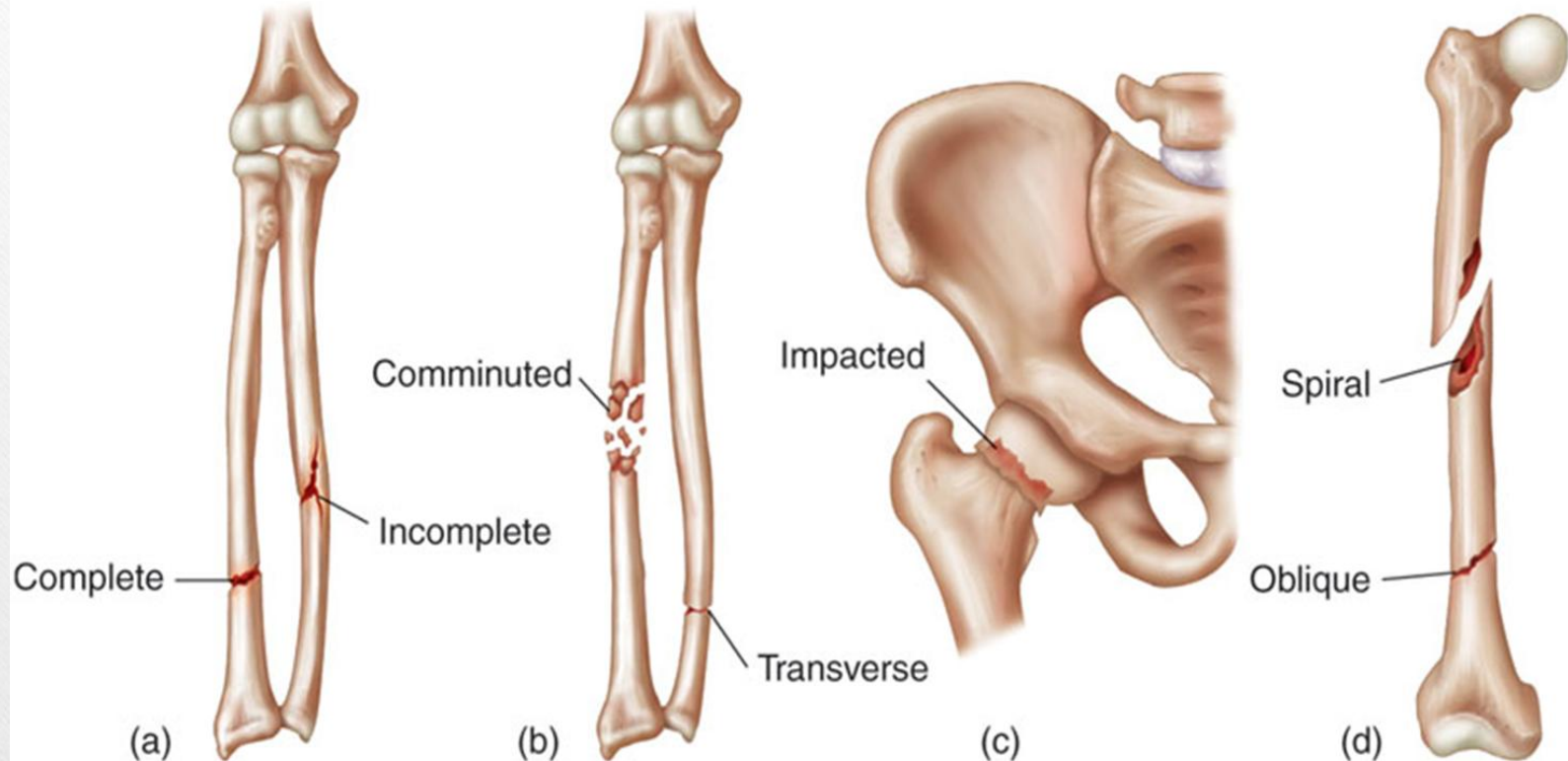
Simple



Compound

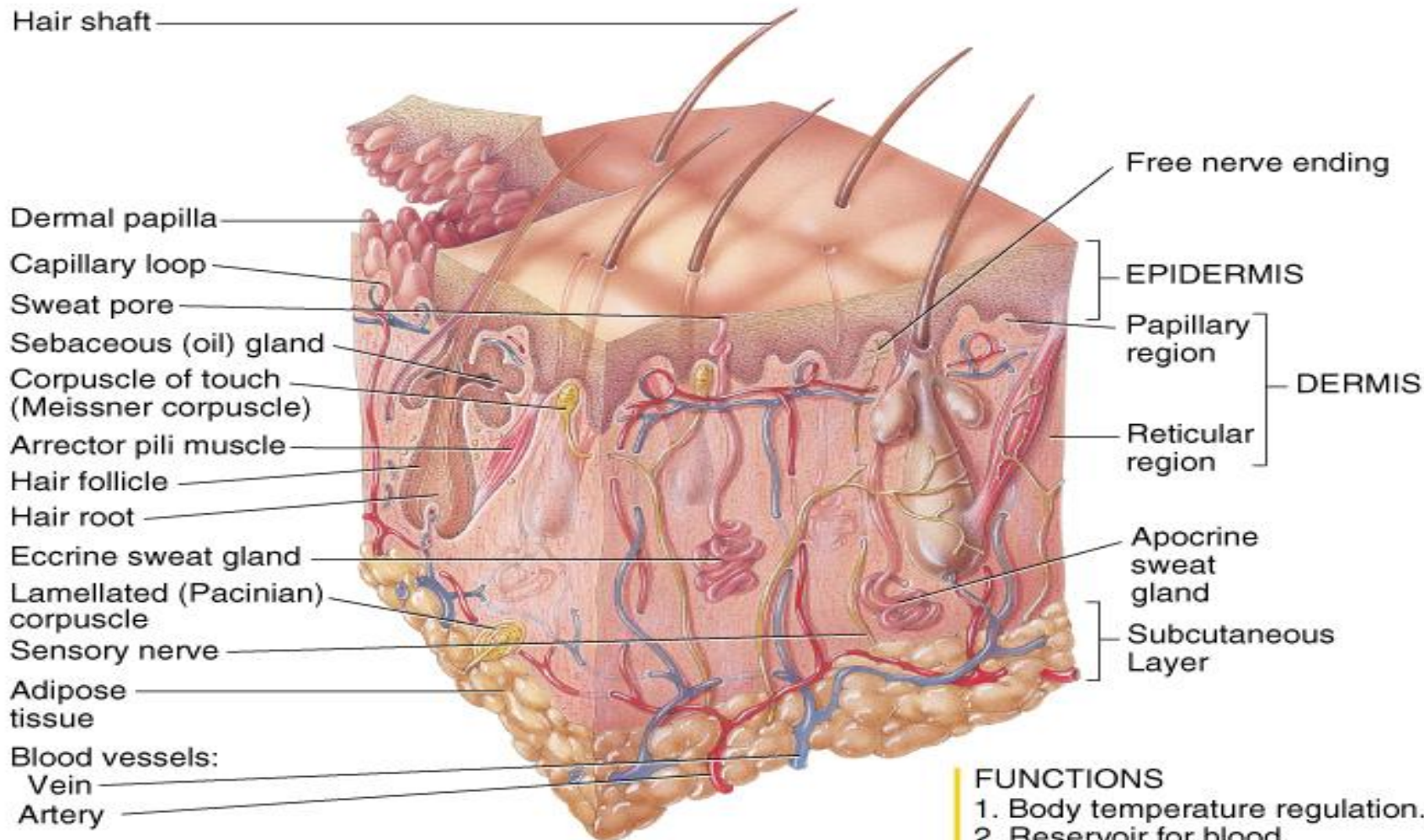
Clinical condition: Common Fractures

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Thank you!

The Integumentary system



FUNCTIONS

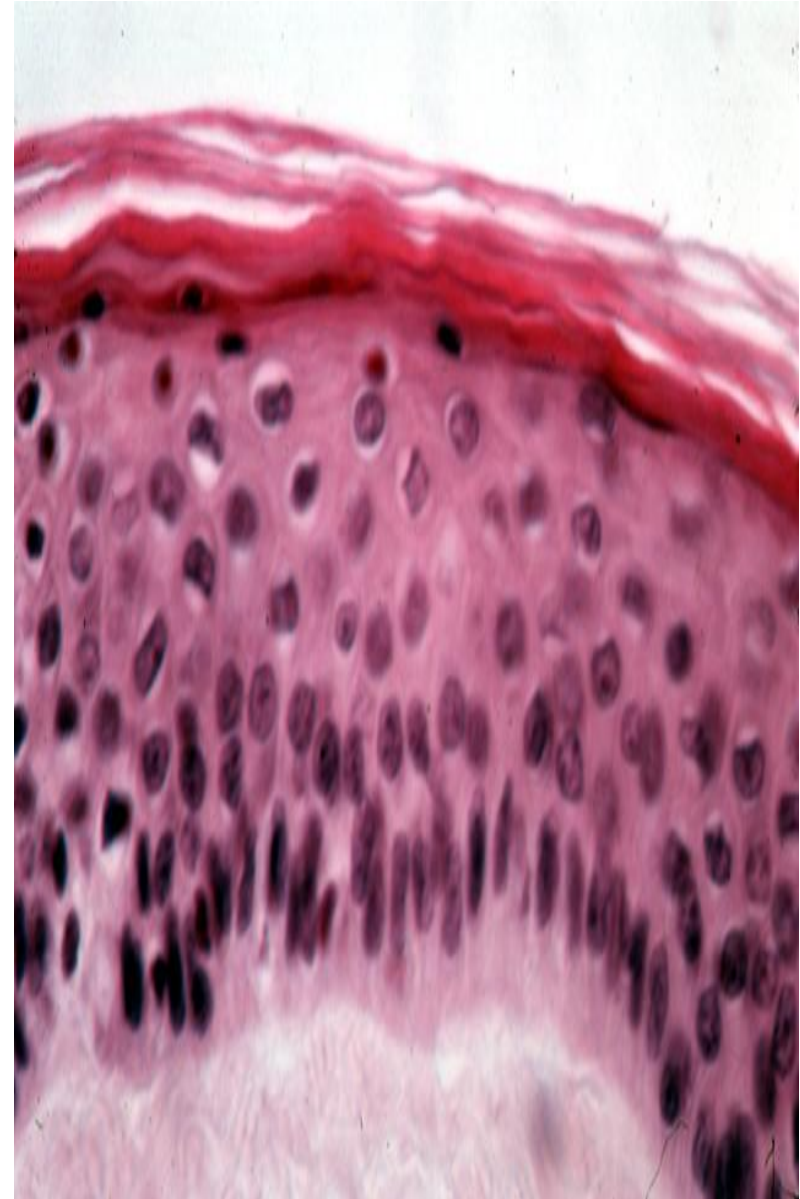
1. Body temperature regulation.
2. Reservoir for blood.
3. Protection from external
4. Cutaneous sensations.
5. Excretion and absorption.
6. Vitamin D synthesis.

Some Facts about Skin

- ◆ Constitutes 16% of the body weight and largest system.
- ◆ Surface area: 1.5-2.0 sq. meters
- ◆ Thickness: 0.5-3.0 mm
- ◆ Growth rate of nail: 0.1mm per day
- ◆ Growth rate of hair: 1.5-2.2 mm per week
- ◆ Life span of hair:
 - ✓ Eyelashes, axilla- 4 months
 - ✓ Scalp – 4 years

General Functions of the Integumentary System

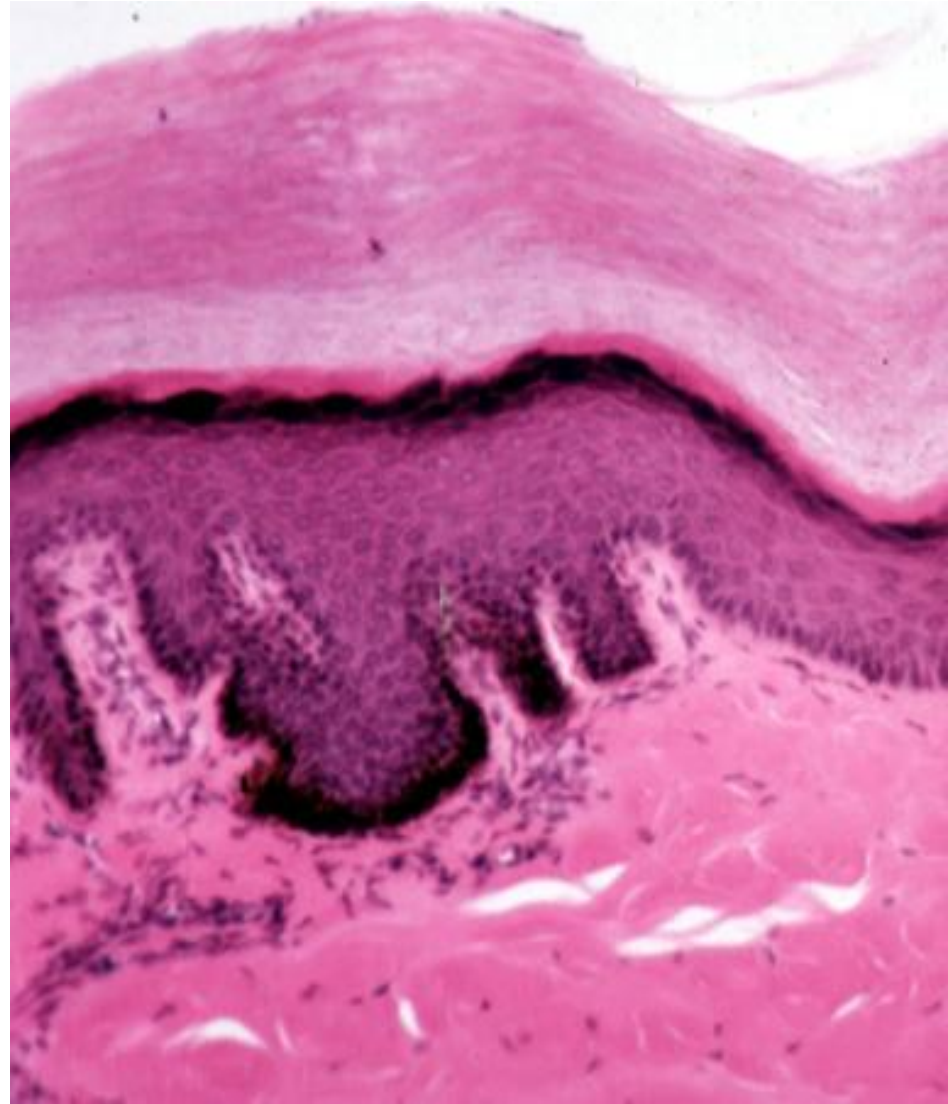
- Protection from
 - mechanical injury
 - Physical protection
 - Chemical prevention of pathogen entry
- Sensation
- Thermoregulation
- Excretion
- Metabolic functions
- Looking good/cosmetics



Skin Types

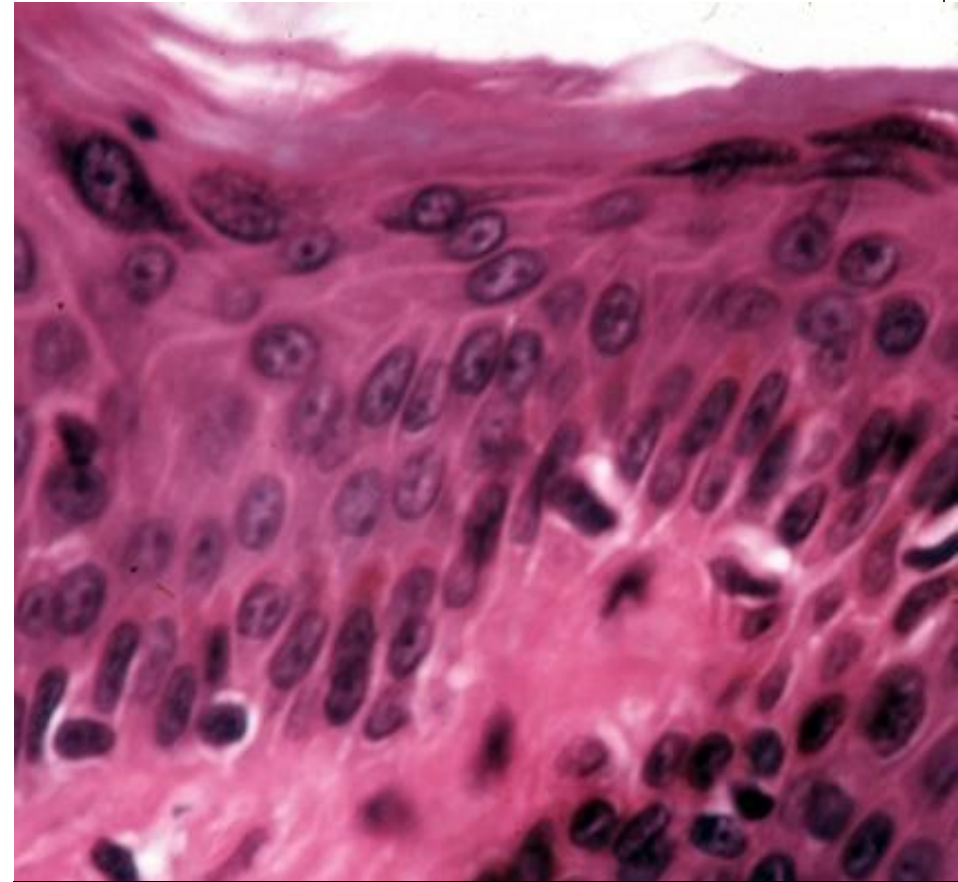
✓ Thick Skin

- Found on soles of feet and palms of hands and corresponding parts of fingers and toes.
- Contains 5 epidermal layers or *strata* (“sheets”):
 - ✦ Stratum basale
 - ✦ Stratum spinosum
 - ✦ Stratum granulosum
 - ✦ Stratum lucidum
 - ✦ Stratum corneum

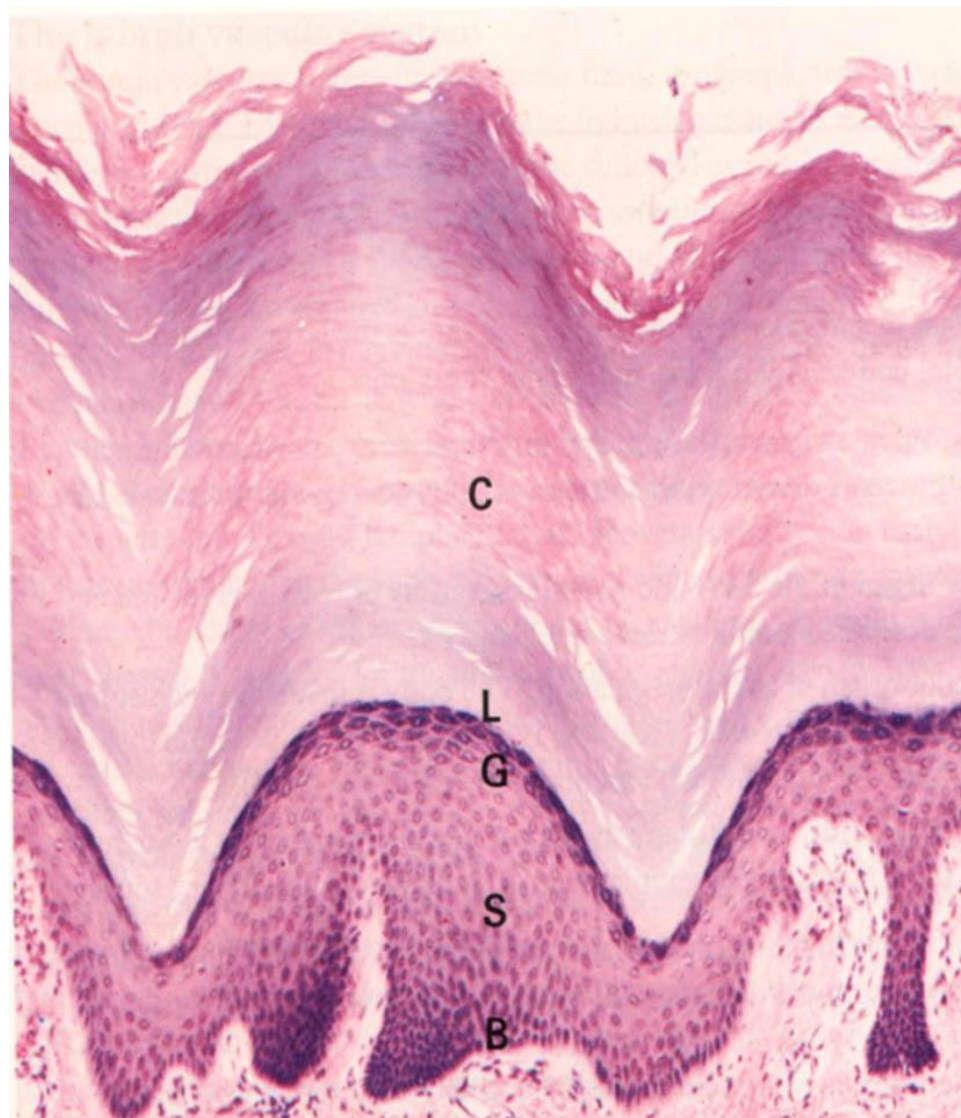


✓ Thin skin

- Found everywhere else on the body.
- Contains only 4 layers. (lacks a stratum lucidum).
- The remaining 4 layers are thinner than those of thick skin.
- Why is thick skin found on the palms and soles? What is the advantage of that?



Note: this slide is at a higher mag. than the thick skin slide on the previous page



Thick skin

Thin skin

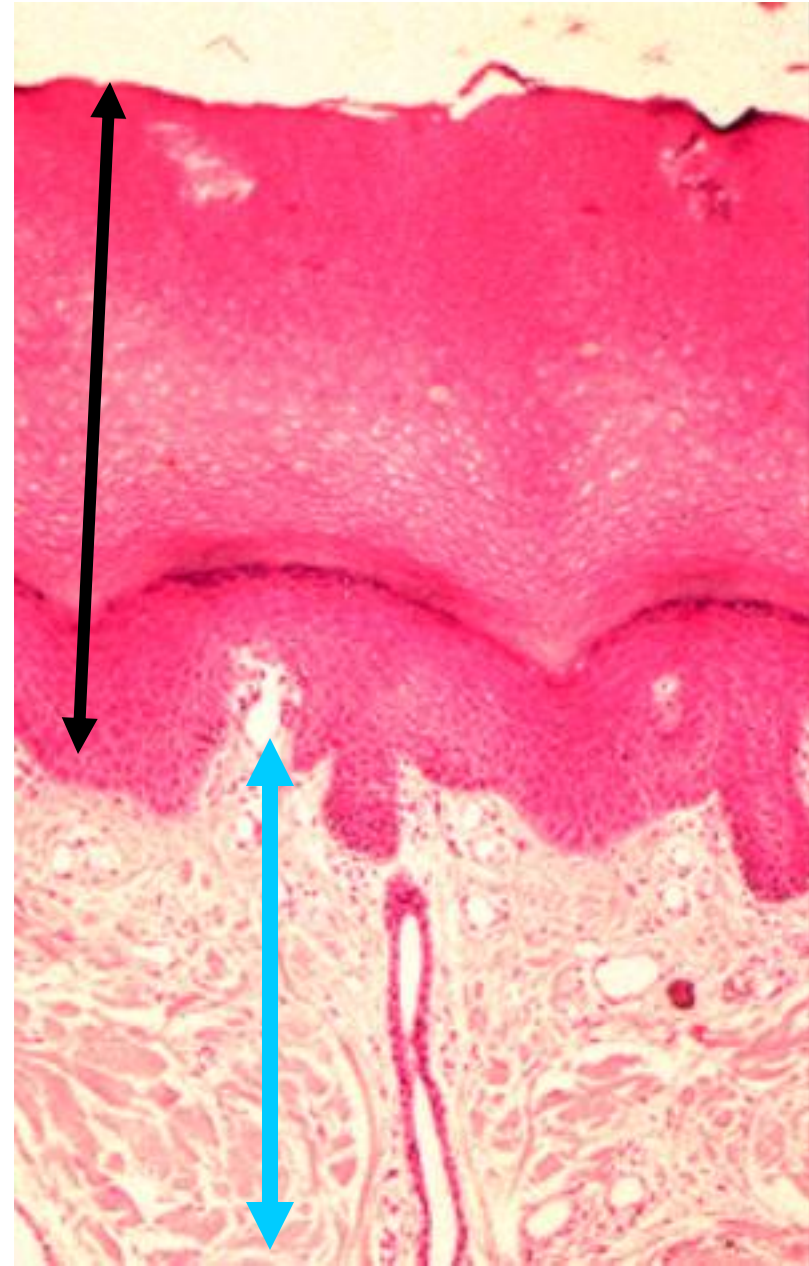


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Basic Skin Structure

The skin has 2 main layers:

- A. The **superficial, avascular epidermis** (*epi* means “above” and *dermis* means “skin”) consisting of 4-5 layers of epithelial cells resting upon a basement membrane.
- A. The **deep, vascular dermis** consisting of fibrous connective tissue. Contains multiple blood vessels, and the accessory appendages.

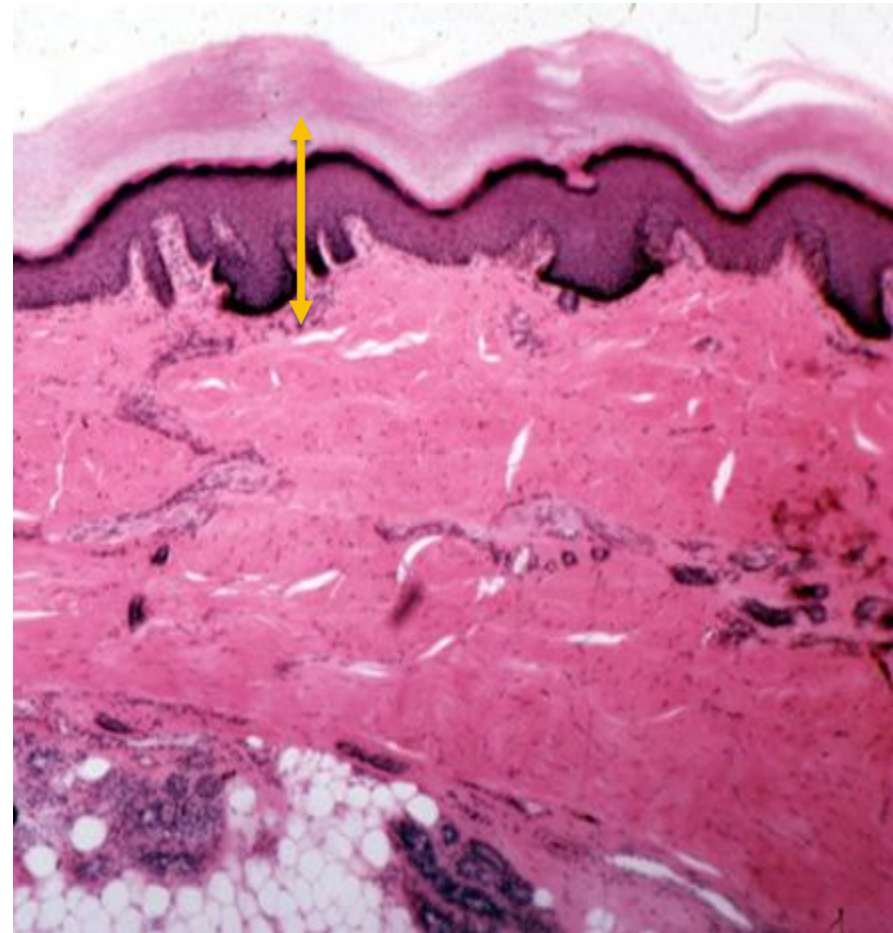


The Epidermis

- ✓ Keratinized stratified squamous epithelium.
- ✓ Avascular.
- ✓ Consists of 4 distinct cell types arranged in 4 or 5 distinct layers.

Epidermal cell types:

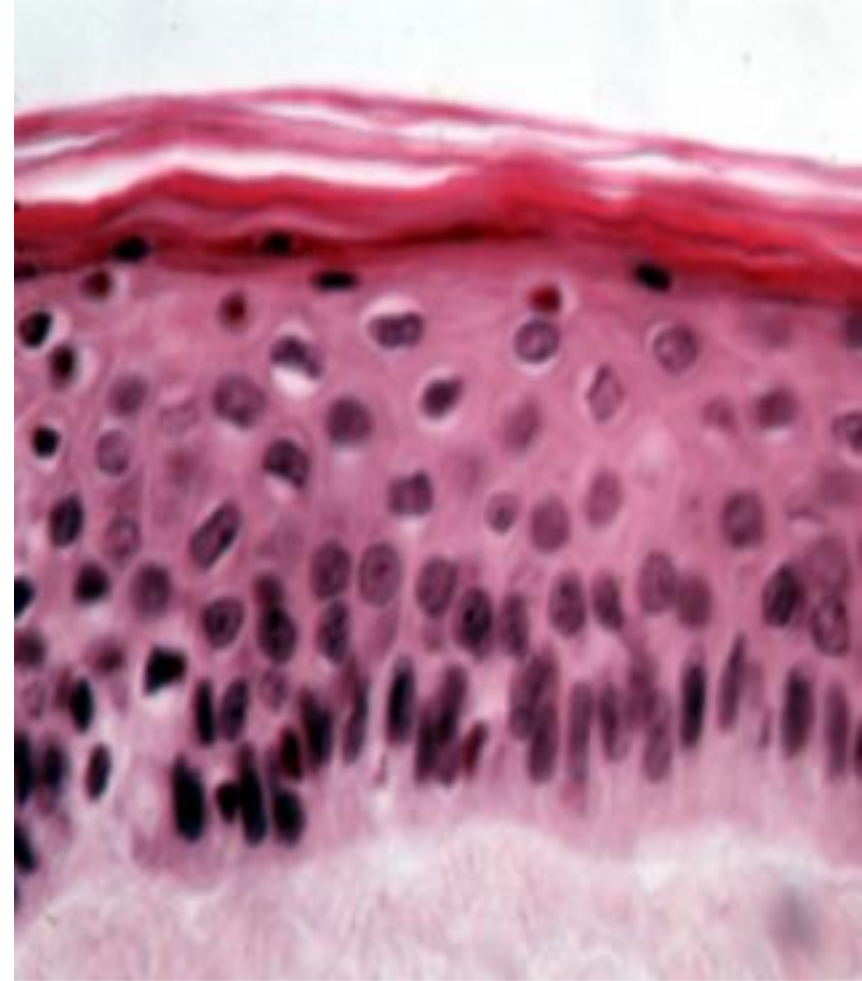
1. **Keratinocytes** (*structure*)
2. **Melanocytes** (*pigment*)
3. **Merkel cells** (*sensation*)
4. **Langerhans' cells** (*phagocytes - immune cells*)



Yellow arrow indicates the epidermis of thick skin

Keratinocytes

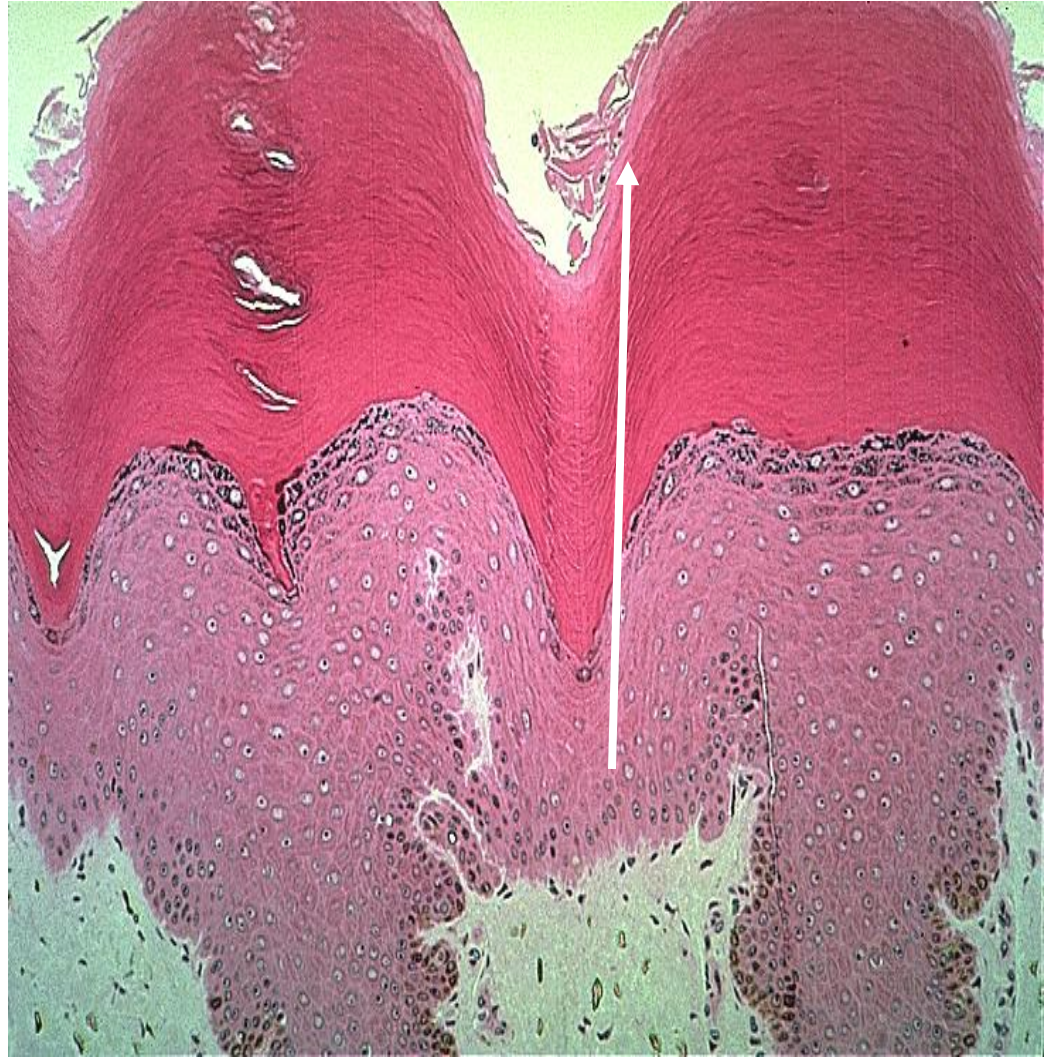
- Most numerous epidermal cell – found in all layers of the epidermis.
- Chief function is the production of keratin – a tough fibrous protein that gives strength and confers a lot of protective ability.
- Tightly connected to one another by desmosomes.
 - Provides continuity, strength, and protection.



Almost all of the epidermal cells in this slide are keratinocytes

Keratinocytes

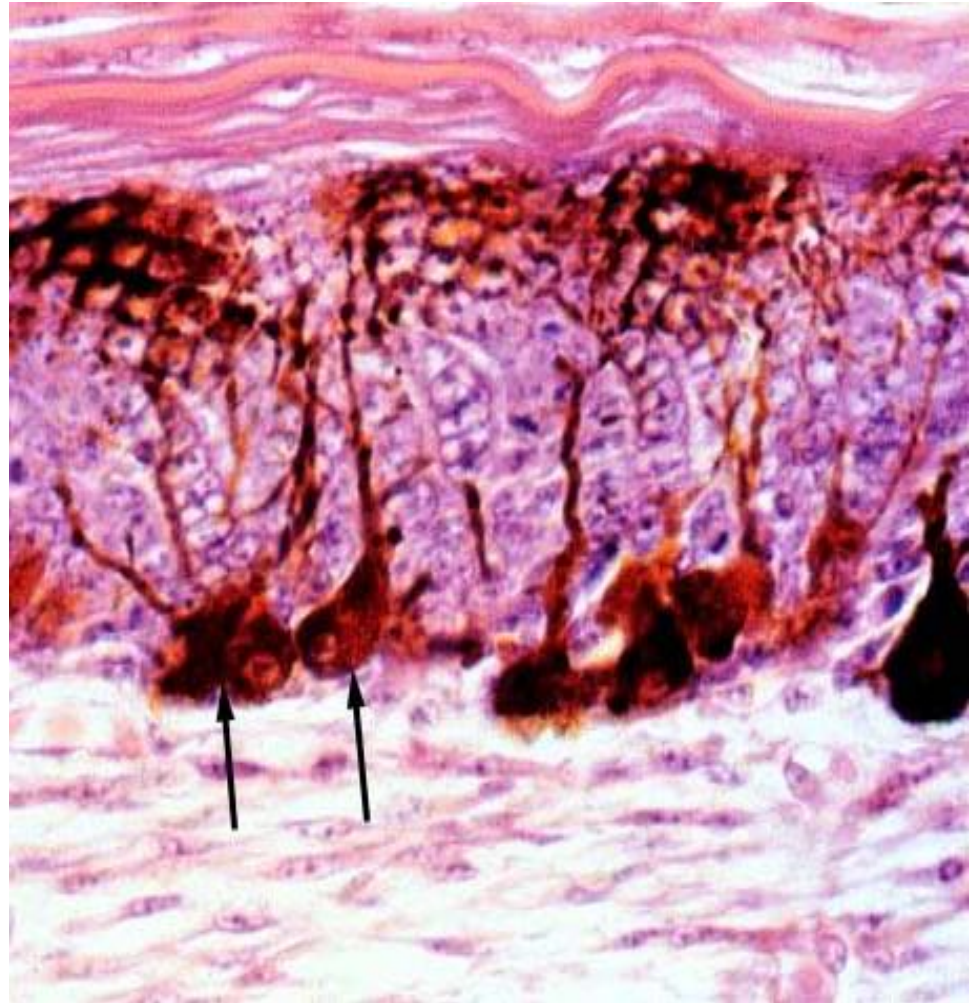
- New cells are continuously made in the deepest layer pushing the older cells up.
- As the keratinocytes move farther from the deepest layer, they make the keratin that eventually dominates their cell contents.



When they have reached the upper layer, they are nothing more than scale-like bags of keratin.

Melanocytes

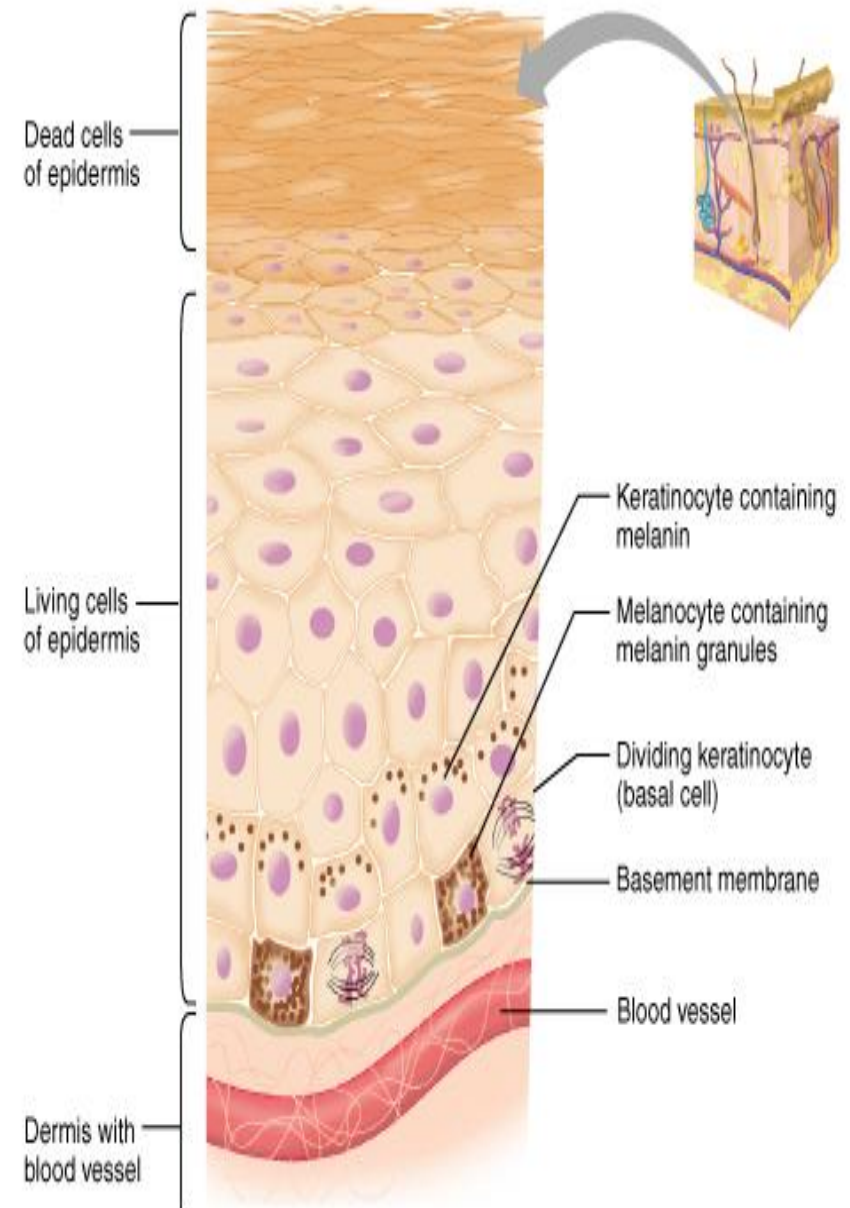
- Spider-shaped epithelial cells that synthesize the protein pigment melanin.
- Found in the deepest layer of the epidermis.
- Melanin is made and then packaged into membrane-bound granules called melanosomes.
- Granules are transferred to the keratinocytes in the 2 deepest layers of the epidermis.



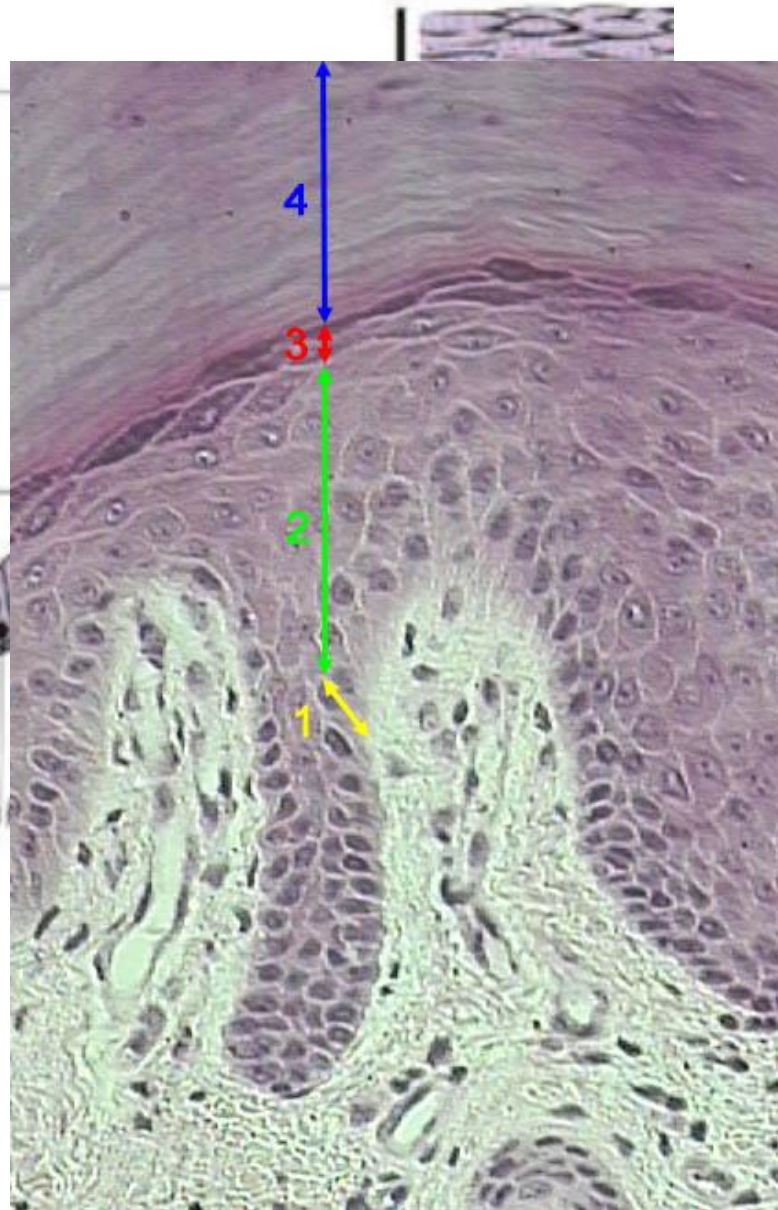
Arrows indicate 2 melanocytes.

Melanocytes

- Melanin granules accumulate on the “sunny side” of the nucleus of the keratinocytes.
 - Would that be the apical or the basal side?
- Melanin granules protect the DNA within the nucleus from being damaged by the ultraviolet radiation from the sun.



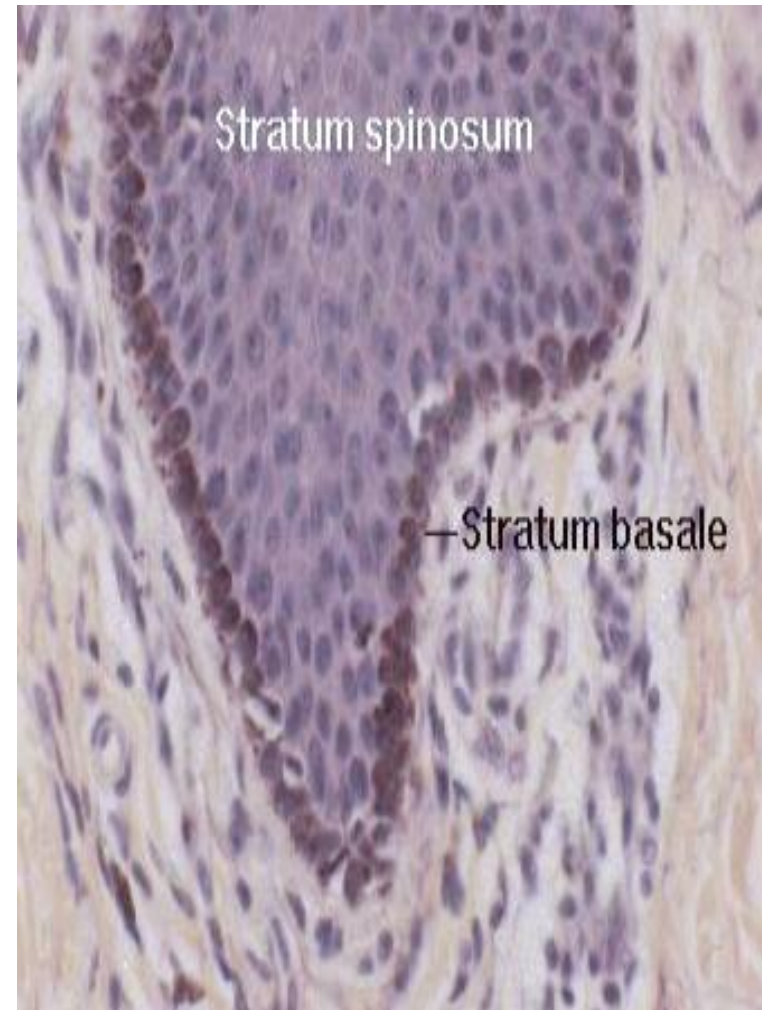
- 4 Stratum corneum
- 3 Stratum granulosum
- 2 Stratum spinosum
- 1 Stratum basale
- Basement membrane



111 Notice the 4 layers of thin skin in both the cartoon and the photomicrograph.

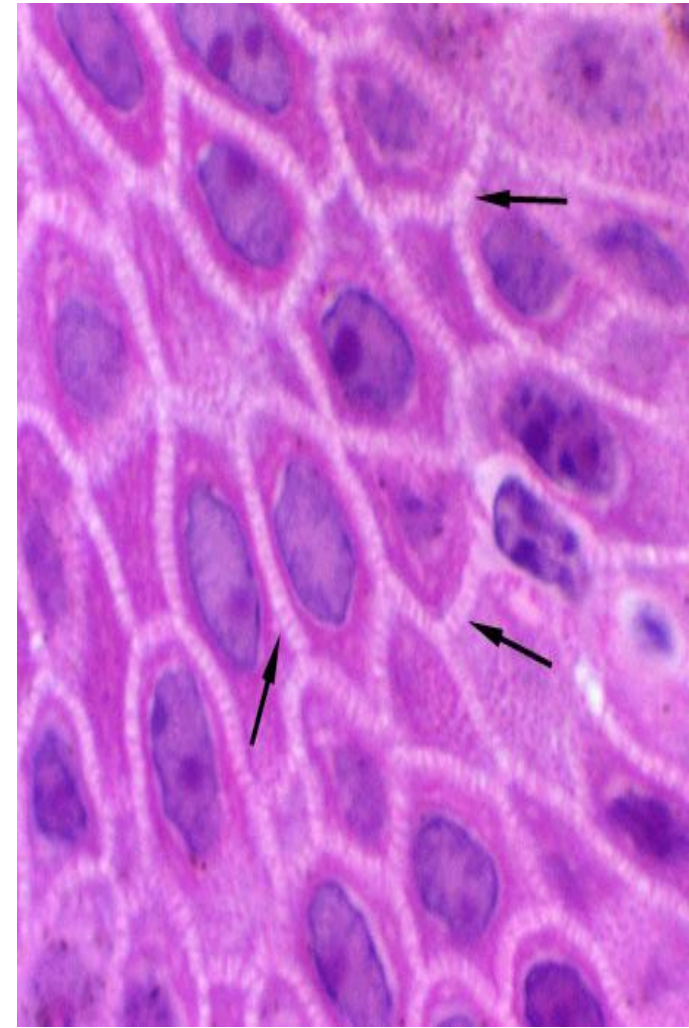
Stratum Basale

- Single row of cuboidal
 - keratinocytes with melanocytes
- Merkel cells interspersed.
- Cells in this layer are highly mitotic – they're dividing often.
 - Due to this fact, this layer is a.k.a. the **stratum germinativum**.



Stratum Spinosum

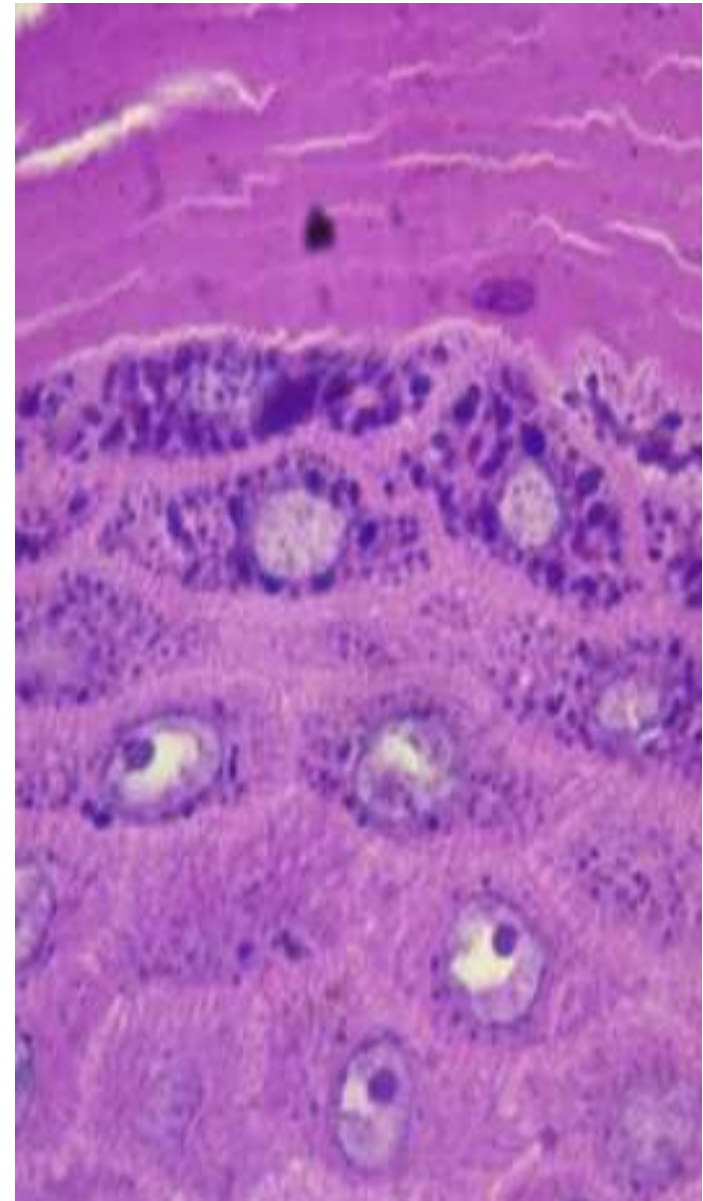
- 2nd deepest layer. Consists of 8-10 layers of cells.
- Cells of the lower layers can still be mitotic.
- As cells get pushed upward, they begin to flatten and begin to make the precursors of keratin.
- A.k.a. the “**prickly layer**” because in tissue sections, they shrink and pull back.
- This makes their exposed desmosomes connecting adjacent cells appear to resemble spikes or spines.



Do you see the spines?

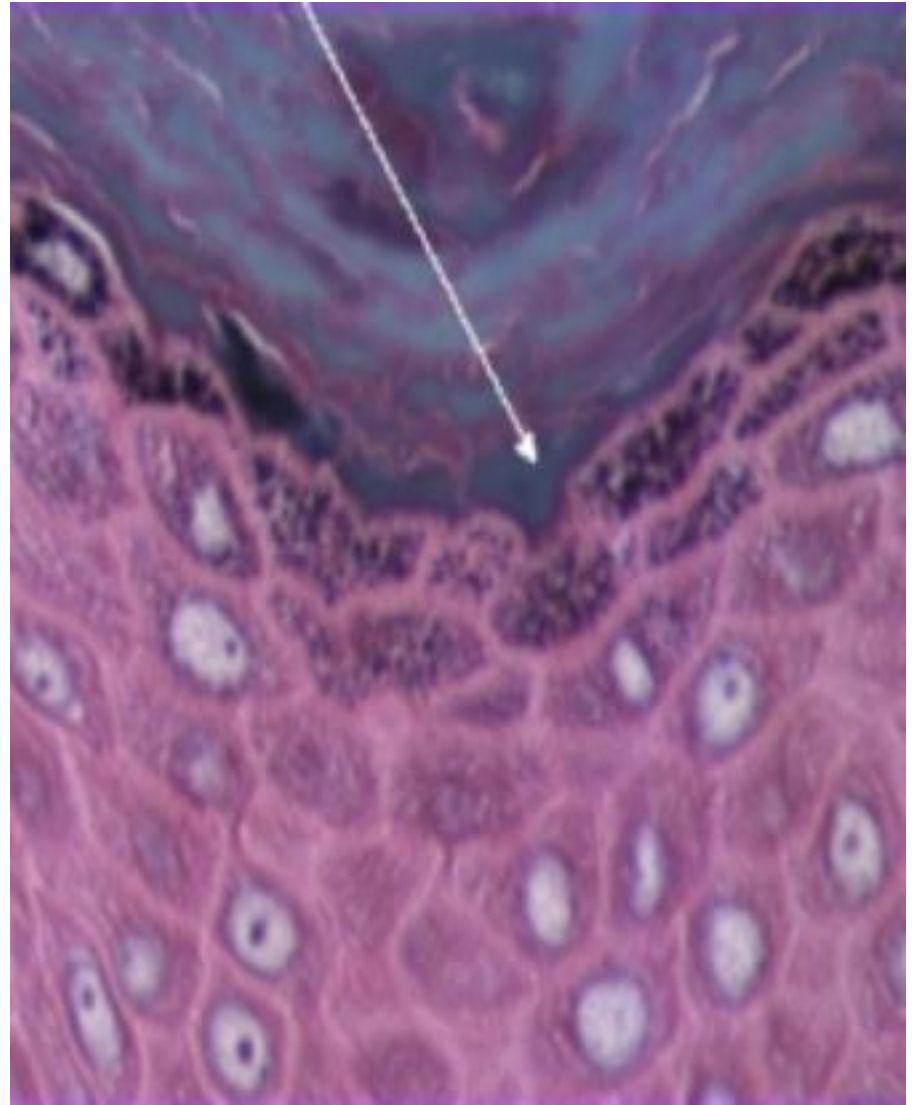
Stratum Granulosum

- ◆ 3-5 cell layers thick.
- ◆ Cell morphology begins to change dramatically as cells continue to flatten and their nuclei and organelles disintegrate.
- ◆ Accumulate granules containing a precursor of keratin and granules containing a waterproofing agent.
- ◆ if water can't diffuse upward, how would the cells above this layer receive nutrients?
What would happen to them?



Stratum Lucidum

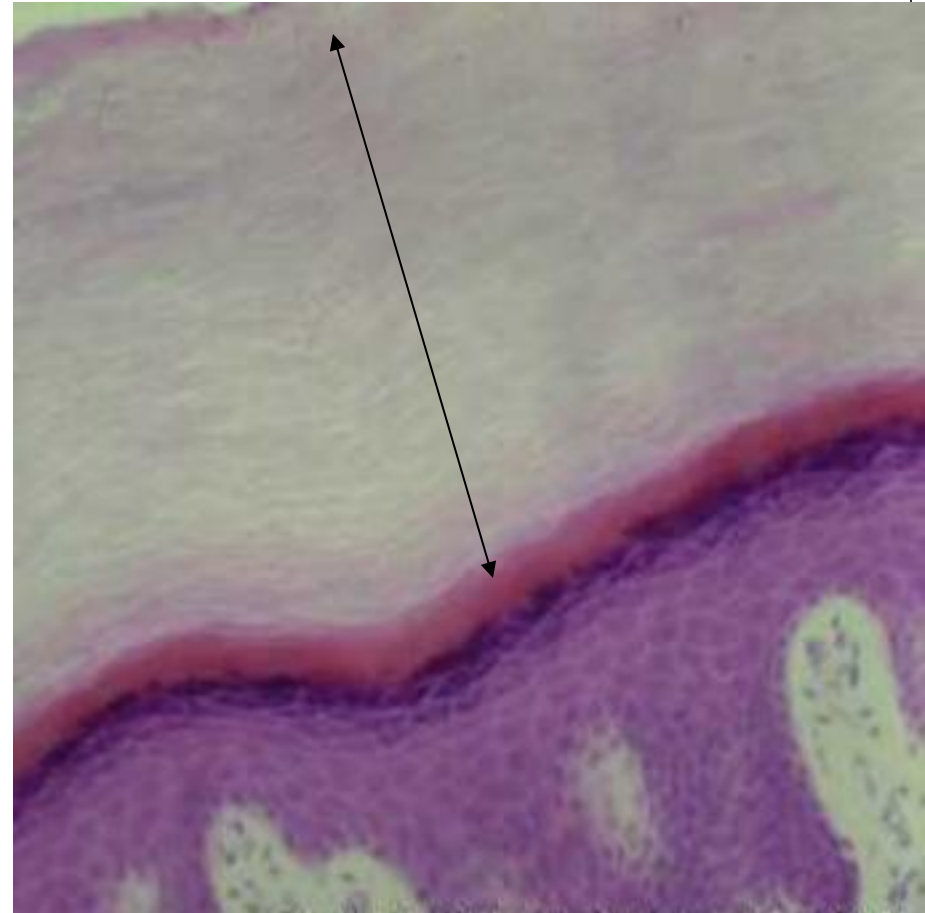
- ◆ flat, dead keratinocytes appears clear in the light microscope
- ◆ because it lacks nuclei and organelles which typically stain well.



Stratum Corneum

Outermost stratum. 20 – 30 layers of flat (squamous), highly keratinized, dead cells.

- ◆ Protects against mechanical abrasion
- ◆ cells can absorb impacts and simply flake off if necessary. Prevents pathogen entry. Prevents desiccation (drying out).
- ◆ A.k.a. the **cornified** layer.
- ◆ The process by which cells in the stratum basale divide and then advance upward

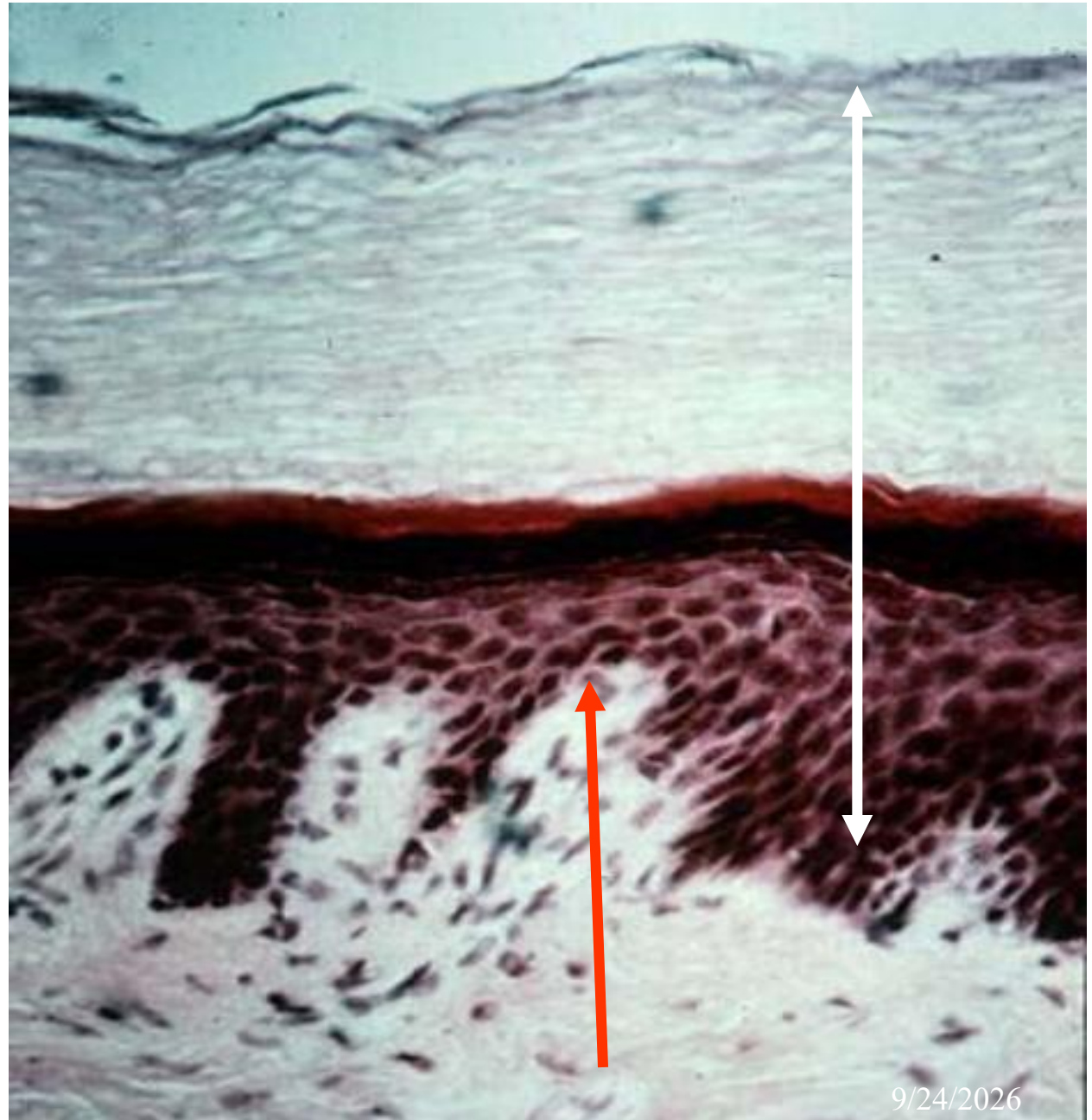


becoming more and more keratinized and less and less alive as they go is known as **cornification**.

Thick and thin skin

	Thin Skin	Thick Skin
Layers of epidermis	St. corneum & spinosum are thin while lucidum is absent.	St. corneum & spinosum are thick while lucidum is present.
Thickness of epidermis	0.10-0.15 mm	0.6-4.5 mm
Epidermal ridges	Absent	Present (well developed dermal papillae)
Hair follicles, arrector pili muscle & sebaceous gland	Present	Absent
Sweat glands	Few	Many
Sensory receptors	Less	More
Distribution	Covers all parts of body except palms & soles	Present in palms, palmar surface of digits & soles

Identify the 2
layers indicated by
the arrows



Skin Color

Due to 3 pigments:

1. Melanin
2. Carotene
3. Hemoglobin

Of these, only melanin is made in the skin.

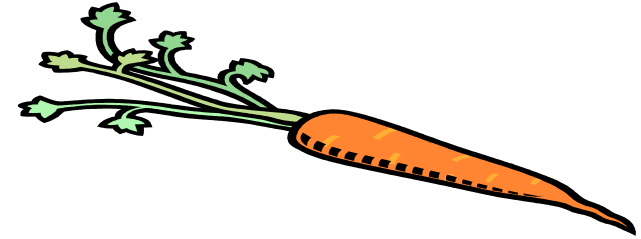


1. Melanin:

- ◆ Polymer of tyrosine amino acids. Its synthesis is catalyzed by an enzyme called tyrosinase. Albinos lack this enzyme.
- ◆ Ranges in color from yellow to reddish brown to black.
- ◆ All people have the same # of melanocytes, individual variations in skin color are due to how much and what type of melanin is made.
- ◆ **Freckles** and **moles** are local accumulations of melanin.

2. Carotene

- Yellow to orange pigment found in plant products such as carrots.
- When large amounts are eaten, it can be deposited in the stratum Corneum of thick skin.



3. Hemoglobin

- Pigmented protein that transports oxygen within the blood.
- In Caucasians, the fair skin allows the crimson color of oxygenated blood to make the skin have a somewhat pinkish hue.

Dermis

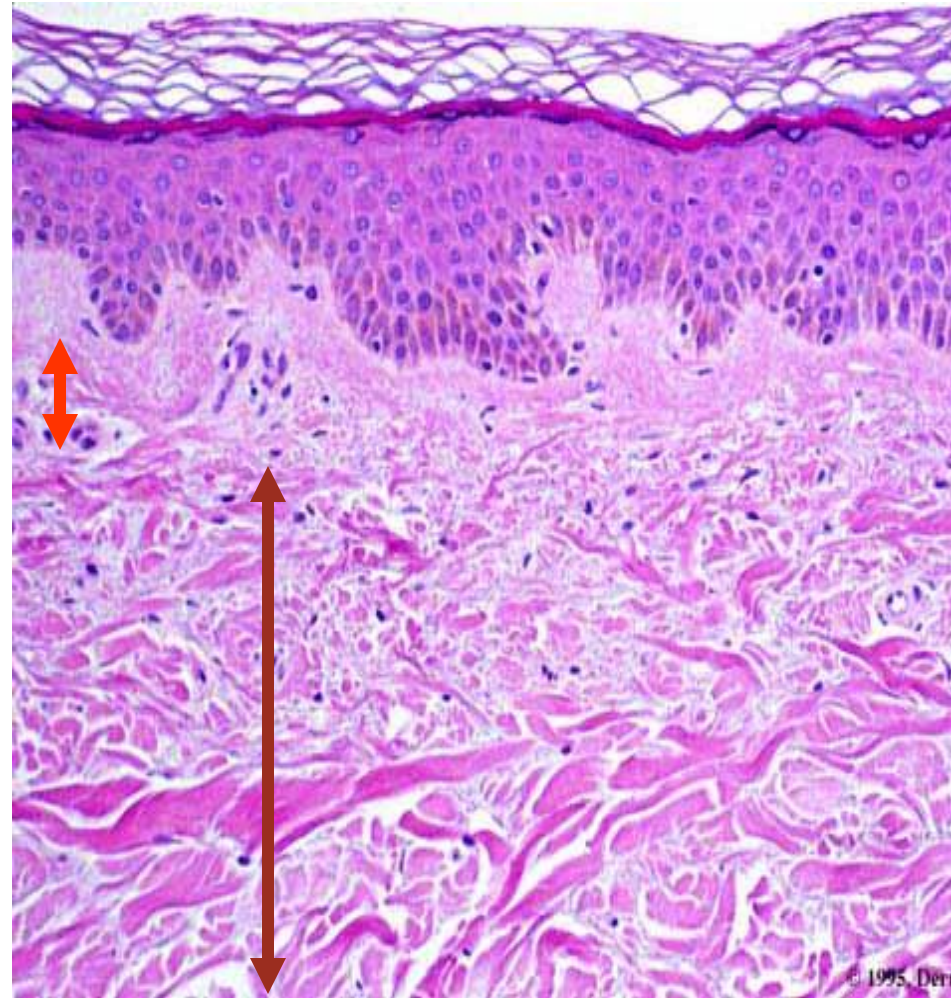
- Strong, flexible fibrous connective tissue.
- Divided into **papillary dermis** and **reticular dermis**.
- **Papillary dermis** is the upper 1/5 of the dermis and consists of loose (areolar) CT.
 - Provides an area for immune cells to fight invaders.
 - Projects upward (as) to interdigitate and form a strong connection with the epidermis.
 - Heavily invested with blood vessels – they constrict in cold weather and dilate in warm weather. Why?
 - Also contains multiple sensory receptors.

dermal papillae



Dermis

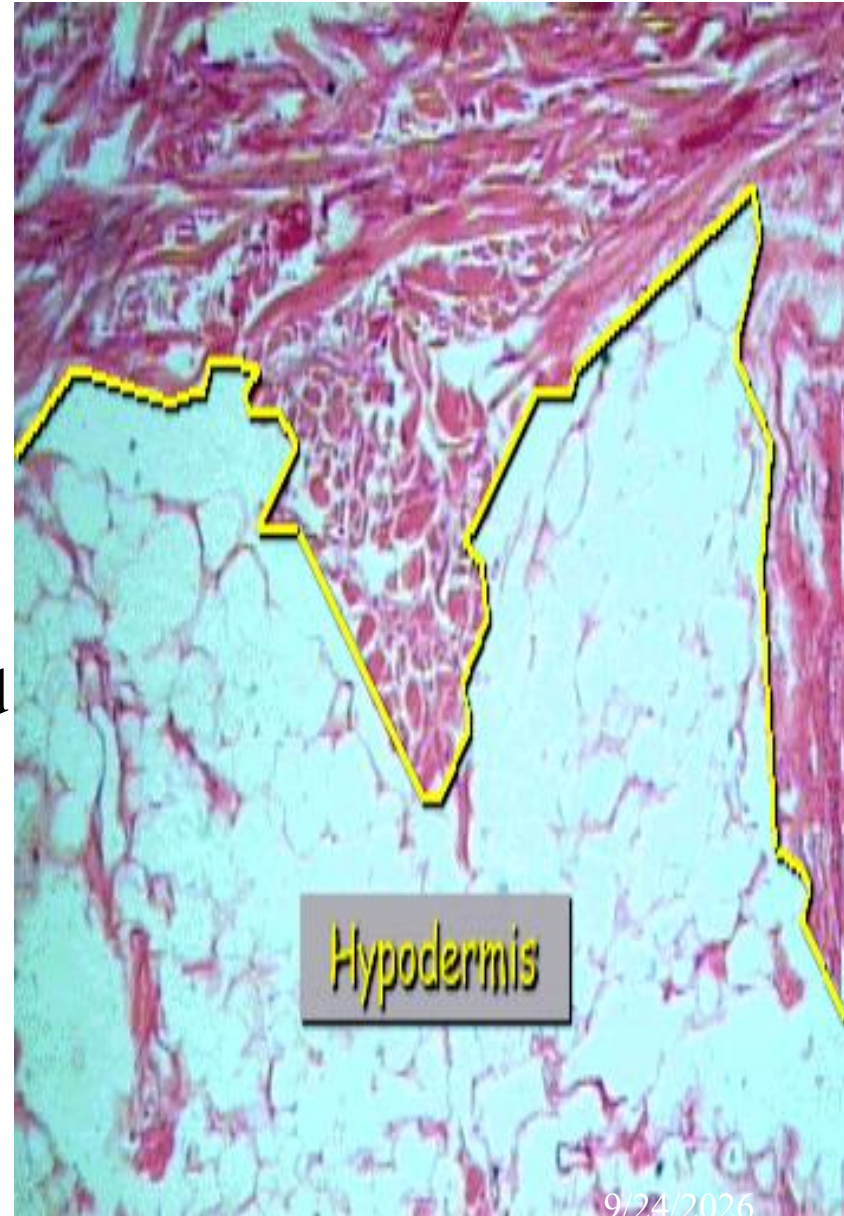
- **Reticular dermis** is lower 4/5 and consists of dense irregular connective tissue.
- The prominent, horizontally running collagenous fibers give the skin strength and resiliency.
- Elastic fibers give the skin the ability to stretch and recoil.
- The majority of the appendages of the skin are contained within the dermis.



Red arrow indicates the papillary dermis and blue arrow indicates the reticular dermis

Subcutaneous Tissue

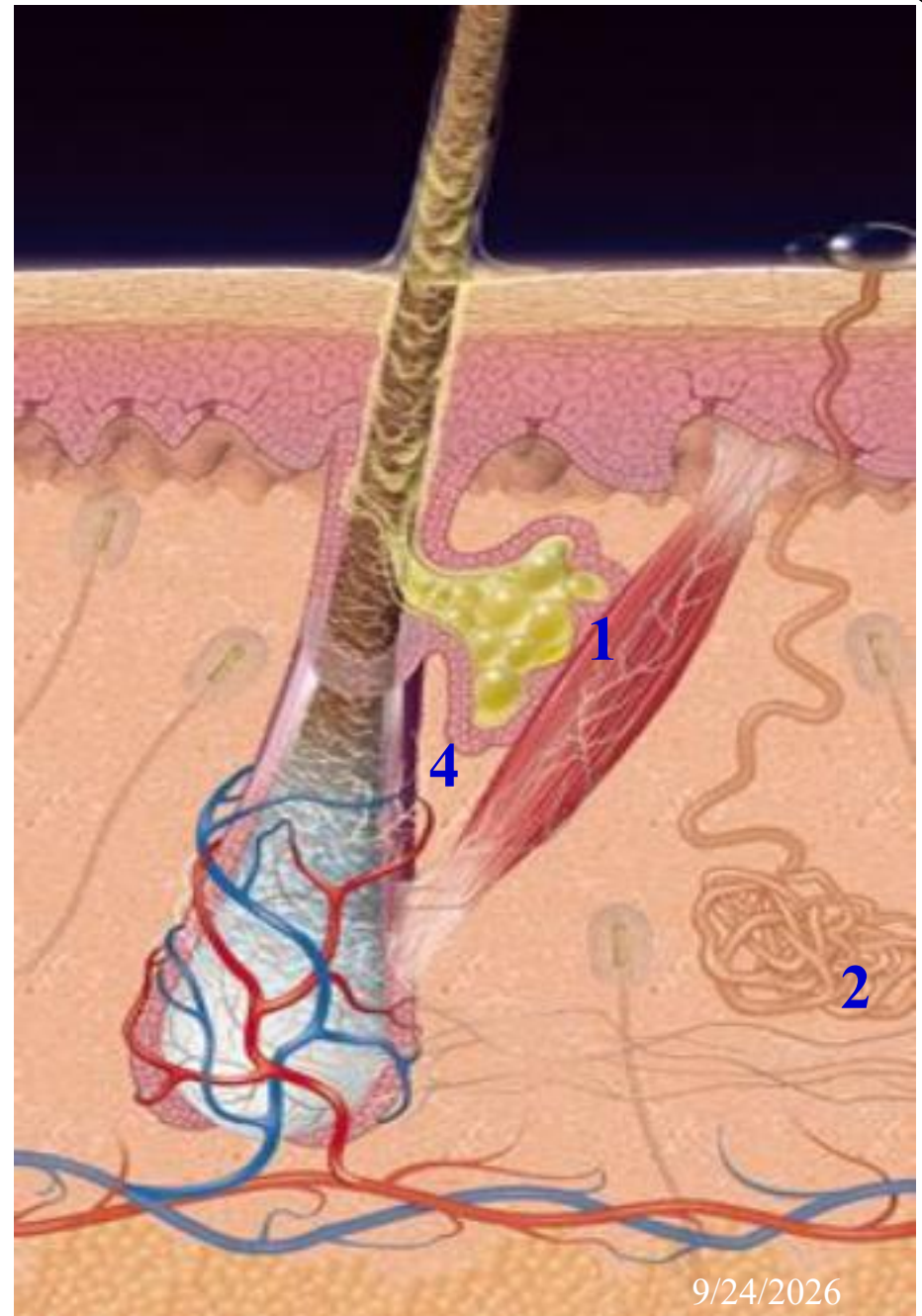
- ◆ A.k.a. the **hypodermis** or the superficial fascia.
- ◆ Deep to the dermis.
- ◆ Consists of primarily adipose tissue plus some loose areolar connective tissue.
- ◆ Stores energy, provides insulation and padding.
- ◆ Anchors the skin to underlying structures, especially muscles.



◆ Different distribution between the

Appendages of the Skin

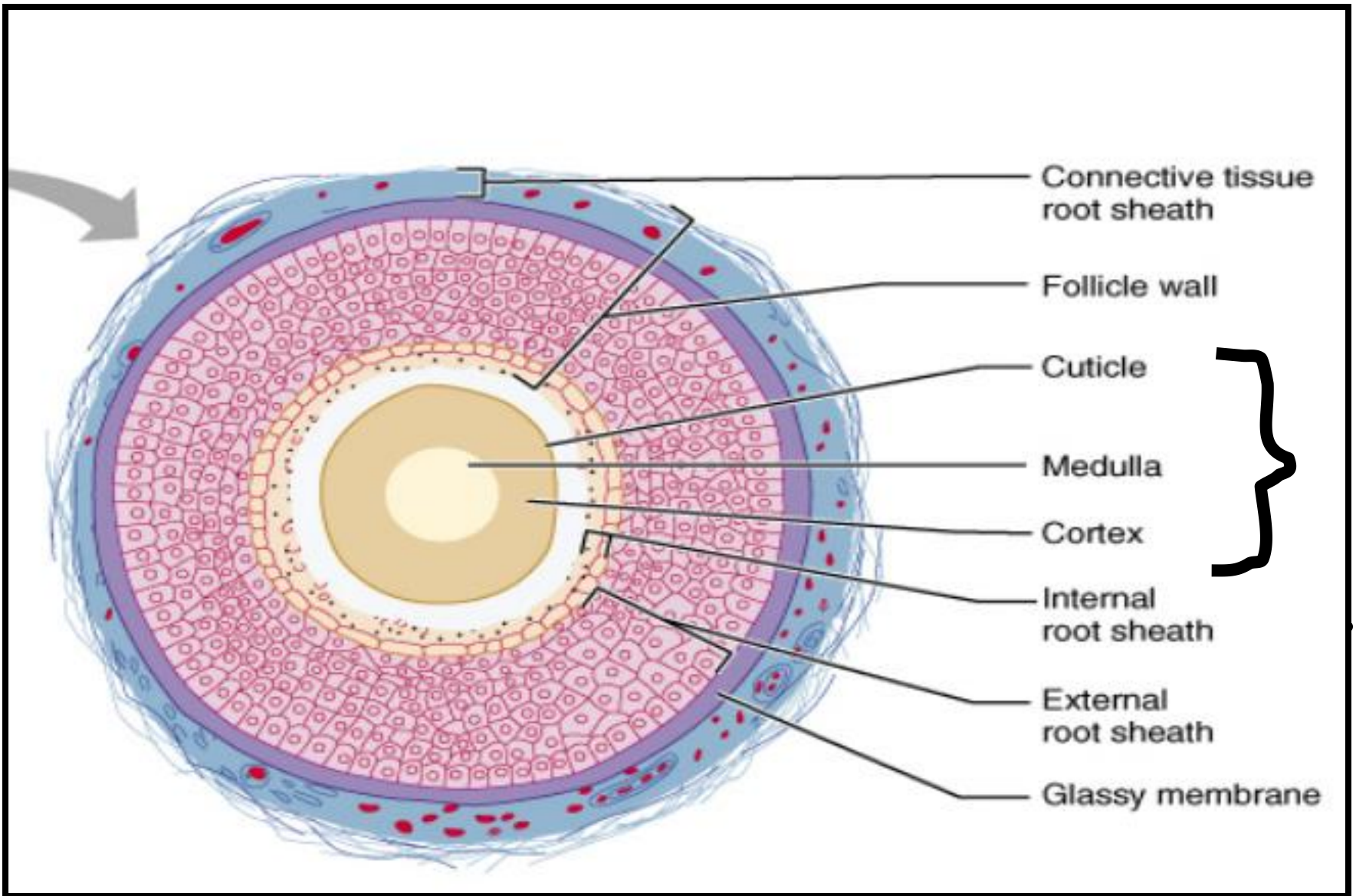
- A. Hair
- B. Nails
- C. Skin glands
 - I. Sebaceous glands
 - II. Sudoriferous glands
 - III. Apocrine sweat glands (e.g. Ceruminous glands)



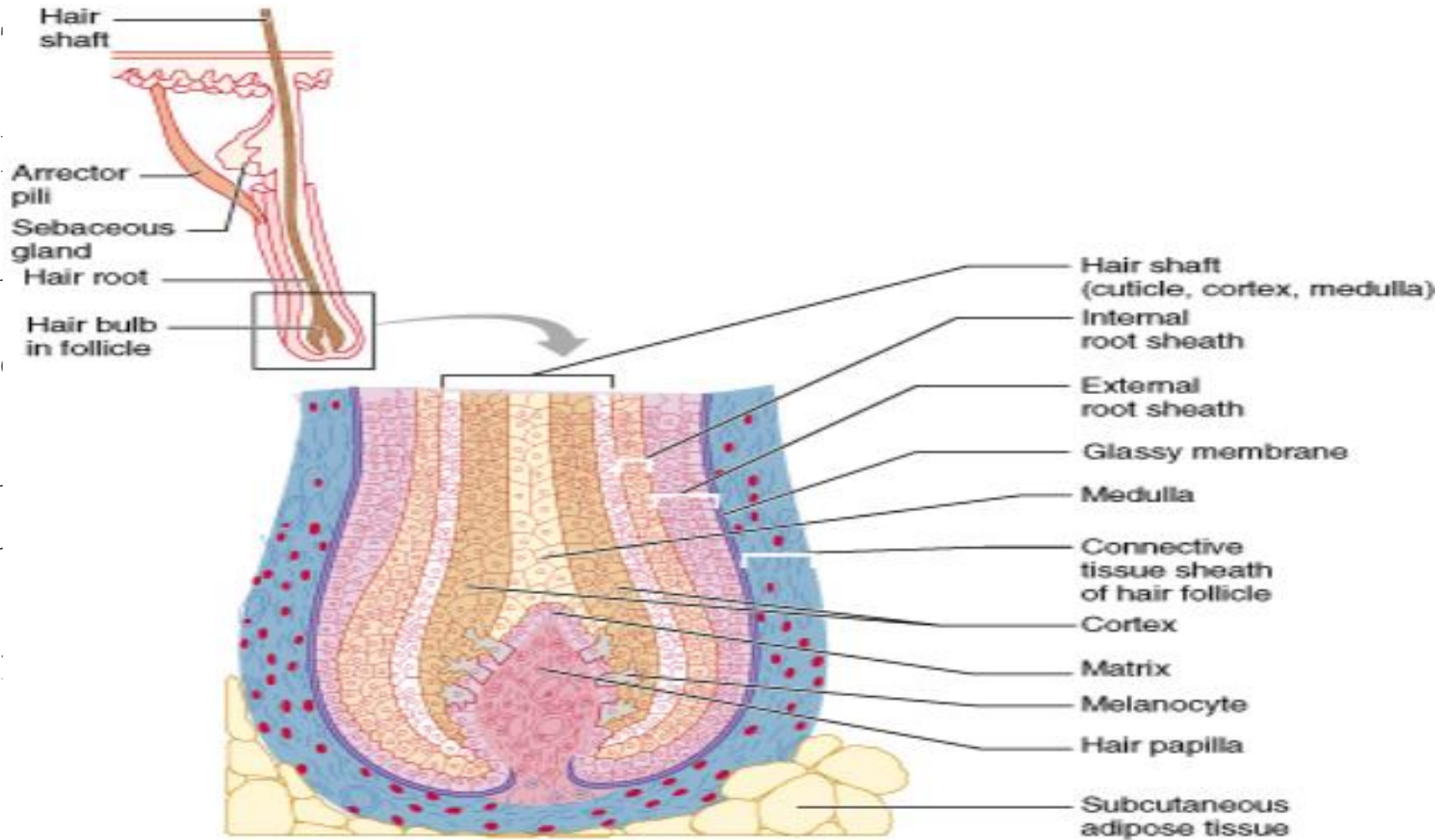
Hair and Hair Follicles

- ◆ Hair is a flexible strand made of highly keratinized dead cells.
- ◆ The keratin in both hair and nails is hard keratin, a stiffer, more compact version of the soft keratin that dominates the epidermis.
- ◆ It is tougher and its cells do not individually flake off.
- ◆ The hair is made by the living hair follicle.

Hair and Hair Follicles

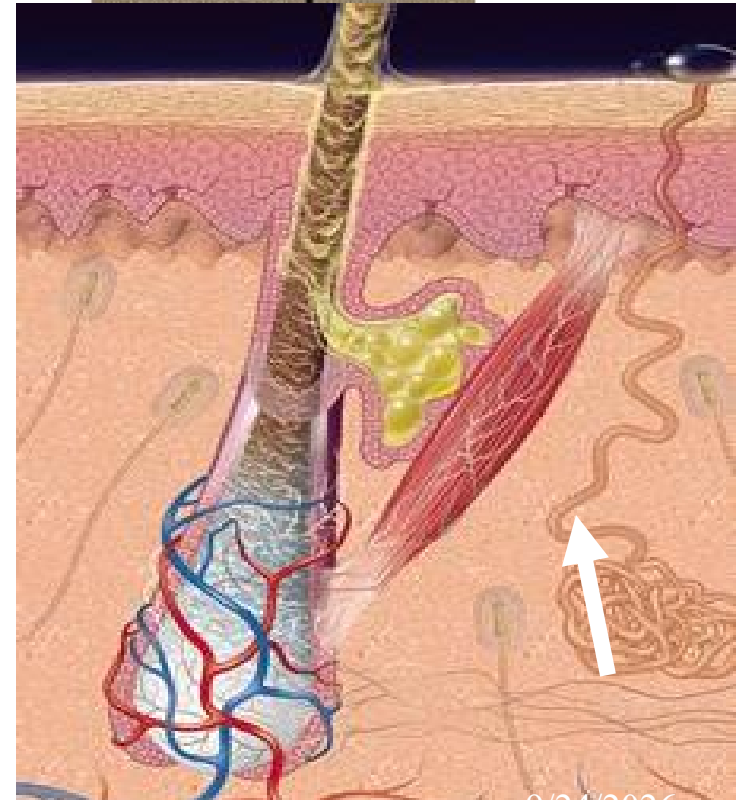
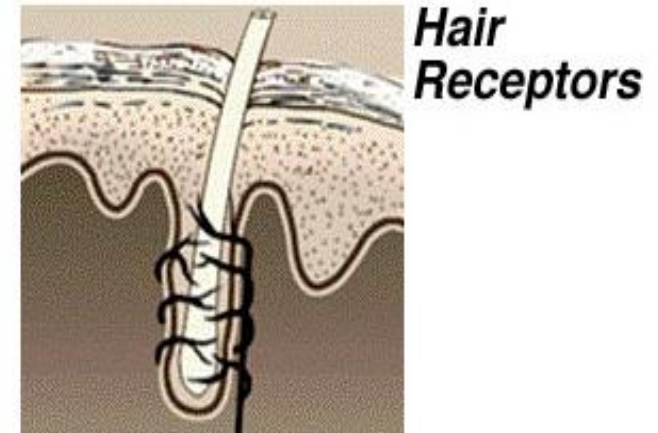


Hair and Hair Follicles



Hair and Hair Follicles

- ◆ Wrapped around the bulb of the follicle is a network of sensory nerve endings known as the hair root plexus. Allow the hairs to serve a sensory function.
- ◆ Attached to each hair is a bundle of smooth muscle known as an arrector pili muscle.
- ◆ In times of fright or cold, these muscles contract and cause the hair to stand on end – and produces goose bumps.



The arrow indicates an arrector pili muscle. In this picture, you should also try to identify the shaft, root, follicle, hair papilla, and sebaceous gland.



Types of hair

- ◆ Not all hair is alike, even on one person. Over the course of our lives, we grow three kinds of hair; **lanugo**, **vellus** and **terminal hair**.
- ◆ **Lanugo** is fine, downy, unpigmented hair that appears on the fetus in the last 3 months of development.
- ◆ **Vellus** by the time of birth, most of it is replaced by similar fine, unpigmented hair called **vellus**.
- ◆ Except for the eyebrows, eyelashes, and hair of the scalp, all of the hair of children, two-thirds of the hair of women and one-tenth of the hair of men is **vellus**.
- ◆ **Terminal hair** is longer, coarser, and pigmented. It occurs on the scalp, eyebrow, and eyelashes; at puberty it replaces the vellus in the axillary and pubic regions, on the face of males (to form the beard), and to varying degrees on the trunk & limbs.)

Color of Hair

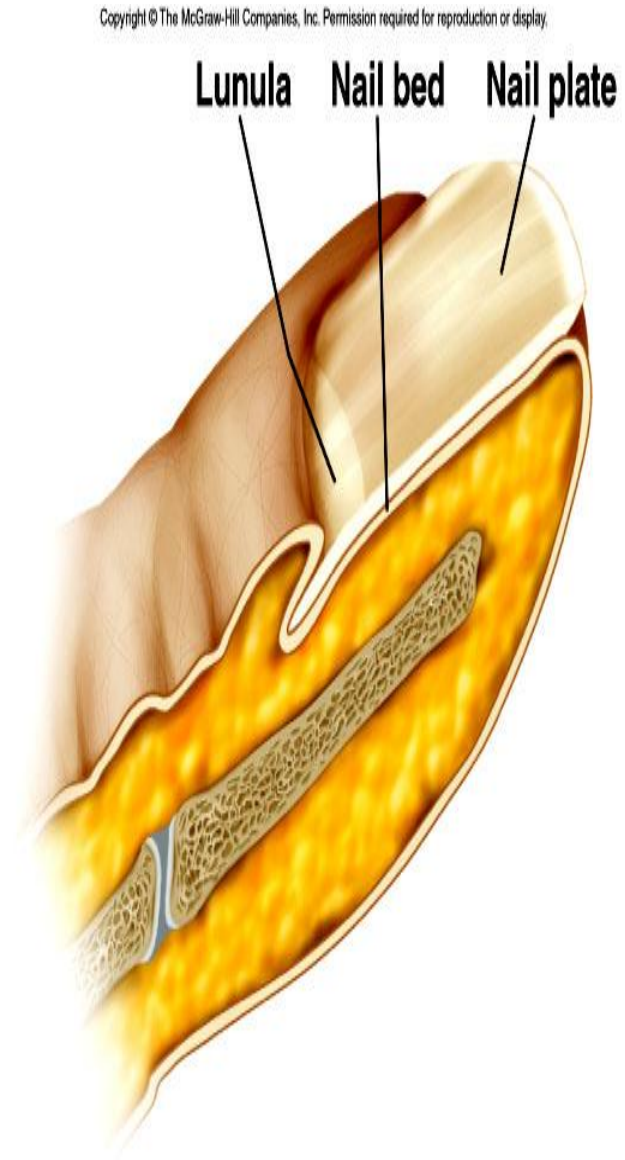
- ◆ Hair color is due to pigment granules in the cells of the cortex.
- ◆ **Brown and black** hair are colored by a form of melanin called Eumelanin.
- ◆ **Blond hair** has a scanty amount of eumelanin but a moderate amount of sulfur-containing pigments called Pheomelanins.
- ◆ **Red** has a slight amount of eumelanin but an abundance of pheomelanins. White hair results from a lack of pigments in the cortex and a presence of air in the medulla.

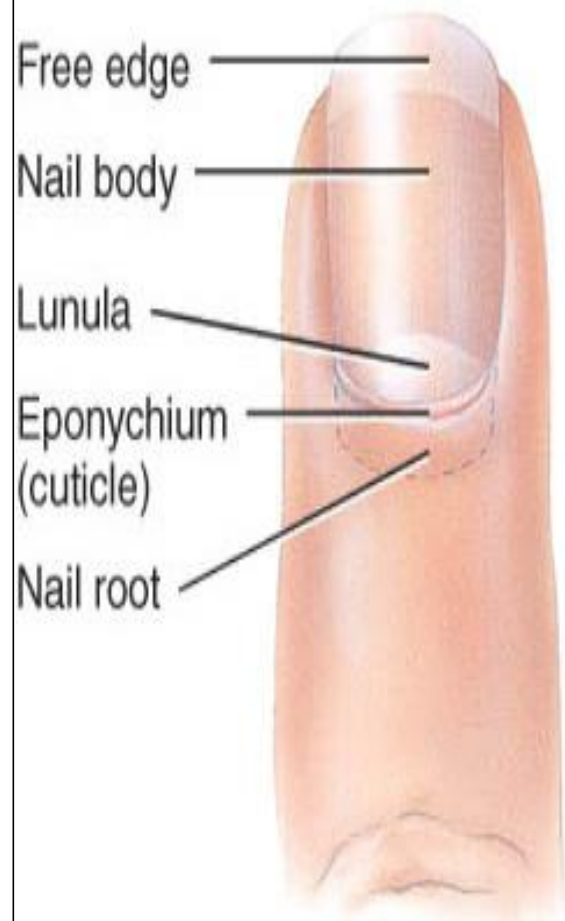
NAILS

- ◆ Plates of stratified squamous epithelial cells with hard keratin
- ◆ Protect distal ends of phalanges
- ◆ Cells are keratinized in the nail root
- ◆ Nail growth occurs in the lunula
- ◆ Cuticle is a fold of stratum corneum on the proximal end of nail

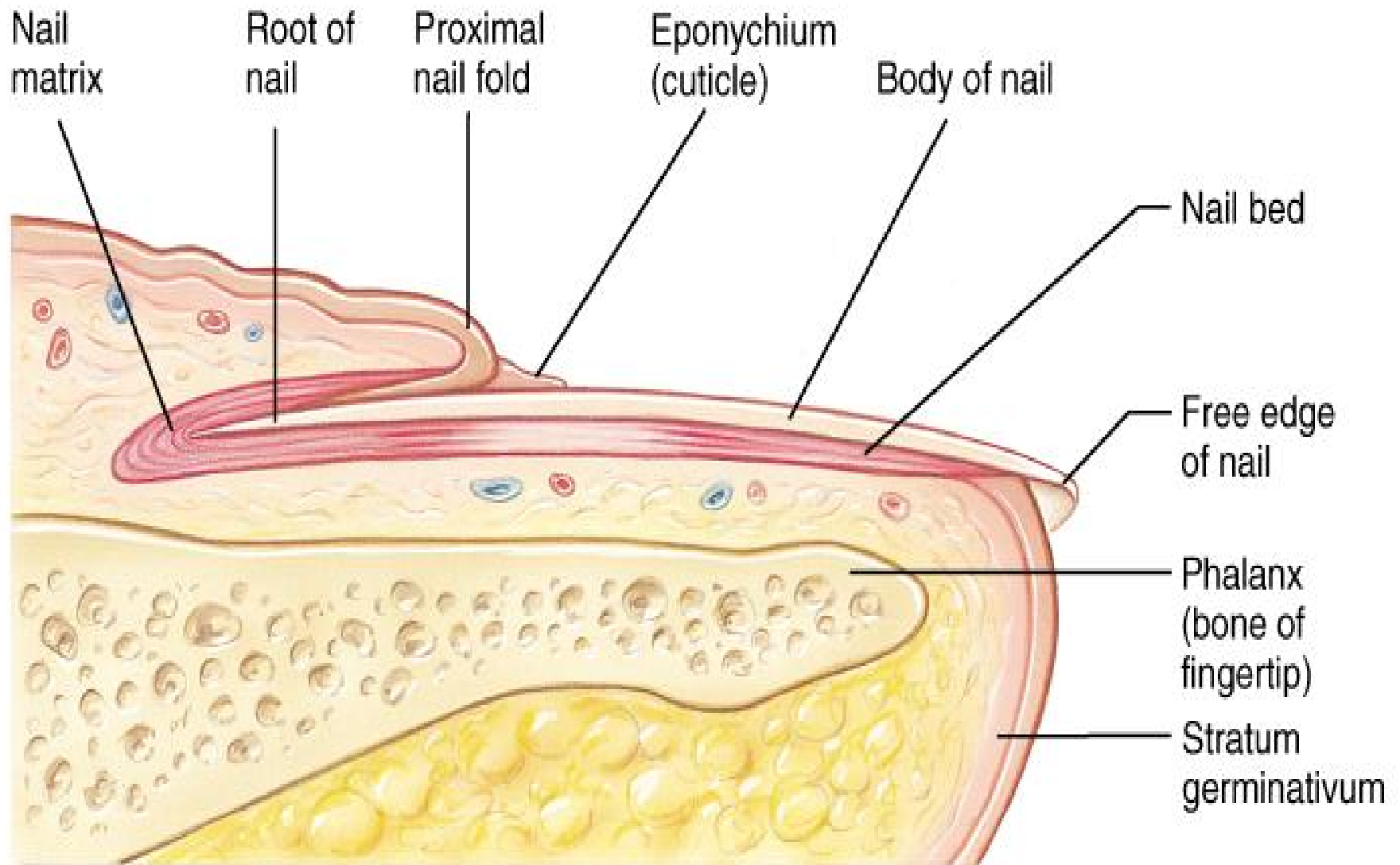
Nails

- **Nail bed** overlaid by nail plate
- Nail growth originates from **root**; cells become keratinized
- Cuticle is formed from stratum corneum





- Tightly packed keratinized cells
- **Nail body:** visible portion pink due to underlying capillaries
- **free edge** appears white
- **Nail root:** buried under skin layers
- **Lunula:** is white due to thickened stratum basale
- **Eponychium (cuticle):** stratum corneum layer
- **Hyponychium** –the lower cuticle

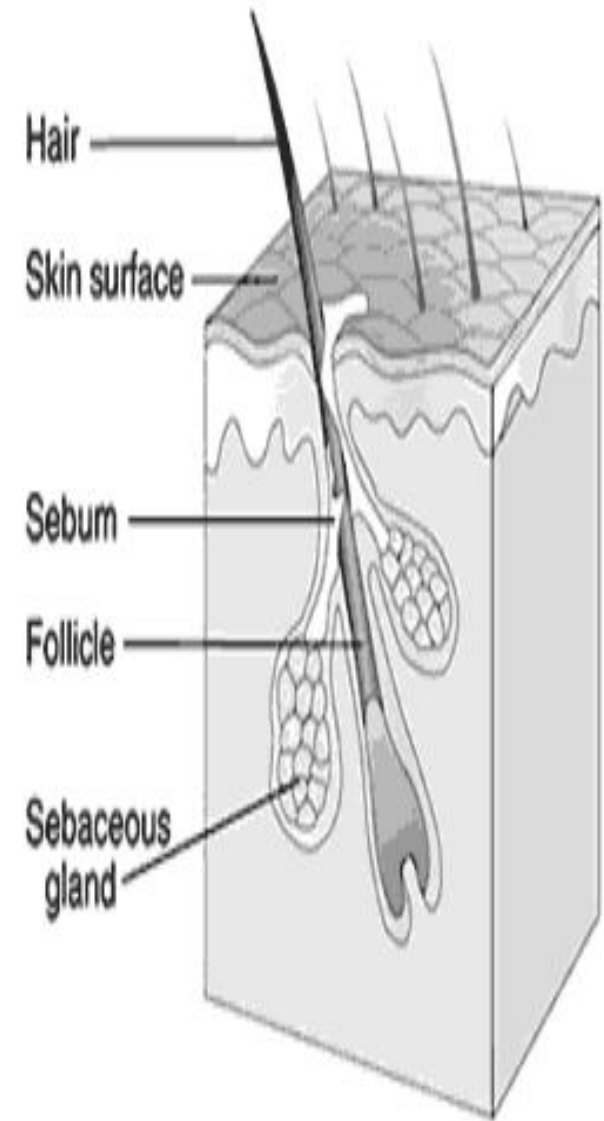


(b)

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Sebaceous Glands

- ◆ Simple alveolar glands found everywhere except palms of the hands and soles of the feet.
 - Do the ducts of these glands branch?
 - Are these glands exo- or endocrine?
- ◆ Secrete an oily, lipid-rich secretion called sebum.
- ◆ Sebum is typically secreted into a hair follicle or occasionally onto the body surface.
- ◆ Sebum softens and lubricates the skin.
- ◆ It also decreases the skin's permeability to water and is quite bactericidal.

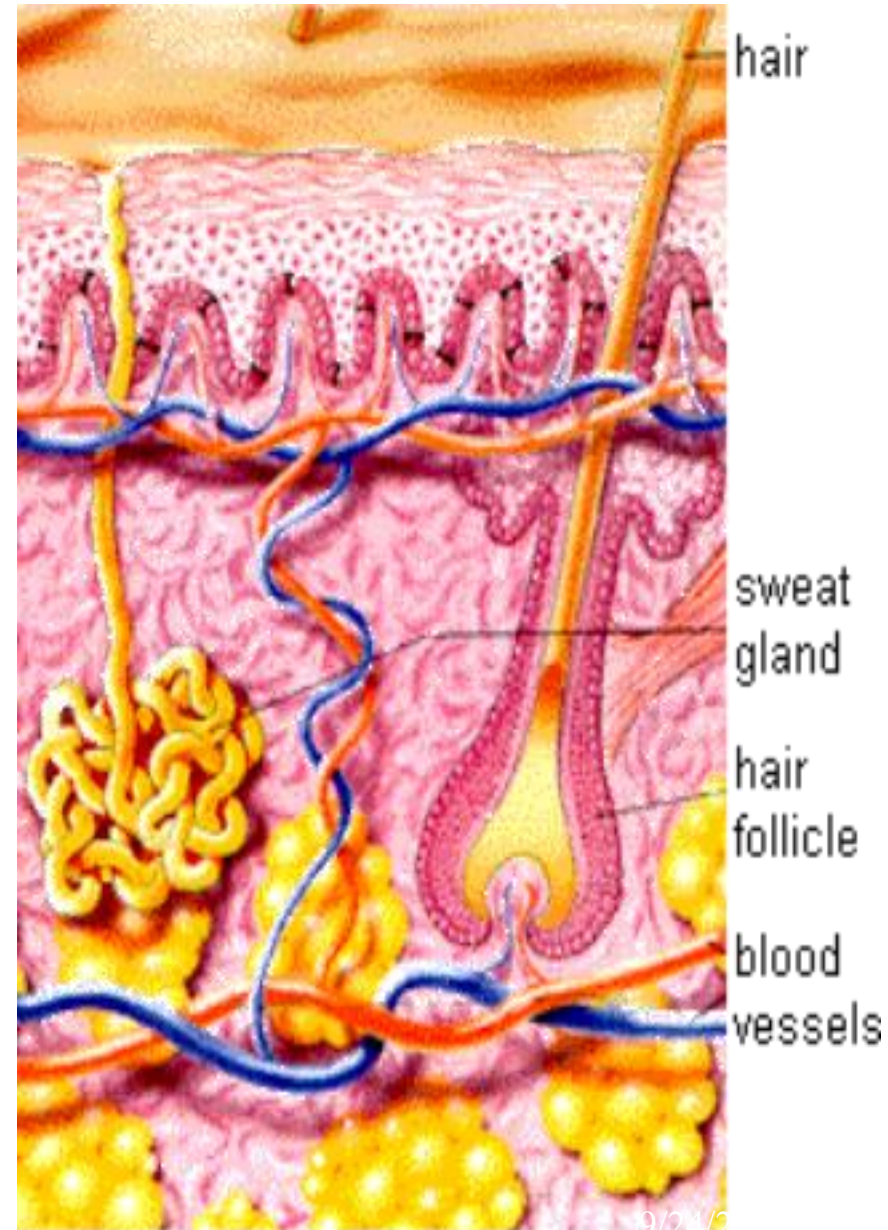


The sebaceous gland is indicated by the arrow. Note how its duct is unbranched and how it empties into a hair follicle.



Sudoriferous Glands

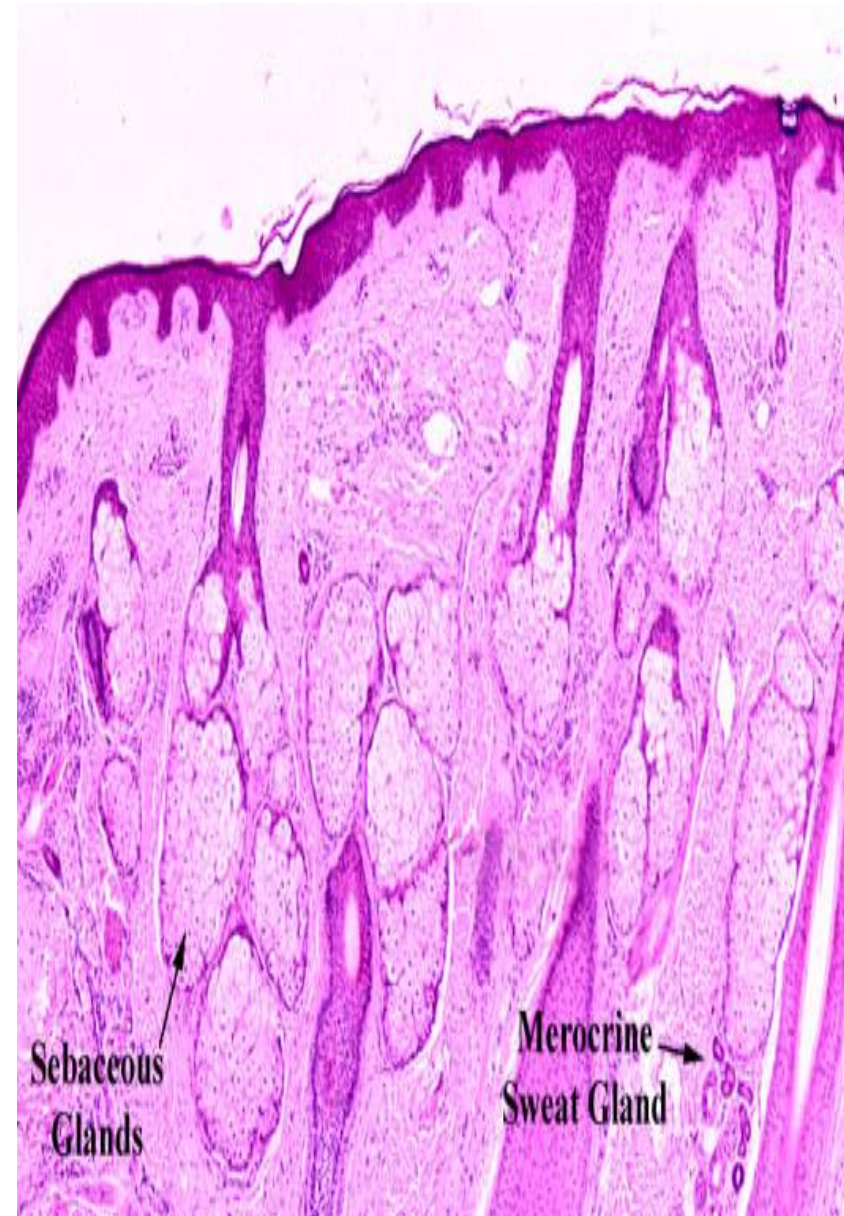
- ◆ Sweat glands.
- ◆ Distributed over the entire body except the nipples and portions of the external genitalia.
- ◆ Over 2.5 million per person.
- ◆ Are 2 types:
 1. **Merocrine** sweat glands
 2. **Apocrine** sweat glands



Merocrine Sweat Glands

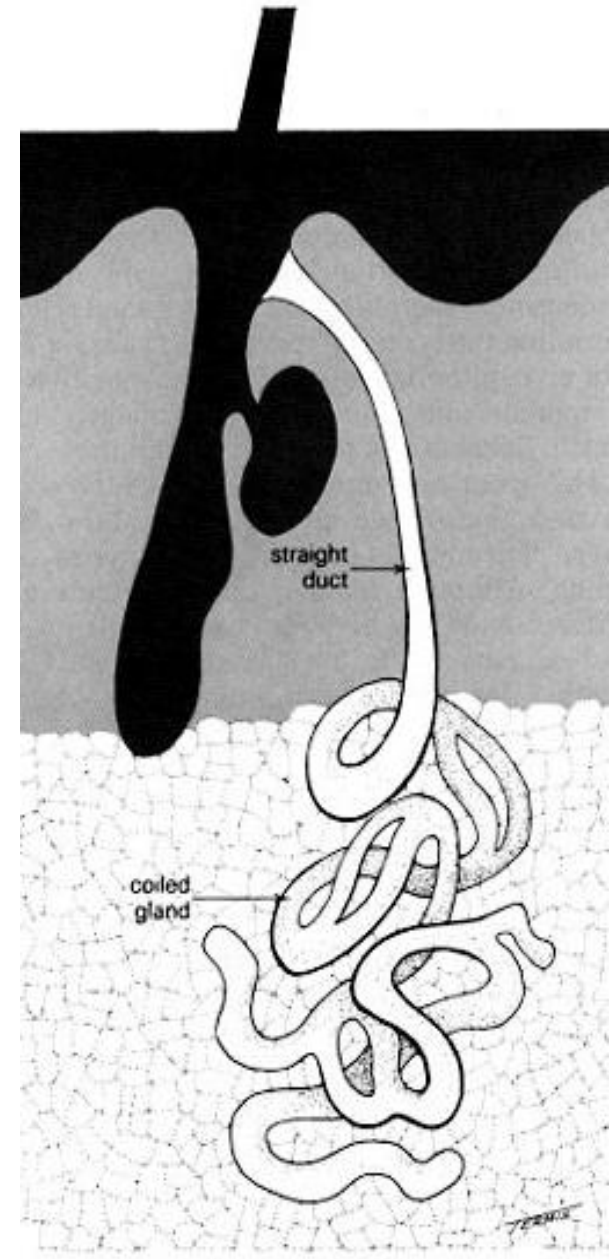
- More numerous than apocrine sweat glands.
- Especially prominent on the palms, soles, and forehead.
- Simple, coiled, tubular glands.
- Duct empties into a funnel-shaped pore at the skin surface.
- Major function of merocrine sweating is to cool the body **thermoregulation**.

- **Merocrine sweat** is a dilute watery solution of some salts (including NaCl), vitamin C, antibodies, small amounts of nitrogenous wastes (urea, uric acid, and ammonia), and lactic acid.
- pH of sweat is 4-6 creating a film on the body known as the acid mantle.
- Such an acidic environment is bacteriostatic – prevents bacterial reproduction and growth.



Apocrine Sweat Glands

- ◆ Found primarily in the axillary, pubic, and anal regions of the body.
- ◆ Also found in the facial region in men only.
- ◆ Larger than merocrine sweat glands.
- ◆ Ducts empty into hair follicles.
- ◆ Apocrine sweat is thicker than merocrine sweat since it contains more lipids and proteins. When it's degraded by epidermal bacteria, foul odors can be produced.



Apocrine Sweat Glands

- ◆ Become active at puberty.
- ◆ Secrete sweat during times of pain, stress, and sexual activity.
- ◆ May (??) have a possible pheromone-like function.
- ◆ Play no role in thermoregulation.

Remember: *Glands are derivatives of epithelial tissue. So even though many reside in the dermis, they are not made from connective tissue.*

Ceruminous Glands



- Modified apocrine glands found in the inner lining of the external ear canal.
- Secrete a sticky waxy substance called **cerumen (earwax)**.
- It combines with sebum and dead epidermal cells to form a bitter compound that offers a barrier to entry of the ear.

Disorders of the Integumentary System

Acne vulgaris

- Caused by increased secretion of oil by sebaceous glands related to increased hormones during puberty

Albinism

- Inherited disorder in which melanin is not produced

Alopecia

- Baldness

Athlete's foot

- Contagious fungal infection of the foot Cellulitis
- Bacterial infection of the dermis and subcutaneous layer of the skin

Chloasma

- Patchy discoloration of the face

Contact dermatitis

- Allergic reaction that may occur after initial contact or as an acquired response

Dandruff

- White flakes of dead skin cells from the scalp

Decubitus ulcers

- Sores or areas of inflammation that occur over bony prominences of the body

Eczema

- Group of disorders caused by allergic or irritant reactions

Fungal skin infections

- Skin infections that live on dead outer surface or epidermis

Furuncle

- Boil, or bacterial infection of a hair follicle

Hirsutism

- Abnormal amount of hair growth in unusual places

Impetigo

- Very contagious bacterial skin infection that occurs most often in children

Kaposi's sarcoma

- Form of cancer that originates in blood vessels and spreads to skin

Vitiligo

- Condition that causes loss of pigment in the skin

Wart

- Papule caused by a viral infection

Skin cancer

- Three forms are basal, squamous, and melanoma

Skin lesions

- Differ in texture, color, location, and rate of growth

Streptococcus

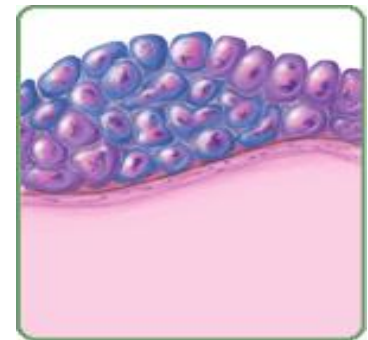
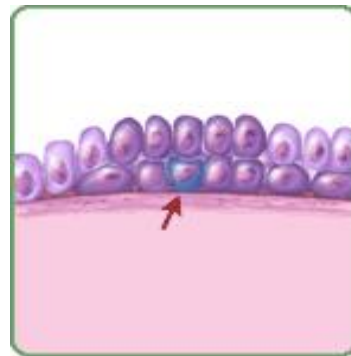
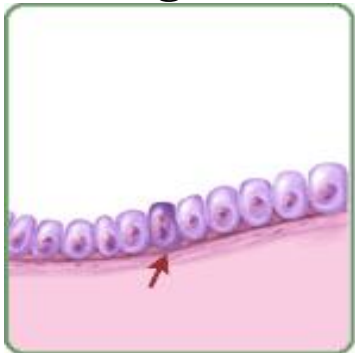
- Non-motile bacteria that affect many parts of the body

Issues and Innovations

- Skin and hair care
 - Skin care products include soap, astringents, and cosmetics
 - Ways to remove hair include wax, depilatory creams, and electrolysis
- Sun and skin cancer
 - Damage by ultraviolet rays
 - Types of skin cancer

Skin Cancer

- Because of its role as our external covering, the skin takes a tremendous amount of abuse.
- One serious disorder that can result is skin cancer.
- Cancer can be thought of as uncontrolled cell division and growth.
- There are **3 types of skin cancers** we will discuss:
 - **Basal cell carcinoma**
 - **Squamous cell carcinoma**
 - **Malignant melanoma**



An abnormal cell develops

From abnormal cells, a cancerous cell develops

Cancerous cells spread, forming a tumor

Skin Cancer

- **Basal cell carcinoma**

- Most common (70% of skin cancers)
- Least vicious
- Usually cured via surgical removal
- Consists of uncontrolled growth of cells of the stratum basale. They'll proliferate and invade the dermis and hypodermis.
- Often occurs on sun-exposed areas of face and neck

Squamous cell carcinoma

- Arises from keratinocytes of stratum spinosum.
- 25% of cases.
- Good prognosis if caught and treated early (surgical excision or radiation).
- Can be fatal if it metastasizes to the lymph nodes.



Malignant melanoma

- Least common and most dangerous.
- Cancer of melanocytes.
- Often arises from a pre-existing mole.
- Follow the ABCD rule for early detection:

Asymmetry (2 sides do not match)

Border irregularity

Color (multiple)

Diameter (>6mm is bad!)



Burns

■ First-degree

- ✓ only the epidermis is damaged.
- ✓ Symptoms include localized redness, swelling, & pain

■ Second-degree

- ✓ the epidermis and upper regions of dermis damaged.
- ✓ Symptoms mimic first degree burns, but blisters also appear

■ Third-degree

- ✓ involve entire thickness of the skin, and may extend into the subcutaneous tissue, exposing the victim to fluid loss & infection.
- ✓ Burned area appears gray-white, cherry red, or black, and there is no initial edema nor pain (since nerve endings are destroyed)

First-degree
(epidermis only; redness)



(b)

Second-degree
(epidermis and dermis, with blistering)



(c)

Third-degree
(full thickness, destroying epidermis, dermis, often part of hypodermis)

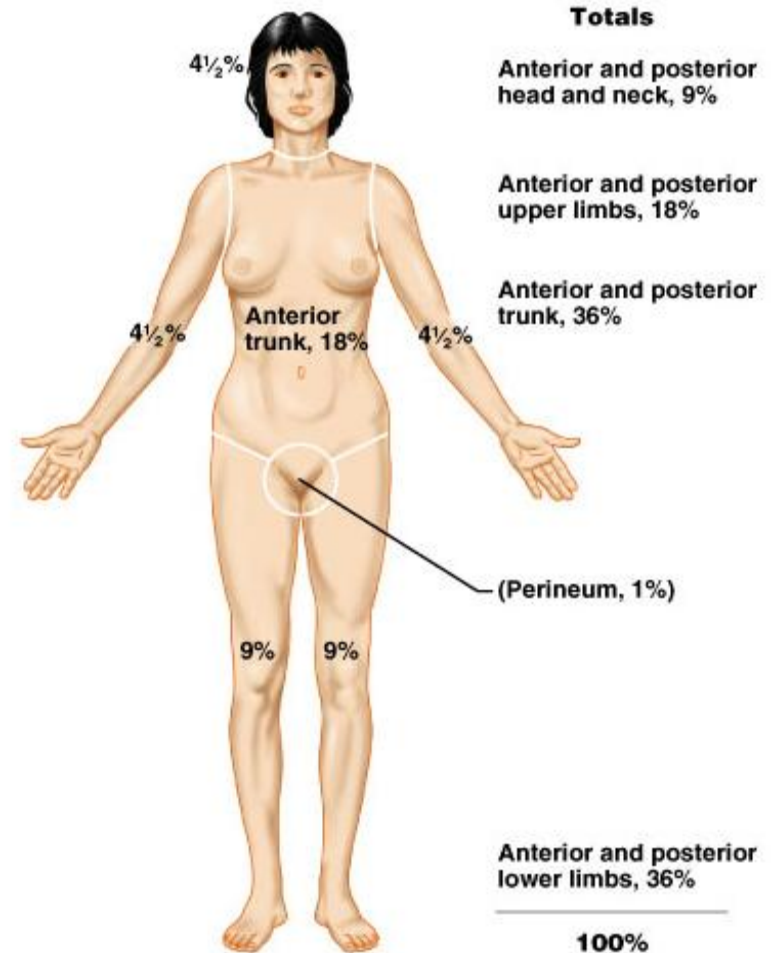


(d)

Critical burns

- Over 10% of the body has third-degree burns
- 25 % of the body has second-degree burns
- Third-degree burns on face, hand or feet
- When more than 20% of the body has been burned, the injury is often fatal.

Estimate by “rule of 9’s”



(a)

Tattooing, Albinism and Vitiligo



THANK YOU!